

IEEE Guide for Diagnostic Field Testing of Fluid-Filled Power Transformers, Regulators, and Reactors

IEEE Power and Energy Society

Sponsored by the
Transformers Committee

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USA

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(Revision of
IEEE Std 62™-1995)

IEEE Guide for Diagnostic Field Testing of Fluid-Filled Power Transformers, Regulators, and Reactors

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Abstract: Diagnostic tests and measurements that are performed in the field on fluid-filled power transformers and regulators are described. Whenever possible, shunt reactors are treated in a similar manner to transformers. Tests are presented systematically in categories depending on the subsystem of the unit being examined. A diagnostic chart is included as an aid to identify the various subsystems. Additional information is provided regarding specialized test and measuring techniques. Interpretive discussions are also included in several areas to provide additional insight on the particular test or to provide guidance on acceptance criteria. These discussions are based on the authors' judgment of accepted practice. It should be noted that the results of several types of tests should be interpreted together to diagnose a problem. Manufacturers' acceptance criteria should also be consulted as it may take precedence over the criteria in this guide.

Keywords: bushing, core, diagnostic evaluation, field testing, fluid-filled transformer, IEEE C57.152™, insulating liquid, off-line testing, reactor, regulator, safety, tank, tap changer, winding

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Introduction

This introduction is not part of IEEE Std C57.152-2013, IEEE Guide for Diagnostic Field Testing of Fluid-Filled Power Transformers, Regulators, and Reactors.

Power transformers usually represent one of the most important and single most costly items in substations. Furthermore, particularly for large transformers, their failures usually result in lengthy outages or downgrading of electric service reliability. For these reasons, a high degree of care is required to properly field test this equipment to confirm equipment status and identify problems.

Because of these considerations, IEEE and other standards development organizations have published, since at least the early 1920s, various recommendations for testing and maintaining transformers. This guide replaces IEEE Std 62™-1995 [B33], since it primarily deals with power transformers, regulators, and reactors, which are devices covered by the Transformers Committee.^a

New sections have been added on safety; tank vacuum testing; visual inspection; a chart providing commissioning, routine, and after-fault testing guidance; and informational annexes. Also, new technologies have been identified that are available for use in field testing.

^a The numbers in brackets correspond to those of the bibliography in Annex J.

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1. Scope

This guide describes diagnostic field tests and measurements that are performed on fluid-filled power transformers and regulators. Whenever possible, shunt reactors are treated in a similar manner to transformers. The tests are presented systematically in categories depending on the subsystem of the unit being examined. A diagnostic chart is included as an aid to identifying the various subsystems. Additional information is provided regarding specialized test and measuring techniques.

Interpretive discussions are also included in several areas to provide additional insight on the particular test or to provide guidance on acceptance criteria. These discussions are based on the authors’ judgment of accepted practice. It should be noted that the results of several types of tests should be interpreted together to diagnose a problem. Manufacturers’ acceptance criteria and other standards in the IEEE C57™ series take precedence over the content of this guide.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 4TM, IEEE Standard Techniques for High-Voltage Testing.^{1, 2}

IEEE Std 510TM, IEEE Recommended Practices for Safety in High-Voltage and High-Power Testing.

IEEE Std C57.12.80TM, IEEE Standard Terminology for Power and Distribution Transformers.

IEEE Std C57.12.90TM, IEEE Standard Test Code for Liquid-Immersed Distribution, Power and Regulating Transformers.

IEEE Std C57.93TM, IEEE Guide for Installation and Maintenance of Liquid-Immersed Power Transformers.

3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* [B32] and IEEE Std C57.12.80 should be consulted for terms not defined in this clause.^{3, 4, 5}

apparent charge (terminal charge): A charge that, if it could be injected instantaneously between the terminals of the test object, would momentarily change the voltage between its terminals by the same amount as the partial discharge (PD) itself. The apparent charge should not be confused with the charge transferred across the discharging cavity in the dielectric medium.

NOTE 1—Apparent charge, within the terms of this guide, is expressed in coulombs (C). One picocoulomb (pC) is equal to 10^{-12} C pulse charge transferred from the PD source to the terminals of the test object.⁶

NOTE 2—The **apparent charge** is measured in terms of picocoulomb (pC) using a calibrated PD measuring circuit as specified in IEEE Std C57.113TM-2010 [B41].

NOTE 3—The **apparent charge** is different from the PD charge because that charge originated at the PD site, which cannot be measured directly.

NOTE 4—The **apparent charge** is sometimes referred to as *terminal charge*.

partial discharge (PD): An electric discharge that only partially bridges the insulation between conductors, and which may or may not occur adjacent to a conductor.

NOTE—PD events may ignite in gaseous dielectrics due to a localized electrical field enhancement. Generally, PDs are caused by dielectric imperfections, such as gaseous inclusions in solid and liquid dielectrics as well as protrusions on electrodes in ambient air.

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³ *IEEE Standards Dictionary Online* subscription is available at:
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⁴ The numbers in brackets correspond to those in the bibliography in Annex J.

⁵ Information on references can be found in Clause 2.

⁶ Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

partial discharge (PD) level: Mean peak value of the apparent charge of PD pulses, whose magnitudes are randomly distributed and are evaluated by a PD measuring instrument specified in IEEE Std C57.113-2010.

NOTE—See IEEE Std C57.113-2010 [B41].

partial discharge (PD) measuring instrument: Analog or digital equipment for wideband measurement of the apparent charge as specified in IEEE Std C57.113-2010.

NOTE—See IEEE Std C57.113-2010 [B41].

radio interference voltage (RIV) level: Mean peak value of the RIV evaluated by an RIV measuring instrument specified in NEMA 107.

NOTE—See NEMA 107 [B57].

radio interference voltage (RIV) measuring instrument: Analog or digital equipment for narrowband measurement of the RIV of partial discharge (PD) events as specified in NEMA 107.

NOTE 1—See NEMA 107 [B57].

NOTE 2—**RIV measuring instrument** is sometimes referred to as an *RIV meter*.

4. Purpose of tests

Transformers are critical components within the overall architecture of a power system network and represent a substantial investment on the part of the user. The life cycle of such devices encompasses manufacture, transport, installation, and in-service aging and maintenance. Each period in the life of a transformer includes unique challenges to its integrity of which the manufacturer and user must be aware. Undetected damage or degradation of the transformer at any stage of this process can predispose the unit to failure. Because the premature loss of a large power transformer can impose significant fiscal, logistical and operational challenges, IEEE guides and standards have been developed to assist in its assessment during each life cycle period.

Large power transformers are devices that generally provide many years of service when well built and maintained. Collectively, the life of transformers roughly follows the classical “bathtub” curve, with a small number of units subject to premature failure, followed by a long period with a low failure rate and then a period of increasing failures as they approach end-of-life. Each set of tests described in IEEE guides and standards is intended to help detect and thereby reduce failures during the first two periods and help the user to predict the outcome and take actions to delay the onset of the final phase.

Factory tests (routine, design, and conformance) such as those described in IEEE Std C57.12.90 are intended to verify that the units are designed and manufactured to meet customer and industry specifications. Such tests are intended to reduce failures during every segment of the life curve. Field tests described herein can be divided into several categories associated with stressors that are unique to each period in the unit’s life cycle (transportation, installation, in-service aging, and maintenance). Such tests seek to identify deviations from the unit’s original condition at the factory. Therefore, optimum interpretation of the field tests described herein requires access to the original tests to quickly identify deviations or trends.

Power transformers, regulators, and reactors are installed in a wide variety of applications. Users need to evaluate a number of parameters, whether selecting tests in response to a specific need for a single transformer or establishing a life cycle management program for an entire fleet. Such considerations include, but are not limited to the following:

- Cost of the transformer(s)
- Criticality of the connected load(s)
- Vintage of the unit(s)
- Loading
- Manufacturer(s)
- Service history of the unit(s) (or of units of similar design)
- Accessories
- Service environment (lightning exposure, through-fault exposure, etc.)
- Availability of a spare(s) or lead time to acquire a new replacement(s)
- Insurance costs

The transportation phase in the life of a large power transformer is brief but may present significant structural and environmental challenges to the unit (see IEEE Std C57.150™-2012 [B48]). Field tests sensitive to the shifting of internal components and those sensitive to adverse environmental exposure during shipment should be selected to identify changes to the unit's integrity since leaving the factory or other point of origin.

The installation phase of a transformer's life is also brief but requires certain select field testing (beyond that which confirms the lack of transportation damage) to validate the correct configuration of the transformer and its accessories, to confirm the proper processing and liquid filling of the tank, and to establish a baseline for future condition assessments (see IEEE Std C57.93).

The service period of the transformer's life cycle is the bottom of the bathtub curve, a long period having a low failure rate. Field testing during this period is intended to identify adverse trends in the aging of the transformer and its accessories. Such testing may be time or condition based, depending on the recommendation of the transformer manufacturer and the philosophy of the user. When indicated, the user may elect to perform additional "special" tests to confirm the onset of significant aging. Such confirmation permits the user to take actions to refurbish select components or begin the process to procure replacement transformers (see IEEE Std C57.140™-2006 [B44]). Service period field testing may also be performed to validate the integrity of the transformer following exposure to normal and abnormal service events such as through faults and lightning surges or any event that causes actuation of the transformer's protective relaying.

During the service period of the life cycle of a transformer, maintenance activities are performed to help preserve its integrity and prolong its useful life. Field testing following certain maintenance activities is not intended to identify aging but seeks to confirm that those activities achieved the desired result, to confirm that new or modified components or accessories are properly functioning, to verify that the unit is in its proper configuration prior to being returned to service, and to obtain data that serves as the new baseline for future evaluations.

The following subclauses describe the fundamentals of the individual tests and provide the user with guidance regarding their applicability and interpretation. Users should be aware that the described tests vary widely in the complexity and cost of the equipment involved and the skill of the operator. Users should carefully consider this data (and that in the references), as well as the condition of the transformer, to determine whether tests should be performed by in-house staff or by a testing service organization.

Given the critical nature of power transformers and the advanced age of many of those assets, significant efforts are underway to advance condition monitoring (see IEEE Std C57.143™-2012 [B45]) and diagnostic technologies. It is recommended that users keep abreast of such developments through other IEEE guides and standards, technical literature, and conferences. The accuracy of test results is critical when comparing them with the results of benchmark tests. It is imperative that the tests be conducted in a

manner consistent with previous tests and while following the instructions for the test device(s) being used to perform the tests.

5. Maintenance tests and information

5.1 Recommended, as-needed, and optional maintenance tests

Table 1 is a compilation of the recommended, as-needed, and optional maintenance tests typically performed on liquid-filled power transformers during their commissioning, while they are in service, and after protection trips caused by either a system fault or an internal fault.

Table 1—Maintenance test chart

Maintenance test	Liquid-filled power transformer			
	Commissioning ^a	In-service ^b	After protection trip due to system fault ^c	After protection trip due to internal fault ^d
Main tank				
Tank pressure	Opt	Opt	Opt	REC
Core ground test	REC	AN	AN	REC
Insulating liquid quality tests and dissolved gas analysis (DGA)	REC	REC	AN	REC
Furan test	Opt	Opt ⁵	Opt	REC
Vacuum	REC	Opt	Opt	REC
Insulation resistance	REC	AN	AN	REC
Winding resistance	REC	AN	AN	REC
Turns ratio (DETC taps)	REC	AN	AN	REC
Excitation current	REC	AN	AN	REC
PF/Tan-Delta	REC	AN	AN	REC
Partial discharge (PD)	Opt	Opt	Opt	Opt
Induced voltage	Opt	Opt	Opt	Opt
Frequency response analysis (FRA)	REC	AN	AN	REC
Dielectric frequency response (DFR)	Opt	Opt	Opt	Opt
Infrared	N/A	REC	N/A	N/A
Bushing				
Contact resistance	Opt	N/A	N/A	Opt
Infrared	N/A	REC	N/A	N/A
PF/Tan-Delta	REC	REC	AN	REC
Continuity	REC	N/A	N/A	REC

Table 1—Maintenance test chart (continued)

Maintenance test	Liquid-filled power transformer			
	Commissioning ^a	In-service ^b	After protection trip due to system fault ^c	After protection trip due to internal fault ^d
Load tap changer (LTC) and de-energized tap changer (DETC)				
Insulating liquid quality tests and DGA for LTC	REC	REC	AN	REC
Contact continuity for LTC	REC	AN	AN	REC
Infrared for LTC	N/A	REC	N/A	N/A
Motor current signature analysis for LTC	REC	AN	AN	REC
Vibration and acoustic measurement for LTC	Opt	Opt	Opt	Opt
Voltage dynamic testing for LTC	Opt	Opt	Opt	Opt
Ancillary equipment				
Gauges calibration	REC	REC	Opt	REC
Gas pressure relay calibration	REC	REC	Opt	REC
Pressure relief vent	REC	REC	Opt	REC
Cooling fan controls	REC	REC	Opt	REC
Cooling pump controls	REC	REC	Opt	REC
Arresters	REC	REC	REC	Opt
Bushing CTs	REC	AN	AN	AN

REC = Recommended

AN = As needed based on the REC Test results

Opt = Optional based on the AN test results

N/A = Not Applicable

^a Newly installed or repaired units prior to energization.

^b In-service transformers may need to be de-energized and properly set up, depending on the test to be performed. Condition-based maintenance practice—oil quality, DGA, and Furan tests—may be carried out at a regular interval and the necessity of other tests depend upon the assessed condition for power and distribution transformers. For hermetically sealed distribution transformers, the first round of tests after commissioning may be time based, and thereafter, the frequency should depend on the assessed condition.

^c After tripping of transformer due to system faults such as overcurrent.

^d After tripping of transformer due to internal faults such as differential tripping (before repair).

^e Furan Testing recommended for generator step-up (GSU) transformers and units operated above nameplate.

5.2 EPRI Power Transformer Maintenance and Application Guide

The EPRI Power Transformer Maintenance and Application Guide [B26] provides maintenance information pertaining to power transformers at nuclear plants. This public document incorporates a technical overview of transformers and a maintenance program designed to help plants avoid transformer failures. The guide covers component design and construction as background for personnel involved with transformers. It also provides warnings and precautions related to temperature rise, loss of cooling, low liquid level, low gas pressure, and static electrification mitigation to assist users in understanding how these factors affect transformer operability and life.

6. Safety

6.1 General

Considerations of safety in electrical testing apply not only to personnel, but also to the test equipment and apparatus being tested. The following guidelines cover many of the fundamentally important procedures that have been found to be practical. Since it is impossible to cover all aspects in this guide, test personnel should also consult IEEE Std 510; manufacturers' instruction manuals; and union, company, and government regulations.

Users of this guide are responsible for determining and complying with appropriate safety, security, environmental, and health and welfare practices, laws, and regulatory requirements applicable to their location, systems, equipment, and operations.

6.2 Types of hazards

6.2.1 Electrical hazards

There are three main electrical hazards—shock, arc flash, and arc blast—to which test personnel may be exposed, particularly if the transformer is not electrically isolated following proper procedures as mentioned in Clause 6.3. Other hazards may be present, and the user should take necessary precautions. Electric shock is contact with energized electrical equipment, conductors, or circuit parts that causes the flow of electrical current through the body. The severity of the shock is determined by the amount of electrical current, the total time that it flows through the body, and where it flows through the body. Humid or wet conditions or sweaty skin increase the potential for electrical shock. At a minimum, the shock hazard must be considered at any voltage greater than or equal to 50 V. Burns to the skin are also another result of an electrical shock. Internal damage caused by the electrical current flowing through the body is also possible; if shocked, the worker should report the incident and seek medical attention.

Electrical equipment that faults and creates an arc flash can expose a worker to extreme heat causing severe burn. Some secondary hazards related to an arc flash are the following:

- Fire
- Toxic smoke inhalation from vaporized copper
- Sound pressure that could damage hearing
- High intensity, ultraviolet, and infrared light that may damage eyesight
- Flying molten metal that may cause injury

Arc blast is associated with the release of tremendous pressures as a result of an arc fault where current flows through the air between two conductors or a conductor and ground. Vaporized copper, molten metal, pressure waves, shrapnel, intense noise, and toxic smoke/gases are some of the resultants of an arc blast. Dangers associated with an arc blast event are high pressures, sound, and shrapnel.

Users of this guide should follow fire safety and other safety requirements and precautions, including, but not limited to, personal protective equipment (PPE) and facility protections, in connection with any testing or evaluation of the transformer equipment.

6.2.2 Other hazards

When working on a transformer, the following additional hazards should also be taken into account when planning the work and completing a Pre-Job Hazard Analysis:

- Falling from heights—Proper harness to provide fall protection should be considered.
- Confined space—Prior to entry, confirmation should be made that the atmosphere inside the tank is adequate to support life. This should be checked according to company guidelines and procedures or manufacturer's instructions.
- Outdoor and wet environments.
- Certain test procedures could result in fire; therefore, non-contaminating fire-fighting equipment should be available before beginning tests that apply dielectric stress to the transformer insulation system.
- The voltage may accidentally exceed the desired maximum during the conduction of high voltage (HV) tests. A sphere gap, adjusted to spark over at a voltage slightly above the desired maximum, may be connected across the voltage source (refer to IEEE Std 4). By selecting the proper value of series resistor, the gap may be used to provide a warning signal, to inhibit further rise in the test voltage, or to activate an overcurrent circuit breaker in the power supply circuit.
- Tests being performed on the transformer while the equipment is under vacuum should only be done with low applied voltages. The dielectric strength of the system is significantly reduced under these conditions. See 6.6.3 for more details.

6.3 Creating an electrically safe work condition

6.3.1 Generic safety principles

WARNING

Electrical equipment should be considered energized until it is proven de-energized and grounded. No person should begin work on de-energized parts until this verification has been completed.

To dissipate residual charges, all terminals should be discharged to ground after test voltages have been removed.

The following generic safety principles should be followed during the execution of transformer testing:

- Do not rush when planning or carrying out the testing work.
- No worker should begin any electrical work until the worker fully understands the instructions received, and in no circumstances should that person exceed those instructions. Should any person consider that the instructions given cannot be carried out safely, that person should refer the matter immediately to an appropriate supervisor.
- No worker should interfere with ground connections, locks, tags, danger or warning signs, safety barriers, flags or other safety devices.
- Do not work on any electrical equipment or circuit where the area is damp or wet until insulated rubber matting is put into place and the circuit is isolated and grounded.
- Only use tools that are properly insulated and approved.

- Conductive articles of jewelry and clothing (such as watchbands, bracelets, rings, keys, chains, necklaces, metalized aprons, cloth with conductive thread, or metal headgear) shall not be worn.
- Near misses and electrical incidents (arc flash and shock) shall be reported immediately. These incidents should be fully investigated, lessons learned, and recommendations implemented.
- During the execution of a task, if any changes are noticed from the planned procedures, then immediately stop the task, think and analyze, assess the risk, control the risk, and then resume work if appropriate.
- Safety signs, safety symbols, or accident prevention tags shall be used where necessary to warn personnel about electrical hazards that may endanger them. Non-conductive barricades shall be used in conjunction with safety signs where it is necessary to prevent or limit individual access to work areas exposing individuals to non-insulated energized conductors or circuit parts. If signs and barricades do not provide sufficient warning and protection from electrical hazards, a standby/signal person shall be stationed to warn and protect employees from entering the area.

6.3.2 Steps to be followed for creating an electrically safe work condition

Before any testing work is performed, an electrically safe work condition shall be evaluated in accordance with the following steps:

- Determine all possible sources of electrical connections to the specific transformer. Check applicable as-built up-to-date single line drawings, diagrams, and identification tags.
- After properly interrupting the load current, open the disconnecting device(s) to the specific transformer.
- Where it is possible, visually verify that blades of the disconnecting devices are fully open or draw-out circuit breakers are withdrawn to the fully disconnected position.
- Apply lockout/tagout devices in accordance with an established company procedure.
- Use an adequately rated voltage detector to test each phase conductor or circuit part to verify they are de-energized. Before and after each test, determine that the voltage detector is operating satisfactorily. Test-before-touch.
- Where the possibility of induced voltages or stored electrical energy exists, ground the phase conductors or circuit parts before touching them. Where it could be reasonably anticipated that the conductors or circuit parts being de-energized could contact other exposed energized conductors or circuit parts, apply portable ground connecting devices rated for the available fault duty.
- Use of working grounds should comply with established company guidelines. For further information, see ASTM F855-2009 [B20].
- The transformer owner should issue a Safe Work Permit to the maintenance and testing personnel.
- Before any test is performed on the transformer, there should be a meeting at the work site of the people who are involved or affected by the test. The test procedure should be discussed so there is a clear understanding of all aspects of the work to be performed. Particular emphasis should be placed on personnel hazards and the safety precautions associated with these hazards. In addition, procedures and precautions should be discussed to ensure the production of meaningful test results without subjecting the test specimen to unnecessary risks.
- Responsibilities for the various duties involved in performing the test should be assigned.

6.4 General practices for internal inspection

When it is necessary to work inside power transformers, the workers should be aware of the rules and requirements of their company and the local, state, and national codes with jurisdiction over the area in which the work is performed. The work should be done in accordance with such applicable rules, codes, and guidelines.

It is recognized that different organizations have interpreted such rules, codes, and guidelines for confined space entry in different ways and that workers may also interpret such codes in slightly different ways. In the absence of such guidance, or as a minimum level of safe practices, the following items are recommended:

- Only workers who have been trained and are familiar with confined space entry procedures should work inside transformers. Verification of the atmosphere is detailed in 6.5.4.
- Bushing terminals and the transformer tank must be securely grounded and current transformer leads shorted.
- A least one person should remain outside the transformer while others are working inside. This person should keep in visual or audible contact with the workers inside. If a worker inside loses consciousness, the outside worker should call emergency rescue workers and never go into the transformer to attempt to remove the fallen worker.
- Risks of engulfment should be eliminated. Entering a transformer tank without first completely draining it of insulating liquid is not recommended. If an inspection is carried out without the liquid removed, steps must be taken to eliminate the possibility of falling into the insulating liquid. If conservator tanks, radiators, coolers, pipes, or other sections of the transformer have been isolated by valves but not drained, and the quantity of liquid in these areas is sufficient to engulf the worker, the valves used to isolate these sections should be locked in the closed position.

6.5 Suggested general control measures

Consistent with the international and national standards on occupational health and safety management systems, the following principles should be practiced as preventive and protective control measures to help protect personnel from hazards of electricity:

- Eliminate the hazard—de-energize all possible sources.
- Use engineering controls wherever possible.
- Substitute safer work systems, e.g., other materials, processes, or equipment.
- Maintain the transformer following preventive, predictive, and reliability centered maintenance strategies.
- Provide administrative controls—electrical technical and safety training, permitting process, and safe work procedures.
- Provide PPE, including measures for its appropriate use and maintenance.

Electrical workers shall be shielded from injury due to electric shock and arc flash hazards by protective equipment rated for the work to be performed. Electrical-specific PPE and other protective equipment shall be of a safe design and constructed for the specific part(s) of the body to be protected. Electrical-specific PPE and other protective equipment should only be considered as a last line of defense when it comes to mitigating exposure to electrical hazards.

Overhead power lines may present a challenge when work is conducted in proximity to them. It is critical that workers understand that even close limits of approach can produce fatal shock hazards. With this understanding, it should be noted that overhead power lines should be considered energized until otherwise confirmed and directed by the electrical utility.

WARNING

Do not attempt to rescue a victim of an incident without de-energizing the electrical system first and suitably protecting the person who will attempt to rescue the victim.
Use of a rescue hot stick may be necessary.

Workers should regularly receive training in testing procedures. Training of personnel in approved methods of resuscitation, including cardiopulmonary, is needed.

6.5.1 List of potential standard operating practices of a single organization and reference codes

- Standard operating practice of a single organization on Information Handling System (IHS)
- Standard operating practices of a single organization on Electrical Specific Personnel Protective Equipment
- Standard operating practices of a single organization on Safe Work Permit
- Standard operating practices of a single organization on Lockout/Tagging of Equipment and Systems
- Standard operating practices of a single organization on Incident Reporting and Investigation
- National Electrical Code® (NEC®) (NFPA 70®, 2011 Edition) [B60]
- OSHA 29CFR1910, Occupational Safety and Health Standards [B64]
- NFPA 70E®-2012 Standard for Electrical Safety in the Workplace® [B61]
- Applicable ASTM standards

6.5.2 Precautions

When testing, precautions shall be taken to prevent personnel from contacting energized circuits. An observer may be stationed to warn approaching personnel and may be supplied with means to de-energize the circuit. The means may include a switch to shut off the power source and ground the circuit until stored charges are dissipated.

6.5.3 Warning signs and barriers

The test area may be marked off with signs and easily visible tape. Warning signs shall conform to the requirements of governing bodies such as the Occupational Safety and Health Administration (OSHA) in the United States. Danger, Warning, Caution signs should follow the format and convention provided by the NEMA Z535.4-2011 [B58] or OSHA 1910.145 rules [B64].

6.5.4 Atmosphere inside tank

Prior to entry, confirmation should be made that the atmosphere inside the tank is adequate to support life. This should be checked according to company guidelines and procedures or manufacturer's instructions.

WARNING

After the access/manhole cover is removed the transformer should not be entered until the shipping gas (including dry air) is completely purged with breathable dry air that has a maximum dew point of -45°C . The oxygen content must be between 19.5% and 23% before entering the tank. Carbon monoxide levels should also be monitored at a level less than 25 ppm. The lower explosive level should be less than 20%. The replacement of gas with dry air is necessary to provide sufficient oxygen to sustain life. If the unit was initially shipped in dry nitrogen, there is a possibility of trapped nitrogen pockets. In this case, a sufficient vacuum should be held for a predetermined period of time and vacuum released with and refilled with dry breathable air.

6.6 Apparatus

6.6.1 Fire-fighting equipment

Certain test procedures could result in fire; therefore, non-contaminating fire-fighting equipment should be available before beginning tests that apply dielectric stress to the transformer insulation system.

6.6.2 Overvoltage

The voltage may accidentally exceed the desired maximum during the conduction of HV tests. A sphere gap, adjusted to spark over at a voltage slightly above the desired maximum, may be connected across the voltage source (refer to IEEE Std 4). By selecting the proper value of series resistor, the gap may be used to provide a warning signal, to inhibit further rise in the test voltage, or to activate an overcurrent circuit breaker in the power supply circuit.

6.6.3 Testing under vacuum

Some users have a practice of taking dc resistance measurements under vacuum while performing dry outs to determine the insulation temperature. Caution should be used when tests are performed on the transformer while the equipment is under vacuum. The dielectric strength of the system is significantly reduced under these conditions—only sufficiently low voltage should be used; consult with manufacturer to obtain recommended voltage level or actions.

6.6.4 Surge arresters

If the test voltage is expected to approach or exceed the operating voltage of any transformer-mounted surge arresters, the arresters should be disconnected before energizing the transformer with test voltage. This avoids arrester damage and limitation of the test voltage due to arrester operation.

7. Tests and test techniques

7.1 Periodic general inspections

The purpose of a maintenance inspection is to evaluate the transformer's condition and make these findings known to appropriate personnel. Periodic inspection of power transformers and their accessories

contributes to their trouble-free operation. These procedures may identify potential problems before they become serious enough to cause equipment outages.

The following are the routine and scheduled inspection items for transformers and the associated equipment in the substation. Inspection frequencies vary based on manufacturer's suggested care, equipment risk and complexity, and the user's own maintenance practices and procedures.

WARNING

Some inspection items may be near the transformer line connections. Only electrically-qualified personnel should be allowed in this area.

7.1.1 Routine inspections

Check and record the following:

- Line voltage and load current
- Insulating liquid temperature, winding temperature, and ambient temperature, as applicable. Peak indicators should be recorded and reset.
- Insulating liquid levels of the main tank and liquid-filled compartments
- Nitrogen gas pressure for blanketed transformers
- Possible leaks in the transformer if the liquid level gauge remains at or near zero but the actual liquid level varies. This is an important maintenance check that verifies the integrity of the transformer seal.
- Radiators for cleanliness and freedom from obstructions
- Radiator connections, bolted pipe joints, bolted access ports, and valves for signs of insulating liquid leakage.
- Grounding and copper bus bars are still in place and have not been stolen
- Condition of controls, relays, and wiring
- Condition of desiccant gel
- Counter readings from load tap changer, circuit breakers, automatic reclosures, and disconnects
- Operation of cooling fans and insulating liquid circulating pumps, where installed. When automatic controls are installed they should be left in the automatic setting.
- Evidence of animal activity
- Results obtained from performing an insulating liquid combustible gas and *dissolved gas analysis* (DGA)
- Infrared temperature evaluation on tank, bushings, LTC, and control cabinet

7.1.2 Scheduled de-energized inspections

WARNING

Do not attempt any of the following with the transformer still in operation. Always de-energize the transformer and the auxiliaries (fans, pumps, and control cabinet) before conducting these inspections. Failure to follow these precautions may cause equipment damage and personal injury.

- Visually examine bushings, arresters, and interconnecting hardware for cracks, dirt, insulating liquid leaks, excessive corrosion, and signs of overheating or electrical tracking. Clean any contaminated areas with a soft cloth and suitable solvent. Then, wipe the area dry.
- Check radiators connections, bolted pipe joints, bolted access ports, and valves for signs of insulating liquid leakage. Tighten any loose fittings and repair any insulating liquid leaks.
- Examine the pump valves for evidence of leaking around the gland seals. Close and open the flapper operating arm. There should be some restriction to the flapper arm movement if the packing is properly tightened. Tighten the gland nut if necessary.
- If the transformer is equipped with a load tap changer, inspect the tap changer for proper operation. Detailed information for the inspection procedures and the frequency of inspection for the tap changer is usually supplied by the manufacturer.
- Inspect any breathers and small screen openings in pressure-relief valves or a pressure-vacuum breather to be certain they are clean and in operating condition.
- If the transformer is equipped with a conservator, or insulating liquid preservation system, remove the expansion tank breather and check for insulating liquid leakage into the bladder. The procedure for making this inspection is usually explained in the manufacturer's instructions.
- Examine the paint finish on the main tank (particularly around welded joints) and on accessory items such as radiators, coolers, and associated piping. Check for paint peeling or cracking and evidence of rust. Clean the affected areas by wire brushing, then wipe with a clean dry cloth. Paint the area with a touch-up primer and a suitable exterior finish coat.

7.2 Main tank (active part)

7.2.1 General

Almost all electrical equipment is contained in some type of tank. This tank provides mechanical protection for the equipment and also acts as a reservoir for the insulating liquid surrounding the equipment. Attached to the tank are a number of bushings, fittings, and associated devices. The types and number of these devices attached to the tank vary with the size, voltage class, and use of the equipment. Generally a device provides one of three or more functions. The most common of these are the following:

- A visual indication of a condition or state
- An alarm indication of some abnormality
- A benefit to the electrical performance of the equipment

7.2.2 Conservators

Conservators are vessels normally located at an elevation higher than the cover of the tank. They can, however, be located on a structure immediately adjacent to the tank. The bottom of the conservator is elevated above the top of the turrets and is connected to the tank by piping. This positioning allows the insulating liquid in the tank to remain at a positive pressure with respect to the atmosphere. There is usually a valve in-line with the transformer tank and a liquid level indicator on the side of the vessel. The oil level should be above turrets that have vent piping, or the tester should verify bushing turrets are bled to confirm no trapped gasses. A conservator provides a fluid reservoir for variations in the insulating liquid as the insulating liquid temperature rises. It functionally acts as an expansion vessel for the tank's insulating liquid.

There are basically three types of conservator systems. The “free-breathing” system is the oldest of these types. The liquid level rises and falls with the temperature of the equipment, and the insulating liquid is constantly exposed to the atmosphere. Some free-breathing conservators may employ dehydrating breathers of either the desiccant or refrigerant type. Both of the other two types of conservators prevent the insulating liquid from coming in contact with the atmosphere. The newer type uses an air cell (sometimes referred to as a *bladder*), which is a large balloon-like envelope located inside the conservator. As the liquid level in the conservator rises and falls, air is expelled or drawn into the air cell. The older type has a diaphragm attached to the inside of the conservator vessel wall that rises with the expansion of the equipment’s insulating liquid. Dehydrating breathers can be used to prevent moisture from collecting in the air space of the conservator with a bladder or diaphragm.

Checks should be carried out according to the following procedure:

Procedure: The liquid level indicated on the liquid level gauge on the side of the conservator vessel should be recorded. This reading should be made with respect to the 25 °C mark on the gauge. The top oil temperature of the equipment should then be recorded. The top oil temperature reading should be used to correct the liquid level gauge reading. The resulting corrected level should be in the normal (25 °C) range. If gauges are not available, a “dipstick” method to use an external site gauge can be used to confirm the oil level. When dipsticking the conservator bag, there should be no oil present on the air side or the outside of the bladder. If this exists, the bladder is breached and should be replaced.

Interpretation: If the corrected level is normal, no additional action should be required. If the corrected level is substantially above or below the normal level, the measurements and calculations should be rechecked. If the results are the same, it may be necessary to add or remove, as the case may be, some of the insulating liquid of the equipment. The user should refer to the manufacturer’s recommendations. In addition, the cause of any incorrect level should be determined and corrective steps should be taken prior to taking any other action. Generally the corrected level should remain fairly constant unless there is an insulating liquid leak, etc.

Precautions: Insulating liquid for diagnostic testing is typically sampled on an energized transformer. Otherwise, insulating liquid should never be added or removed from an energized transformer, except in the most extreme circumstances, and then only with great knowledge and care.

7.2.3 Tank vacuum testing

7.2.3.1 General

Transformer tanks are designed to withstand a specified level of vacuum. This level of vacuum withstand depends on the type and size of the unit. Prior to the factory tests, transformer tanks are subjected to vacuum during the fabrication stage and again verified during the final insulating liquid filling stage. Large transformers shipped with dry air or nitrogen again need vacuum application when received at the site and as part of the energizing processing. The principal function of vacuum application is to remove the trapped air and moisture from the insulation and enable the insulation to attain its full dielectric strength. Before vacuum application, the manufacturer’s instruction manual should be referred to and necessary precautions followed as suggested.

7.2.3.2 Precautions

Ensure that the tank, including fittings (including conservator tank, radiators, pumps, associated valves, and monitoring equipment), are suitable for vacuum application. If any fittings are not designed to withstand vacuum, they need to be removed and blanked off.

Ensure the suitability of LTC application or pressure equalization. LTCs with separate barrier boards, if not suitable for full vacuum, should also be pressure equalized to avoid any damage to the barrier boards.

Make sure that no rigid connections have been made to bushings, insulators, and lightning arresters.

7.2.3.3 Vacuum leak rate test

After all parts have been assembled, seal the tank so that there are no leaks during vacuum application. Apply vacuum as specified by the manufacturer; large units may be subjected to a vacuum level of 1 torr or below. Hold the vacuum for 4 h. Secure the vacuum valve at the transformer tank. Wait 5 min and record any rise in pressure. At 10 min, measure the change in tank pressure. If the vacuum rise exceeds the manufacturer's limits or the operating company limits, the leaks should be corrected and the test repeated. After a successful vacuum leak rate test, continue with the designated vacuum time for liquid filling.

At the end of 4 h, the tank plate walls should not have any appreciable deflection. If observed, deflection should be discussed with the manufacturer.

7.2.4 Dew point

Dew point is covered in Annex D.

7.2.5 Insulating liquid

7.2.5.1 General

The insulating liquids used in transformers, regulators, and reactors act as both an insulating fluid and a heat transfer medium to carry off excess heat generated by the losses of the power equipment. The tests listed in Table 2 measure the properties used to determine a liquid's condition. ASTM publishes a summary of these tests and their usefulness (see ASTM D117 [B1]). The IEEE publishes guides that use the results of these tests to determine the condition of service-aged mineral oil, silicone, less-flammable hydrocarbon, and natural and synthetic ester insulating liquids and the diagnosis of power equipment based on insulating liquid condition (see IEEE Std C57.106™ [B39], IEEE Std C57.111™ [B40], IEEE Std C57.121™ [B42], IEEE Std C57.147™ [B46]).

Sampling techniques for these test methods (see ASTM D923 [B3]) should ensure that the specimen taken is representative of the insulating liquid contained within the equipment. Natural contaminants exist within the body of sampling valves; therefore, to maintain sample integrity, the valves should be flushed before the extraction is performed.

The existence of a positive tank pressure should be confirmed before attempting to obtain a sample. Failure to do so may result in a gas bubble entering the tank and creating an immediate dielectric breakdown while moving upward or a latent breakdown by becoming lodged in the windings. This condition may result in the premature and violent failure of the equipment.

A sufficiently large sample should be withdrawn so that enough insulating liquid is available to perform the desired tests. Typically 1 quart (0.95 L) is enough. Table 2 gives the insulating liquid volumes needed for individual tests. Proper sample containers and sampling procedures should be used to ensure a representative test sample (see ASTM D923 [B3]).

Table 2—Minimum volume of liquid for each test

Property	ASTM standard test	Quantity of insulating liquid (mL)
Visual examination	D1524 [B10]	10
Sediments and sludge	D1698 [B12]	50
Color	D1500 [B9]	125
Dielectric breakdown voltage	D1816 [B13]/D877 [B2]	500
Dissipation factor	D924 [B4]	250
Water content	D1533 [B11]	50
Acid number	D974 [B6]	20
PCB content	D4059 [B16]	10
Interfacial tension	D971 [B5]	20
Relative density	D1298 [B8]	125
Furan	D5837 [B17]	40
Particle count	D6786 [B18]	100
Corrosive sulfur	D1275 [B7]	500
Oxidation inhibitor	D2668 [B14]	20
Dissolved gases	D3612 [B15]	50
		Total: 1870
NOTE—The quantities listed have generally been found to be needed for the test procedures. Since some equipment manufacturers make larger containers, the test laboratory should be consulted prior to sampling to ensure that the sample volume is adequate.		

In most cases, the sample should be transported to the laboratory in a clean, dry container. Prolonged exposure to direct sunlight or contamination by excessive atmospheric moisture should be avoided. Many of the liquid volumes for measurements specified in Table 2 are not standardized. However, the values listed have been found to be practical and are commonly used.

Mineral oil in service may be placed into the following groups based on the evaluation of the characteristics; more details are in 7.2.5.2 through 7.2.5.11:

- Group I—Mineral oil that is in satisfactory condition for continued use
- Group II—Mineral oil that requires only reconditioning for further service
- Group III—Mineral oil in poor condition (such insulating liquid should be reclaimed or disposed of depending on economic considerations)
- Group IV—Mineral oil in such poor condition that it is technically advisable to dispose of it

Tests should be performed at least annually, but more often if the equipment is strategically located in the system.

7.2.5.2 Acid number

The acid number test (ASTM D974 [B6]) determines the acidic degradation constituents in service-aged insulating liquid.

This test should be used to indicate the relative change in an insulating liquid during use under oxidizing conditions. Acidity is gauged by the acid (neutralization) number, expressed as the number of milligrams of potassium hydroxide required to neutralize the acid in a gram of insulating liquid. Transformer grade insulating liquids contain only trace levels of acidic constituents when new; the acid number increases as the insulating liquid degrades. A used insulating liquid having a high acid number indicates that the insulating liquid is either oxidized or contaminated with materials such as varnish, paint, or other matter. In some insulating liquids, this condition may be indicative of sludge formation. There is no direct correlation between the acid number and the corrosive tendency of the insulating liquid towards metals in electrical

power equipment. Short chain acids are detrimental to insulation systems and can induce oxidation of metals when moisture is also present. Changes occur over long periods of time. Elevated levels are not indicative of a problem in the equipment, but of a potential threat to the internal components of the equipment.

Maximum recommended values of acid number for different types of insulating liquid are given in Table 3.

Table 3—Acceptable acid number values for new and in-service insulating liquids by voltage class

Type of insulating liquid	Voltage class (kV)	Acid number (mg KOH/g), maximum			
		Mineral oil ^a	LFH ^b	Silicone ^c	Natural ester ^d
New insulating liquid in new equipment	—	0.015	0.03	0.01	0.06
Service-aged insulating liquid	—	—	0.20	0.2	—
	≤ 69	0.20	—	—	0.3
	> 69 to < 230	0.15	—	—	0.3
	≥ 230	0.10	—	—	0.3

^a See IEEE Std C57.106-2006 [B39].

^b See IEEE Std C57.121-1998 [B42].

^c See IEEE Std C57.111-1989 [B40].

^d See IEEE Std C57.147-2008 [B46] (some of these values are provisional; see standard for additional information).

7.2.5.3 Visual inspection and color

Visual inspection and color tests cover estimating, during a field inspection, the color and condition (free water or sediment such as metal particles, insoluble sludge, carbon, fibers, dirt, etc.) of a sample of insulating liquid (ASTM D1524 [B10]) and a more precise laboratory determination of color (ASTM D1500 [B9]).

The observation of cloudiness, particles of insulation, products of metal corrosion, or other undesirable suspended materials, as well as any unusual change in color, should be followed up with a laboratory examination and analysis for proper diagnosis. If insoluble contaminants are present in the insulating liquid, valuable information concerning the condition of the transformer and its components may be obtained by filtering the insulating liquid and identifying the residue. Ultimately, a number of other tests may be incorporated to help in the diagnosis of the potential problem.

Color is used to indicate the relative change in insulating liquid during use and is expressed by a numerical value or color description based on comparison with a series of color standards. There should be no direct correlation between a change in the color of the insulating liquid and a specific problem within the equipment. Changes normally occur over long periods of time. A rapidly increasing number should be indicative of a dramatic change in operating condition and generally precedes other indications of a problem. A high color number occurs in combination with the presence of insulating liquid deterioration or contamination, or both.

The color interpretation for mineral insulating oil is given in Table 4.

Table 4—Relative condition of mineral oil based on color

Color comparator number	ASTM color	Mineral oil condition
0.0 to 0.5	Clear	New
0.5 to 1.0	Pale yellow	Good
1.0 to 2.5	Yellow	Service-aged
2.5 to 4.0	Bright yellow	Marginal
4.0 to 5.5	Amber	Bad
5.5 to 7.0	Brown	Severe (reclaim)
7.0 to 8.5	Dark brown	Extreme (scrap) ^a

^a Retest to confirm reading prior to scrapping oil.

7.2.5.4 Dielectric breakdown voltage

Dielectric breakdown voltage tests (ASTM D877 [B2], ASTM D1816 [B13]) determine the dielectric breakdown voltage of service-aged insulating liquid. Table 5 lists acceptable breakdown values.

Table 5—Acceptable dielectric breakdown values for new and in-service insulating liquids by voltage class

Test method	Voltage class (kV)	Dielectric breakdown strength (kV), minimum			
		Mineral oil ^a	LFH ^b	Silicone ^c	Natural ester ^d
ASTM D 1816 [B13]—1 mm gap					
New insulating liquid in new equipment	≤ 34.5	—	20	—	—
	> 34.5	—	25	—	—
	≤ 69	25	—	—	25
	> 69 < 230	30	—	—	30
	> 230 < 340	32	—	—	32
	≥ 340	35	—	—	35
Service-aged insulating liquid	Not specified	—	23	—	—
	≤ 69	23	—	—	23
	> 69 < 230	28	—	—	28
	≥ 230	30	—	—	30
ASTM D 1816 [B13]—2 mm gap					
New insulating liquid in new equipment	≤ 34.5	—	40	—	—
	> 34.5	—	50	—	—
	≤ 69	45	—	—	45
	69 < 230	52	—	—	52
	≥ 230 < 340	55	—	—	55
	≥ 340	60	—	—	60
Service-aged insulating liquid	Not specified	—	34	—	—
	≤ 69	40	—	—	40
	≥ 69 < 230	47	—	—	47
	≥ 230	50	—	—	50
ASTM D877 [B2]					
New insulating liquid in new equipment	Not specified	—	30	30	—
Service-aged insulating liquid	Not specified	—	24	25	—

^a See IEEE Std C57.106-2006 [B39].

^b See IEEE Std C57.121-1998 [B42].

^c See IEEE Std C57.111-1989 [B40].

^d See IEEE Std C57.147-2008 [B46] (some of these values are provisional; see standard for additional information).

The recommended method to determine the dielectric breakdown voltage of insulating liquid (ASTM D1816 [B13]) uses spherical capped electrodes of the Verband Deutscher Elektrotechniker (VDE) type in its test cell. Contamination and products of deterioration generally reduce the dielectric strength of insulating liquid. The dielectric breakdown voltage of insulating liquid indicates the insulating liquid's ability to withstand electrical stress without failure. High dielectric strengths do not indicate the absence of

contaminants. There should be no direct correlation between a certain breakdown voltage and failure, except in extreme cases.

This test may be satisfactorily performed in the field but is more controllable in a laboratory environment. A visual test should be performed to verify that the sample does not contain free water or air bubbles caused by agitation during transport.

7.2.5.5 Dissolved gas

The dissolved gas test (ASTM D3612 [B15]) determines the dissolved gas components in service-aged insulating liquid.

This test should be used to determine the amount of specific gases generated by a liquid-filled in-service transformer. Certain combinations and quantities of these generated gases are frequently the first indication of a possible malfunction that may eventually lead to failure if not corrected. Arcing, PD, low-energy sparking, severe overloading, and overheating in the insulation system are some of the mechanisms that can result in chemical decomposition of the insulating materials and the formation of various combustible and noncombustible gases dissolved in the insulating liquid. Normal operation may also result in the formation of some gases, but not to the same extent as when a malfunction exists.

Precautions: The sample should preferably be obtained using a clean, moisture-free, gas-tight container to isolate it from excessive atmospheric moisture and to maintain its quantity of dissolved gases. Care should be taken to purge the container of atmospheric gases but retain all gases from the equipment at the time the sample is taken. See [B6] for additional guidance.

After determining the quantities of key dissolved gases from the sample using this procedure, a prescribed diagnostic routine to assist in interpretation of the analysis should be followed (IEEE Std C57.104™ [B38], IEC 60599 [B30]).

It can be difficult to determine whether or not a transformer is operating normally if it has no previous dissolved gas history. Also, considerable differences of opinion exist for what is considered a “normal transformer” with acceptable concentrations of gases. Many techniques for the detection and measurement of gases have been established. However, it should be recognized that analysis of these gases and interpretation of their significance at this time is not an exact science but an art subject to variability.

7.2.5.6 Interfacial tension

The interfacial tension (IFT) test (ASTM D971 [B5]) determines the IFT of service-aged insulating liquid against water.

This test method should be used to indicate the IFT between an electrical insulating liquid and water. This is a measurement of the molecular attractive force between their unlike molecules at the interface. This test provides a means of detecting soluble polar contaminants and products of deterioration in the insulating liquid. There is a unique relationship between IFT and acid number in that the acid number of the insulating liquid increases and the IFT decreases as a liquid oxidizes. To a certain extent, the IFT is a measure of the remaining useful life of the insulating liquid, short of its being reclaimed. Decreased levels are not indicative of a problem in the equipment, but of a potential threat to the future operating condition of the equipment.

This test may be satisfactorily performed in the field, as well as in a laboratory environment.

Interpretation: Recommended acceptable values of IFT for different conditions of insulating liquids are shown in Table 6.

Table 6—Acceptable interfacial tension values for new and service-aged insulating liquids by voltage class

	Voltage class (kV)	Interfacial tension (mN/m), minimum			
		Mineral oil ^a	LFH ^b	Silicone ^c	Natural ester ^d
New insulating liquid in new equipment	All	38	—	—	—
	Not specified	—	38	—	—
Service-aged insulating liquid	Not specified	—	24	—	60% of original value
	≤ 69	25	—	—	—
	> 69 ≤ 230	30	—	—	—
	> 230	32	—	—	—

^a See IEEE Std C57.106-2006 [B39].

^b See IEEE Std C57.121-1998 [B42].

^c See IEEE Std C57.111-1989 [B40].

^d See IEEE Std C57.147-2008 [B46]. The limit values are provided for reference purposes and more long-term experience is needed to validate the test applicability to alternative liquids.

7.2.5.7 Particle count

The particle count test (ASTM D6786 [B18]) determines the number and size of particles present in mineral insulating oil. Particles in insulating oil may have an impact on the dielectric strength or power factor (PF) of the insulating oil. The source of the particles can be from internal material of the equipment such as carbon, cellulose fibers, metals, and oil degradation. Other external sources are possible when contaminants are introduced when processing the oil or when the equipment is open to the environment.

Particle counts will provide a general sense of the degree of the contamination and may be used to determine the effectiveness of oil filtration. Water in the oil greater than 10 ppm may contribute to increase particle count.

Particle counts may be undertaken to help in the determination of the cause of degradation of standard oil tests. The IEEE has not established a guide with any limits for particle count for insulating oil however, CIGRE [B21], and at least one manufacturer (Service Handbook for Power Transformer [B69]) have suggested points at which further action may be required for in-service insulating mineral oil. At least one manufacturer of transformers has particle count limit requirements for placing a transformer into service and in-service limits.

Particle counting has evolved over time and an interpretation of the present methods used is essential to insure proper interpretation of the reported results. The preferred method is the use of Automatic Particle Counters (APC) calibrated using the ISO Medium Test Dust (MTD). The calibration is performed according to ISO 11171:2010 [B52]. The laboratory conducting the particle count is responsible to report the calibration method used and insure that the repeatability and reproducibility criteria of the test method are in conformance to the test method.

Collecting a sample for particle count is an important aspect to obtain proper results. The test method (ASTM D6786 [B18]) requires a particle clean bottle of at least 100 ml and cleaned to contribute less than 1% of the total particles expected in the sample. Sample collection using a bottle is to be conducted according to ASTM D923 [B3].

An informative description of the methods used for calibration, interpretation, and suggested action points for in-service mineral insulating oil is provided in Annex I.

7.2.5.8 Dissipation factor

The dissipation factor (DF) test (ASTM 924 [B4]) determines the DF of new and service-aged insulating liquid.

This test should be used to indicate the dielectric losses in insulating liquid when used in an alternating electric field and to indicate the energy dissipated as heat. The DF is the ratio of the power dissipated in the insulating liquid in watts to the product of the effective voltage and current in volt-amperes, when tested with a sinusoidal field under prescribed conditions. A low DF indicates low dielectric losses. It is useful as a means to maintain sample integrity and as an indication of changes in quality resulting from contamination and deterioration in service or as a result of handling. Insulating liquid samples that are defective often pass other standard electrical and chemical tests, yet fail this test.

This test may be satisfactorily performed in the field, as well as in a laboratory environment. A visual test should be performed to verify that the sample does not contain air bubbles due to agitation during transport. After allowing the specimen to settle in the test cell, ASTM 924 [B4] should be followed in a laboratory. In the field, the recommendations of the test equipment manufacturer should be followed.

Interpretation: The recommended acceptable values of percentage DF for different categories of new and service aged insulating liquids are shown in Table 7.

The DF limits given for an insulating liquid are based on the understanding that DF is an indicator test for contamination by excessive water (in combination with particulate matter) or polar or ionic materials in the liquid.

High levels of liquid DF are of concern because contaminants may collect in areas of high electrical stress and concentrate in the winding, making transformer cleaning difficult and masking changes in winding PF due to other causes such as changing water content. A very high liquid DF may be caused by the presence of free water, which could be hazardous to the operation of a transformer. Whenever there is high liquid DF, the cause should be sought. Oxidation, free water, wet particles, contamination, and material incompatibility are possible sources of high liquid DF.

Table 7—Acceptable dissipation factor values for new and service-aged insulating liquids

Status of insulating liquid	Dissipation factor (%), maximum							
	Mineral oil ^a		LFH ^b		Silicone ^c		Natural ester ^d	
	25 °C	100 °C	25 °C	100 °C	25 °C	100 °C	25 °C	100 °C
New insulating liquid in new equipment	0.05	0.40	0.1	1	0.1	—	0.5	—
Service-aged insulating liquid	0.5	5.0	1	—	0.2	—	0.5	—

^a See IEEE Std C57.106-2006 [B39].

^b See IEEE Std C57.121-1998 [B42].

^c See IEEE Std C57.111-1989 [B40].

^d See IEEE Std C57.147-2008 [B46] (some of these values are provisional; see standard for more details).

7.2.5.9 Polychlorinated biphenyl content

The polychlorinated biphenyl (PCB) content test (ASTM D4059 [B16]) determines the PCB content of service-aged insulating liquid.

PCBs are regulated substances in many countries. For this reason, it is important to know the present condition of power equipment with regard to its PCB concentration. A transformer having a PCB concentration of < 50 mg/kg (ppm) is classified as non-PCB-contaminated equipment; however, the insulating liquids may still require proper disposal. A PCB concentration ≥ 50 mg/kg but < 500 mg/kg

results in a classification of contaminated. A concentration of ≥ 500 mg/kg is considered PCB material. Because most laws deal with the PCB concentration of the involved insulating liquid, it is most important to be aware of the PCB concentration of the insulating liquids on any given system. National, state or local governmental regulations may require specific actions, records, and disposal methods for values of even < 50 mg/kg.

This testing may be performed in the field prior to laboratory testing. In the field, there are a number of commercially available screening kits. Check the expiration date and the compatibility of the insulating liquid with the kit. These types of tests only estimate the PCB concentration and do not give exact numerical values. It is essential that the manufacturer's recommendations be followed precisely when performing the field screening test. This type of test gives a positive indication for chlorinated compounds, whether they are PCB or not. Therefore, care should be taken not to introduce other chlorinated compounds into the procedure. Laboratory testing is required to obtain an actual PCB concentration.

Interpretation: PCB regulations vary by location. Local regulators should be consulted for appropriate guidelines.

7.2.5.10 Sludging condition

The sludging condition test covers the determination of pentane-insoluble sludge present in service-aged insulating liquid. For mineral oil, this test is generally not performed unless IFT is < 26 mN/m or the acid number is > 0.15 mg KOH/g liquid.

Sludge is a resinous, polymeric substance that is partially conductive, hygroscopic, and a heat insulator. If there is water in the transformer, it is attracted to the sludge. The presence of soluble sludge should be an indication of deterioration of the insulating liquid, presence of contaminants, or both. It serves as a warning that there may be formation of sediment.

This test is intended to determine the extent to which the insulating liquid has begun to sludge, see ASTM D1698 [B12]. The test has value in determining the proper procedure for performing maintenance on a transformer. If the insulating liquid has not started to sludge or is only sludging slightly, the transformer's insulating liquid may be circulated through a reclaiming system, thus extending the life of the insulating liquid and the transformer. If the insulating liquid has progressed into sludging such that sediment exists, more dramatic maintenance procedures may be required, including removing the transformer from service and thoroughly washing down the insulation system, tank, and cooling system. This is necessary since sludge (sediment) and moisture become trapped in cooling systems, thereby reducing effective cooling. There is also a possibility that the moisture-laden sludge will collect in critical regions of electrical stress and result in premature failure or, at the least, reduced heat transfer efficiencies.

NOTE—Tiny, solid particles may not be sludge. They could be clay fines or artifacts.

7.2.5.11 Water (moisture)

Some moisture is always present in any transformer. In addition, since the paper in the insulation system has a great affinity for water, most of the moisture present is in the paper.

The dielectric strength of the paper is very sensitive to the presence of moisture, as is the insulating liquid. Therefore, it is important that the moisture content be known and its concentration controlled.

Water migrates between the solid and liquid insulation in a transformer with changes in load and, therefore, temperature. Consequently, the concentration of water in the insulating liquid alone, expressed in mg/kg (ppm), does not provide sufficient information to obtain an adequate evaluation of the insulation system dryness. Relative saturation provides a better evaluation under a wide range of operating conditions and temperatures. Even using relative saturation to evaluate insulation system dryness has some inherent biases

due to the fact that water never reaches equilibrium between the solid and liquid insulation. The further from equilibrium the system is when the sample is taken, the greater the bias. The bias may be either positive or negative and can be affected by short-term transients at solid/liquid surfaces or by longer-term transitions within the thicker insulation.

7.2.5.11.1 Water content

Table 8 shows the suggested limits for moisture in insulating liquids as determined by Karl Fischer analysis (ASTM D1533 [B11]). Liquids at delivery prior to processing should have moisture content at or below the specification limits for the liquid.

With regard to insulating liquid moisture levels, solubility equations have been used to assess the solubility limit of moisture in insulating liquid. The solubility limit is the maximum amount of moisture that is soluble in the insulating liquid at a specific temperature. It is also referred to as the *moisture in insulating liquid saturation limit* or simply *100% saturation* and is determined when the calculation of the percent saturation equals 100%. The equation is $[(\text{mg/kg water in insulating liquid})/(\text{mg/kg of water in insulating liquid at saturation})] \times 100$.

Table 8—Acceptable water content values for new and in-service insulating liquids by voltage class

Status of insulating liquid	Voltage class (kV)	Water content ^a (mg/kg), maximum			
		Mineral oil ^b	LFH ^c	Silicone ^d	Natural ester ^e
New insulating liquid in new equipment	Not specified	—	25	50	—
	≤ 69	20	—	—	300
	> 69 < 230	10	—	—	150
	≥ 230	10	—	—	100
Service-aged insulating liquid	Not specified	—	35	100	—
	≤ 69	35	—	—	400
	> 69 < 230	25	—	—	200
	≥ 230	20	—	—	150

^a These values are for insulating liquids not insulation dryness or dielectric integrity.

^b See IEEE Std C57.106-2006 [B39].

^c See IEEE Std C57.121-1998 [B42].

^d See IEEE Std C57.111-1989 [B40].

^e See IEEE Std C57.147-2008 [B46] (note some of the values are provisional; see standard for additional information).

7.2.5.11.2 Discussion

IEEE Std C57.106 [B39] discusses the absence of thermodynamic equilibrium in an operating transformer. This leads to the conclusion that a quantitative correlation of moisture in liquid to moisture in paper is not currently feasible. This conclusion is based on the following conditions in an operating transformer:

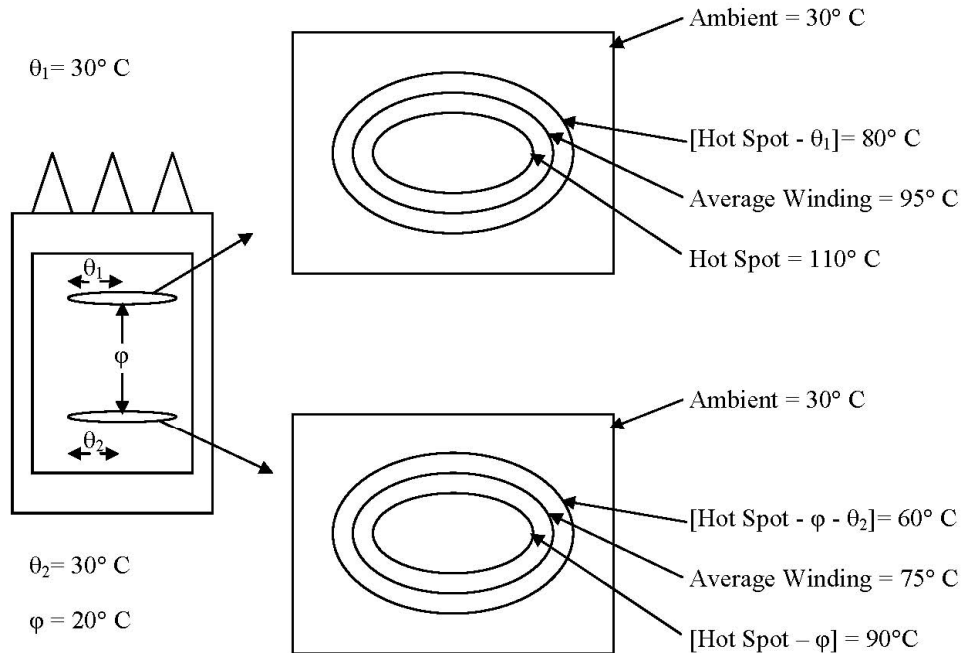
- Temperatures and moisture concentrations vary significantly with location
- Temperatures and moisture concentrations vary significantly with time
- Time variation is rapid for temperature and slow for moisture diffusion
- Solubility of moisture in liquid is not the same for the various insulating liquids

The first three statements can be evaluated quantitatively by using the solid insulation time constants for moisture diffusion and temperature. A model using this evaluation (Thompson [B70]) shows that the moisture diffusion time constants are on the order of magnitude of weeks and months, whereas typical temperature time constants from the IEEE loading guide IEEE Std C57.91™ [B37] for transformers are on the order of magnitude of hours. In addition, in this model, using curves for a flat sheet of paper Oommen [B62], gives an approximate range of 1% to 3.5% for a temperature gradient of 40 K.

CAUTION

Once the water content in insulating liquid reaches the solubility limit (100% relative saturation), free water droplets form, which adversely affects the dielectric strength of the insulating liquid and may result in transformer failure.

Next, considering the thermograph model in Figure 1, there is an even greater temperature gradient (50 K). This hypothetical model serves the purpose of showing possible temperature gradients in an operating transformer that is at constant load and constant ambient conditions (temperature and wind velocity). This model also indicates that the prediction of any single homogeneous moisture-in-paper value is invalid. While an accurate model can be attempted only with temperature probes, nonetheless it is clear here that moisture curves predict a range of moisture values whenever there are temperature gradients. Also, after the publication of moisture equilibrium curves for paper-mineral oil systems, cautions have been issued to not use these curves to determine the dryness level of the solid insulation in an operating transformer (Oommen [B63]). Finally, adding the conditions of changing load and ambient conditions further complicates any quantitative analysis.



$$\phi = (\text{Hot Spot Temp.}) - (\text{Maximum Temp. in Horizontal Plane at bottom of Coil})$$

$$\theta_1 = 2 \cdot [(\text{Hot Spot Temp.}) - (\text{Avg. Winding Temp at Top Coil})]$$

$$\theta_2 = 2 \cdot [(\text{Hot Spot Temp.}) - (\phi)] - (\text{Avg. Winding Temp at Top Coil})$$

Figure 1—Model of isothermal lines and temperature gradients for mineral-oil-filled 65 K rise transformer at constant full load and constant ambient conditions

Caution should be taken when energizing any transformer during cold weather. From a dielectric strength point of view, reduced insulating liquid temperatures in a de-energized transformer could increase the moisture in insulating liquid percentage saturation level enough to significantly reduce the dielectric breakdown strength of the insulating liquid—even at percentage saturation levels below 100%. Consequently, when energizing a new transformer or re-energizing any transformer during cold weather, caution should be taken so that the dielectric breakdown strength of the insulating liquid, referenced to the actual temperature of the insulating liquid in the de-energized transformer, is sufficient for service. Equations for percent saturation values at various temperatures and for various insulating liquids are found in Griffin et al. [B28] and Du [B24].

7.2.6 Core

7.2.6.1 General

Transformer core types are designated as core-form or shell-form. In a core-form transformer, the windings are wound on cylinders and placed around the core legs. In a shell-form transformer, the core is formed like a shell around the windings after the windings are set in the tank. In both types, the core is insulated from the tank and other grounded items. In addition, grounding is installed to help prevent a voltage rise from occurring on the core during operation. Should an inadvertent ground occur while the transformer is in service, a circulating current may be generated in the core. The magnitude of the circulating current is inversely proportional to the resistance of its path. Severe damage may occur to the core if this condition is allowed to persist. The heat produced by this condition may generate large quantities of ethylene gas and, under severe conditions, quantities of acetylene. Under the most extreme conditions, winding insulation can be destroyed, thus causing the transformer to fail. For proper operation of the core system, it should first be determined whether an inadvertent core ground exists. In the absence of an inadvertent core ground, the core insulation resistance should be measured to determine its adequacy.

7.2.6.2 Core insulation resistance and inadvertent ground tests

The resistance of the core's insulation system should be measured regularly. Trends are important to indicate the rate of deterioration of a core's insulation system. This test should be performed prior to placing the unit in service and following modifications to the transformer that could affect the integrity of its core insulation. This test may also be performed at other times, usually when indicated by gas chromatography or during a major inspection.

In addition to measuring the core-insulation-to-ground resistance, the technique may also be used to detect the presence of inadvertent grounds. The only way to be sure that an inadvertent core ground exists is to remove the equipment from service and perform a resistance-to-ground test on the core itself. This test can be successfully performed only after the core grounding strap is disconnected from ground. On shell-form equipment built before 1997, the core grounds may not be easily accessible, while on modern shell-form units, the core grounds are usually brought out for testing and diagnostics. Where core grounds are not accessible, the manufacturer or a qualified consultant should be contacted. Many devices have more than one core or have the core divided into separate units. Cores and units may be tested together, but if an inadvertent core ground is indicated, their straps should be separated for independent testing. When the core clamp or core frame is also insulated, then both core and frame are generally brought out via separate core and core clamping frame test bushings. In this case the Core to Ground, Frame (or Clamp) to Ground, and Core to Frame (or Clamp) should be measured.

Procedure: For insulation resistance and inadvertent ground tests the voltage should not exceed 1000 V. The following sequence should be performed:

- a) The core grounding strap should be located. On modern core-form transformers, the core ground connection may be brought through the cover by means of a small bushing. Thus the transformer need not be opened.
- b) The strap should be disconnected from where it is bolted to the frame, tank, etc.

Precautions: Care should be taken to secure all hardware as the strap is disconnected. Sometimes the fastener is not captive. Dropping a lock washer or nut down into the windings may lead to transformer failure. Thus, care should be taken to secure hand tools and hardware.

- c) A test should be made between the strap and its grounding point to determine whether an inadvertent ground exists. This is usually performed by using a direct current (dc) high-resistance meter. Readings $< 10 \text{ M}\Omega$ should be read on a lower scale for accuracy.
- d) The temperature of the core should be estimated to give a corrected reading. The temperature of the windings and insulating liquid should be near the reference temperature of 20°C .

Interpretation: See Table 9. Consult with manufacturer to provide the necessary clarification for insulating liquids other than mineral oil.

Table 9—Typical insulation resistance ranges for various conditions of core insulation

Type of equipment	Core insulation resistance ($\text{M}\Omega$)	Condition of insulation
New	> 500	Manufacturer to be consulted for values less than $500 \text{ M}\Omega$ for proper course of action.
Service aged	> 100	Normal
	10 to 100	Indicative of insulation deterioration
	< 10	Needs to be investigated

7.2.6.3 Location of an inadvertent core ground

If an inadvertent core ground exists, decisions about whether detection or location and repairs are to be performed in the field depend on the circumstances involved. Visual inspections may typically reveal the source of the inadvertent core ground if that inadvertent ground is along the top core yoke of a core-form transformer. Otherwise, locating it and determining a remedy can be quite difficult. The severity of the problem; importance of the equipment; and its size, type of construction, and other factors should be considered. On shell-form equipment, the ground strap is generally not easily accessible. In this case, the manufacturer or a qualified consultant should be contacted for assistance.

The following procedure should be followed to locate inadvertent core grounds:

- a) The core grounding strap should be located.
 - b) The strap should be disconnected from where it is bolted to the frame, tank, etc.
- Precautions:** Care should be taken to secure all hardware as the strap is disconnected because failure to do so may result in transformer failure.
- c) A current-protected dc supply should be connected across the core from side to side. This connection should cause the voltage to bridge all of the core's laminations.
 - d) The negative lead of a dc voltmeter should be connected to a convenient point of ground inside the tank.
 - e) Contact should be made with the positive lead of the voltmeter to the core's lamination, starting at one side of the core. A voltage should be observed. If not, the contact should be moved to the other side of the core. The contact should be moved gradually across the core, moving at right angles to the core's laminations, until the voltmeter reads zero.

Interpretation: The lamination plane at the zero point is the location of the inadvertent core ground. A visual examination of this plane may or may not reveal the source of the inadvertent ground. Consult the manufacturer for recommended actions for troubleshooting/repair.

Alternatively, if the core ground provided by the manufacturer is grounded externally to the tank, the circulating current may be reduced by connecting a resistor between the ground connection and the tank. The resistor limits circulating currents to a safe level. This provides an opportunity for in-service monitoring by measuring the voltage across the resistor. This technique should only be used after consultation with the manufacturer.

7.2.7 Winding resistance

In general, the windings are checked for evidence of physical displacement or distortion, broken connections or strands, short-circuited turns, and insulation defects. The parameters usually checked are described in this clause along with an indication, where possible, of acceptable limits for the quantities being measured. For the tests and measurements on windings, the insulating liquid pumps should be switched off.

NOTE—In general when performing a series of tests, winding resistance measurements should be done after excitation, FRA and leakage reactance (short circuit impedance) tests to avoid any issues with residual magnetism.

Transformer winding resistances are measured in the field to check for abnormalities due to loose connections, broken strands, and high contact resistance in tap changers. The results are usually interpreted based on comparing measurements made separately on each phase of a wye-connected winding or between pairs of terminals on a delta-connected winding. Comparison may also be made with original data measured in the factory. The resistances between phases should be within 2% of each other. Agreement to within 5% for any of the above comparisons is usually considered satisfactory. It may be necessary to convert the resistance measurements to values corresponding to the reference temperature in the transformer test report. The conversions are accomplished by Equation (1):

$$R_s = R_m \left(\frac{T_s + T_k}{T_m + T_k} \right) \quad (1)$$

where

- R_s = resistance at desired temperature T_s
- R_m = measured resistance
- T_s = desired reference temperature (°C)
- T_m = temperature at which resistance was measured (°C)
- T_k = 234.5 °C (copper)
= 225 °C (aluminum)

NOTE—The value of T_k may be as high as 230 °C for alloyed aluminum.

It is important to determine the transformer winding temperature when measuring resistance. However, it is very difficult to measure the winding temperature accurately under field conditions. Methods commonly used include the following:

- Place a thermometer in contact with the tank wall. If the transformer has been recently removed from service, this does not give an accurate indication of the real winding temperature.
- Use values obtained from the permanently installed temperature indicators. If the transformer has recently been removed from service, this may be the only means available for estimating the winding temperature.

- For nitrogen-blanketed transformers, a thermometer may be placed in the main tank by inserting it in openings in the top of the tank or in the wells for the permanent temperature indicators. Inserting a thermometer into the main tank usually requires breaking the hermetic seal on the transformer and relieving a positive nitrogen pressure. This may introduce moisture into the transformer and carries the risk that a conductive object could accidentally be dropped into the winding. Mercury thermometers should not be used inside a transformer tank due to the consequences of breakage. Using the wells for the permanent temperature indicators requires removing the permanent sensor.

Temperature correction of the winding resistance is not normally required on site because measurement comparisons are made between phases. Normally there is some variation in the values indicated by the different sensors. If the transformer has been out of service long enough to have a uniform temperature throughout its mass, averaging the values from all indicators may yield better results than using a single indicator.

7.2.7.1 Conductor resistance measurement techniques

Transformer winding resistance is usually measured by one of the following: bridge techniques, the voltmeter-ammeter method, or a micro-ohmmeter. When bridges are used, a Wheatstone bridge is preferred for resistance values of megohms. A Kelvin bridge or a micro-ohmmeter is preferable for resistance values of $< 1000 \text{ M}\Omega$.

7.2.7.2 Voltmeter-ammeter (Kelvin) method

The voltmeter-ammeter method also known as the Kelvin 4-wire method is the most common method used to measure transformer winding resistance. The measurement is made using direct current. Simultaneous readings of current and voltage are taken via the connections shown in Figure 2.

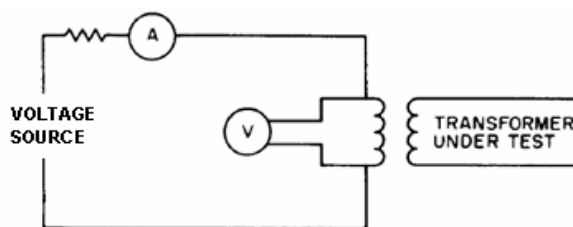


Figure 2—Method for taking simultaneous readings of current and voltage

The required resistance is calculated from the readings in accordance with Ohm's Law.

A regulated dc power source specifically designed for saturation of the inductive load of a transformer winding is recommended. Winding resistance meters for field testing are available with these features.

Precautions: To help minimize measuring errors, the following precautions should be taken:

- a) The measuring instruments or winding resistance meters should have sufficient ranges to allow readings to be made as near full scale as possible and in any case above 70% of full scale.
- b) The polarity of the core magnetization should be kept constant during the resistance measurements.

NOTE—A reversal in magnetization of the core can change the time constant and result in erroneous readings.

- c) The voltage measurement leads should be independent of the current leads and should be connected as closely as possible to the terminals of the winding to be measured. This avoids including the resistances of current-carrying leads and their contacts and the resistances of extra lengths of leads in the reading.
- d) The currents used for these measurements should not exceed 15% of the rated transformer winding current. This avoids heating the winding and possible change in resistance.
- e) Due to the inductance of the winding, a long dc time constant can result in unstable current and voltage readings for some time.

Readings should not be recorded until after the current and voltage have reached steady-state values. Stable readings can be obtained faster by using a dc power source designed specifically to saturate inductive loads.

7.2.7.3 Discharge

WARNING

Safety must be carefully considered before disconnecting test leads. Energy stored in the transformer windings from the applied direct current can create large discharge voltage (inductive kickback). A discharge circuit must be employed that can dissipate the stored energy quickly.

The test system or winding resistance meter should have an indication light or a sound alarm to show when discharge is complete and test leads can be removed safely.

7.2.7.4 Demagnetization

The direct current used to measure winding resistance may cause the core to magnetize (polarize). A magnetized core can cause high inrush currents when the transformer is energized. To help reduce damage to the transformer and associated protection systems, the core may be demagnetized prior to applying full alternating current (ac) voltage.

The demagnetizing procedure consists of applying direct current to the windings and reversing polarity a number of times while reducing the current applied until the core is demagnetized. See 7.2.11.1.1.

CAUTION

This procedure involves disconnecting and reconnecting the test leads many times. Be sure the windings are discharged completely before removing the leads. A test system or winding resistance meter that performs demagnetization without disconnecting and reconnecting the leads is recommended.

The winding resistance meter dc power source can be used for this procedure.

An alternate method is to apply a significant portion of the ac voltage and then gradually reduce the voltage; however this factory method is seldom available in the field.

7.2.7.5 Winding connections

The individual phase or terminal-to-terminal resistance readings should be recorded along with the sum total winding resistance.

7.2.7.5.1 Wye windings

For wye windings, the recorded resistance measurement may be from terminal to terminal or from terminal to neutral. For terminal-to-terminal measurements, the total resistance reported is the sum of the three measurements divided by two.

7.2.7.5.2 Delta windings

For delta windings, the recorded resistance measurement may be from terminal to terminal with the delta closed or from terminal to terminal with the delta open to obtain the individual phase readings. If the delta is open, the reported total winding resistance is the sum of the three phase readings. If the delta is closed, the recorded total winding resistance is the sum of the three phase-to-phase readings times 1.5.

7.2.7.5.3 Autotransformer windings

For autotransformer winding resistance measurements, the following method or an equivalent should be used. For the series winding resistance, the current should be circulated between the HV and neutral terminals and the voltage measured between the HV terminal and the LV terminal. For the common winding resistance, the current should be circulated between the HV and neutral terminals and the voltage measured between the LV and neutral terminals. If needed, for the resistance of the lead and in-line windings (if any) between the LV terminal and the neutral connection, the current should be applied between the HV terminal and the LV terminal and the voltage measured between the LV terminal and the neutral terminal.

7.2.8 Ground connection

Use of working grounds should comply with established company guidelines. This should include both equipment that is out of service and testing equipment such as cranes, trailers, etc. For further information see ASTM F855 [B20].

7.2.9 Graded insulation

When the insulation level of the winding is graded from one end to the other, the magnitude of the applied test voltage should correspond to the lowest insulation level.

CAUTION

When testing multiple apparatus together such as bushings and windings where the voltage rating of the bushing may be the limiting component or when shorting HV and LV windings together, test operators must be aware of the limiting component when selecting the test voltage.

7.2.10 Ratio/polarity/phase relation

7.2.10.1 General

The turns ratio of a transformer is the ratio of the number of turns in a HV winding to that in a LV winding. The voltage ratio of a transformer is the ratio of the root mean square (rms) terminal voltage of an HV winding to the rms terminal voltage of an LV winding under specified conditions of no load.

The polarity of a transformer is determined by the internal connections and is indicated by the nameplate markings. Polarity is of interest if the transformer is to be connected in a parallel manner with one or more other transformers.

Results of the polarity and voltage ratio tests are absolute and should be compared with the manufacturer's nameplate specifications.

Transformer voltage ratio, polarity, and phase connections should be checked before energizing the transformer for the first time on site. The voltage ratio in particular is checked during regular (e.g., annual) inspections, and it is recommended that it also be checked before returning the transformer to service if the unit has been subjected to a through-fault. This ratio test is typically performed in the as-found tap positions.

During field commissioning, if the transformer has taps for changing the voltage ratio, the voltage ratio should be determined for all taps. This means the voltage ratio test should be performed for all taps of the DETC and all taps of the LTC. It is considered best practice to perform the final (DETC) voltage ratio test in the as-left position and to test the voltage ratio after a DETC change.

CAUTION

If the DETC has been left in the same position for a long period of time, extreme caution should be used when changing taps because the drive mechanism may seize or fail from the contacts becoming stuck or possibly coked due to non-operation.

7.2.10.2 Measuring the transformer ratio/polarity/phase relationship

A number of modern commercial transformer ratio measurement test sets are available from manufacturers serving the power industry. These instruments, when operated in accordance with the manufacturer's instructions, provide convenient and accurate readings of the following:

- Voltage ratio
- Polarity verification
- Phase angle
- Test meter current

The term “transformer turns ratio” (TTR) meter is commonly used to describe these instruments even though the actual turns ratio is not being measured.

The principle of operation is to apply a reduced voltage to the HV terminals and produce a resulting voltage at the LV terminals. The two voltages are accurately measured and used to calculate and display the transformer voltage ratio.

A continuous testing mode facilitates measurements on multi-tap windings and quickly measures, displays, and records the ratio and test meter current for each tap. Communication ports are helpful in automating testing and test data recording.

When a three-phase transformer voltage ratio test is performed on a single-phase basis, the proper connections and phase vector relationships should be considered. Detailed connection charts with corresponding formulas referring measured voltage ratio to nameplate voltage ratio are typically provided with the ratio measurement instrument for this method.

7.2.10.3 Other methods for measuring the transformer ratio/polarity/phase relationship

If a commercial test set is not available, the transformer voltage ratio can be measured using the voltmeter method or the capacitance and power factor bridge method. Also, polarity can be verified using the inductive kick method or the alternating voltage test method.

These other methods are described in Annex H.

7.2.10.4 Interpreting the transformer ratio test

The tolerance of the voltage ratio should be within 0.5% of the specified nameplate voltage for all windings and winding taps. The following are some clarifications on how the voltage ratio tolerance should be interpreted and applied.

The voltage ratio tolerance applies to the phase-to-neutral voltage for three-phase, wye-connected windings. If the phase-to-neutral voltage is not explicitly indicated on the nameplate, it may be calculated by dividing the phase-to-phase voltage by $\sqrt{3}$.

The measured voltage ratios of the outer phases of a three-phase transformer are sometimes observed to be slightly different but are not a cause for the transformer to be rejected unless the difference is $> 0.5\%$.

NOTE 1—A difference in the outer phases may be observed when windings and lead arrangements are designed neglecting “half-turn-effect” compensation and may not always represent a problem.

NOTE 2—In some cases, transformers with load taps in the LV winding are not designed with an equal number of turns per tap. This is due to an unevenly divisible number of turns required for voltage regulation on an LV winding having a low overall number of turns. In such cases, the voltage variation per tap changer operation is not uniform and may not be within the 0.5% tolerance of some of the nameplate tap voltages. As long as all three phases have the same measured voltage ratios and the measured voltage ratios at the extreme ends of the regulation range are within the 0.5% tolerance, the transformer should not be rejected.

NOTE 3—In some cases, a transformer LTC switches coarse and fine tap windings in various combinations to achieve the desired voltage ratio. In these cases, it is especially important that the ratios should be observed to have a trend that preserves appropriate step voltage ratio changes throughout the tap range.

NOTE 4—In some cases, a transformer LTC relies on a bridging transformer (autotransformer) to provide an intermediate voltage ratio between two winding tap voltage ratios. Bridging positions should be observed to have an appropriate intermediate ratio.

7.2.11 Excitation current

7.2.11.1 Residual magnetism

7.2.11.1.1 Effect of residual magnetism

CAUTION

This procedure involves disconnecting and reconnecting the test leads many times. Be sure the windings are discharged completely before removing the leads. A test system or winding resistance meter that performs demagnetization without disconnecting and reconnecting the leads is recommended.

The transformer core may have residual magnetism present as a result of being disconnected from the power line or, as is frequently the case, as a result of dc measurements of winding resistance. The residual magnetism results in the measurement of higher-than-normal excitation current.

There is no widely accepted field method for distinguishing between the effect of residual magnetism and the effect of a problem present in the transformer. However, experience shows that although some residual magnetism is almost always present in the core, in most cases it has no significant effect on test results.

In most of the problems detected by using this procedure, the difference between the individual phase currents of three single-phase transformers or between the currents of the outer phases of three-legged, three-phase transformers has exceeded 10%. Exceptions occur when the core is shell-form or five-legged. However, smaller changes in relative currents may also be indicative of problems associated with the core and should be investigated. If a significant change in the test results is observed, the only known reliable method of excluding the effect of residual magnetism is to demagnetize the transformer core.

It is recommended that the dc measurements of the winding resistance be performed after the excitation current tests.

7.2.11.1.2 Methods for demagnetization

Two techniques can be used to demagnetize the transformer core. The first method is to apply a diminishing alternating current to one of the windings. For most transformers, due to the high voltage ratings involved, this method is impractical and involves safety hazards.

A more convenient method is to use direct current. The principle of this method is to neutralize the magnetic alignment of the core iron by applying a direct voltage of alternate polarities to the transformer winding for decreasing intervals. The interval is usually determined when the demagnetizing current reaches a level slightly lower than the previous level, at which time the polarity of the voltage is reversed. The process is continued until the current level is zero. On three-phase transformers, the usual practice is to perform the procedure on the phase with the highest excitation current reading. In most cases, experience has demonstrated that this procedure is sufficient to demagnetize the whole core.

7.2.11.2 Excitation current test

7.2.11.2.1 General

The excitation current test, also known as the *single-phase excitation test*, is most often performed in the field as a diagnostic test to monitor the open circuit characteristics of the windings of a transformer. It may also be used in the factory as a preliminary test to check the transformer circuit before applying full power.

The excitation current test can be useful to locate defects such as parallel windings with unequal turns, problems in some types of shields, and foreign objects that create shorted turns. In some cases, this test can also cause a dielectric breakdown in insulation that has been damaged, such as a slipped or crushed turn that has previously failed or is severely contaminated. It can also be used to detect problems in tap-changing devices and major defects in the magnetic core structure.

The excitation current test consists of a simple open-circuit measurement of the current magnitude and loss, typically on the HV side of the transformer, with the terminals of the other windings left floating (with the exception of a grounded neutral). Excitation current is measured at rated frequency and usually at voltages up to 10 kV. Three-phase transformers are tested by applying a single-phase test voltage to one phase at a time.

The diagnostic analysis of excitation current test results is based largely on pattern recognition. Both current magnitude and loss depend on the transformer design and are unique for each unit. The results can also be used in comparison with previous data from tests performed at the same test voltage.

Load losses and impedance voltage field test data should be comparable to the factory test data.

Factory losses are measured under three-phase excitation, at rated voltage and reported as the sum of three current (I) squared times resistance (R) and stray losses. Field losses are measured under single phase excitation, at current much lower than rated and reported as per phase I^2R and stray losses.

7.2.11.2.2 Measurement

The excitation current measurement consists of two components: the total current magnitude (mA) and loss (W). The magnitude of the excitation current reading depends on the relative values of the inductive and capacitive components of the core and insulation, while the loss component is always dominated by the core loss, primarily generated by eddy currents.

Excitation current should be measured at the highest test voltage possible, but should never exceed the voltage rating of the winding across which the test voltage is applied. Selecting the highest possible test voltage assures maximum turn-to-turn stress. However, due to historical practices and the availability of equipment capable of measuring excitation current, the industry has accepted 10 kV as the maximum for such measurements. For this reason, some utilities specify the inclusion of field excitation current tests at the factory, i.e., single-phase tests performed at 10 kV. The applied voltage up to 10 kV may also be limited by the test set power rating, and the test should be performed at the highest test voltage possible within the range of the test set.

The excitation current test can be performed by applying voltage to either the HV terminals or the LV terminals. However, these tests can be confined to the HV windings. Defects in the LV windings should still be detected, and the charging current required should be reduced. In case of suspected problems, consideration may be given to measuring excitation current in the LV windings.

It is generally desirable to make all excitation current measurements on a given transformer at the same potential; however, there may be certain exceptions in the case of units equipped with an LTC. For example, LTCs equipped with a preventive autotransformer can be operated in either the bridging or the non-bridging position. The LTC position may burden the test set differently, so two different test voltages may have to be applied for these positions (e.g., 2 kV and 10 kV bridging and non-bridging positions, respectively).

NOTE 1—The bridging LTC positions generally require more current and often are the limiting factor for the applied test voltage. If the test voltage can be raised to some moderate level (e.g., 6 kV) with the preventive autotransformer in the circuit (bridging), then perform all tests on all positions of the LTC at a convenient voltage such as 5 kV.

The phase pattern obtained for the magnitude of the excitation current may differ from one obtained for the loss. The loss component is always dominated by the core loss and, as such, normally has an expected phase pattern. The current magnitude and loss pattern relationship becomes very useful in interpreting the current readings. The loss pattern is often used to validate the results when unexpected current magnitudes are obtained.

NOTE 2—Residual magnetism may produce misleading excitation current values when these are used solely as a criterion of the condition of the magnetic circuit of a transformer. Although some residual magnetism is almost always present in the core, in most cases it has no significant effect on test results. If residual magnetism exists, then it is recommended that the core be demagnetized before performing the test. Often, residual magnetism is introduced as a result of a dc winding resistance test. It is recommended to perform excitation current tests before any test that may introduce residual magnetism.

Winding terminals normally grounded in service should be grounded during the excitation current tests, except for the particular winding energized for test. For example, with a wye-wye transformer, the neutral of the winding energized for test would be connected to the return lead (measure) of the measurement circuit, while the neutral of the other winding would be connected to ground.

WARNING

Caution should be exercised in the vicinity of transformer terminals because voltage is induced in all windings during the test.

7.2.11.2.3 Equipment

Equipment required for the excitation current test include a single-phase, 50 Hz or 60 Hz power supply capable of delivering up to 10 kV and a measurement circuit capable of measuring current in milliamperes (mA) and optionally power loss in watts (W). The measurement circuit needs to be capable of operating in an ungrounded specimen test (UST) in the GROUNDED-GUARD Mode (see A.3.3). The return lead from the winding under test must be isolated from ground and from any other current trying to return to the test set. Undesired return currents should return to the test set via the GROUNDED-GUARD terminal. Figure 3 illustrates the basic test circuit.

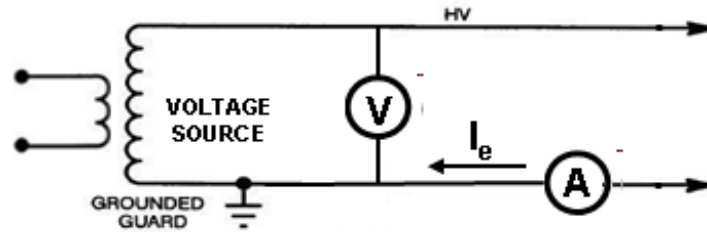


Figure 3—Basic measurement circuit

7.2.11.2.4 Test procedure

The excitation current measurements shown in Table 10 should be performed on transformers being tested.

Table 10—Measurements required for excitation current test

LTC	Required test measurements
Present	Test on the following LTC positions: half of the LTC positions, neutral, and one step in the opposite direction. See NOTE 1, NOTE 1, and NOTE 3.
Not Present	DETC position that is dictated by the in service conditions
NOTE 1—The DETC should be set to the voltage tap position that is dictated by the in service conditions for these tests.	
NOTE 2—The excitation current test is not significantly affected by changing tap positions unless the change results in a change to the magnetic excitation circuit. Therefore, the tap positions listed above are sufficient to demonstrate acceptable performance, subject to the comments in NOTE 3.	
NOTE 3—Some transformers are designed with unequal step voltages. If the step voltage is used to excite an auxiliary transformer, such as a preventive autotransformer or reactor, then an additional excitation current test should be performed at the LTC position that results in the maximum excitation voltage in the auxiliary transformer.	

The typical test connections for various winding configurations are shown in Figure 4/Table 11, Figure 5/Table 12, Figure 6/Table 13, and Figure 7/Table 14.

Excitation current measurement of a single-phase transformer

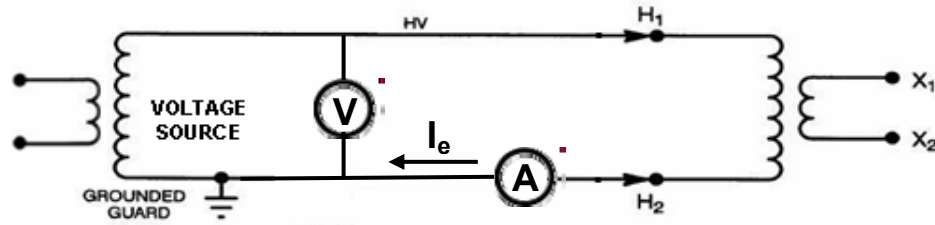


Figure 4—Measuring circuit for a single-phase transformer

Table 11—Test procedure for single-phase transformer

Test no.	Energize	Meter input	Float ^a	Measure (I_e)
1	H ₁	H ₂ ^b	X ₁ X ₂ Y ₁ Y ₂	H ₁ -H ₂
2	H ₂ ^b	H ₁	X ₁ X ₂ Y ₁ Y ₂	H ₂ -H ₁

^a Normally grounded terminals of X and Y must be grounded.

^b H₂ may be designated as H₀.

Excitation current measurement of a single-phase autotransformer

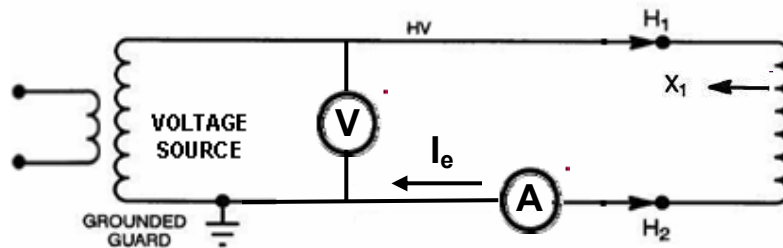


Figure 5—Measuring circuit for a single-phase autotransformer

Table 12—Test procedure for single-phase autotransformer

Test no.	Energize	Meter input	Float ^a	Measure (I_e)
1	H ₁	H ₀ X ₀ ^b	Y ₁ Y ₂	H ₁ -H ₀ X ₀
2	H ₀ X ₀	H ₁	Y ₁ Y ₂	H ₀ X ₀ -H ₁

^a If present, normally grounded terminals of Y must be grounded.

^b H₂ may represent H₀ or X₂ or X₀ (not shown in Figure 5).

Excitation current measurement of a wye-connected transformer

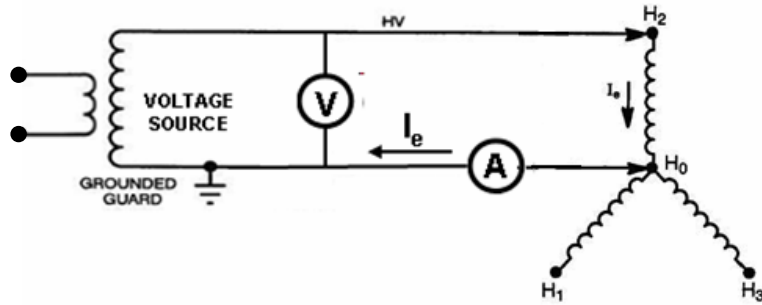


Figure 6—Measuring circuit for a wye-connected transformer

Table 13—Test procedure for wye-connected transformer

Test no.	Energize	Meter input	Float ^a	Measure (I_e)
1	H ₁	H ₀	H ₂ H ₃ X ₁ X ₂ X ₃ Y ₁ Y ₂ Y ₃	H ₁ - H ₀
2	H ₂	H ₀	H ₁ H ₃ X ₁ X ₂ X ₃ Y ₁ Y ₂ Y ₃	H ₂ - H ₀
3	H ₃	H ₀	H ₁ H ₂ X ₁ X ₂ X ₃ Y ₁ Y ₂ Y ₃	H ₃ - H ₀

^a Normally grounded terminals X and Y must be grounded.

Excitation current measurement of a delta-connected transformer

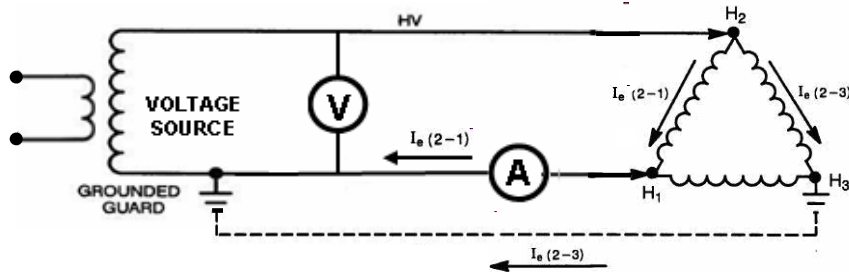


Figure 7—Measuring circuit for a delta-connected transformer

Table 14—Test procedure for delta-connected transformer

Test no.	Energize	Meter input	Ground	Float ^a	Measure (I_e)
1	H ₁	H ₃	H ₂	X ₁ X ₂ X ₃ Y ₁ Y ₂ Y ₃	H ₁ - H ₃
2	H ₂	H ₁	H ₃	X ₁ X ₂ X ₃ Y ₁ Y ₂ Y ₃	H ₂ - H ₁
3	H ₃	H ₂	H ₁	X ₁ X ₂ X ₃ Y ₁ Y ₂ Y ₃	H ₃ - H ₂

^a Normally grounded terminals of X and Y must be grounded.

7.2.11.2.5 Analysis of test results

The usual approach to the analysis of the excitation current test results is to compare the results with the previous tests or with similar single-phase transformers or with phases of a given three-phase transformer. For the great majority of three-phase transformers, the pattern is two similar high readings on the outer phases and one lower reading on the center phase. The recommended initial tests include measurements at half of the LTC positions, the neutral position, and one step in the opposite direction. The results may differ for various LTC positions, but the relationship between the phases is expected to remain unchanged. The understanding of how the LTC affects the current magnitude of individual phases is essential for developing proper analysis.

Three types of patterns can be easily described:

- a) High-low-high (HLH) pattern
 - Expected for a three-legged core transformer.
 - Expected for a five-legged core (or shell) transformer with a delta-connected secondary winding.
- b) Low-high-low (LHL) pattern
 - Obtained on a three-legged core transformer if the traditional test protocols are not followed.
 - Neutral on high side wye-configured transformer is inaccessible.
 - Obtained when the third terminal on a delta-connected transformer has not been grounded.
 - Expected for a four-legged core transformer.
- c) All three similar patterns
 - Expected for a five-legged core (or shell) transformer with a non-delta secondary winding.

References are provided in Annex J (Poulin [B66], Rickley and Clark [B67], Rickley et al. [B68], Lachman [B54], Duplessis [B25], IEEE Std 62™-1995[B33]).

7.2.12 Leakage reactance/short-circuit impedance testing

7.2.12.1 General

The short-circuit impedance, or leakage reactance, of power transformers is often measured in the field to detect physical damage in transformer windings. During the service life of a transformer, a number of overcurrent events can occur. Winding deformation can result from high electro-mechanical stresses imposed by high currents. Transformers can perfectly function in the presence of deformation under normal load conditions. However, the mechanical integrity of such a unit is degraded, making it more likely to lead to an immediate failure during an overcurrent event. Furthermore, the winding damage can occur during transportation and installation. The leakage reactance test is sensitive to winding deformation and, as such, can be used to ascertain the presence of winding distortion and gauge the risk of failure.

Leakage reactance tests should be viewed in context with other tests that are sensitive or symptomatic to winding deformation. These tests are as follows:

- Frequency response analysis
- Low voltage impulse
- Capacitance (as part of PF testing)

7.2.12.1.1 Relationship between leakage reactance and short-circuit impedance and linear relationship between current and leakage flux

The leakage flux path includes the iron core and the unit permeability space (air/liquid). The reluctance of the former is significantly smaller than that of the latter. Therefore, the reluctance of the leakage flux path is determined by the air/liquid space, which produces a linear relationship between current and leakage flux. Thus, change in flux (Ψ) per unit change in current, i.e., leakage inductance ($L = \Psi/I$), remains the same regardless of the current level. This means that leakage reactance measured at several amperes is the same as one measured at full load current (within a margin of error). This allows the comparison of factory-measured short-circuit impedance with field-measured leakage reactance.

7.2.12.1.2 Factory measurement of impedance and load loss versus field measurement of leakage reactance and loss

At the factory:

- The measurement uses three-phase excitation and rated current, and impedance is calculated as the average of all three phases.
- Load loss represents the sum of I^2R and stray losses associated with the full load current.

In the field:

- The measurement *usually* uses single-phase excitation and current created by low voltage (e.g., 2 A to 10 A and 100 V), and per-phase leakage reactance is calculated as a reactive component of the impedance.
- Loss represents the sum of I^2R and stray losses associated with the current much lower than rated load current.

A combined influence of different instrumentation and test setups, difference in flux distribution under three- and single-phase excitation, the presence of the resistive component, and the averaging of factory data can create a diagnostically significant variance between the two values. However, the differences between factory and field test conditions notwithstanding, the factory nameplate impedance can serve as a useful guideline (mostly for initial field measurement) for evaluating the value measured in the field. The factory losses, however, *cannot* be compared with losses measured in the field. Leakage reactance is often referred to in terms of $\%X$ because it includes only the reactive portion of the impedance, whereas short-circuit impedance ($\%Z$) includes the total impedance. Given the interchangeability of these two tests for field diagnostics, applicable diagnostic methods and limits can be applied to both types of measurements.

7.2.12.1.3 Leakage reactance test methods

There are two methods for performing leakage reactance tests, as follows:

- Three phase equivalent test
- Per-phase test

For new or rebuilt three-phase transformers, the results of an initial three-phase equivalent test are compared with the nameplate value. A set of three per-phase tests is also performed for phase comparison and provides a benchmark for future analysis. For single-phase transformers, per-phase tests are performed and the results are compared with the nameplate values. To properly compare leakage reactance test results, it is essential that all tests be performed on the same DETC and LTC positions as indicated by the nameplate or benchmark results.

7.2.12.1.4 Leakage reactance limits

For three-phase transformers, the three-phase equivalent test results should be within 3% of nameplate values. Analysis needs to consider the effects of different test setup, instrumentation, or tap changer position.

For three-phase transformers, the per-phase test results should be within 3% of the average value of all three phases. In some rare cases, it is possible that the three-phase equivalent measurement matches nameplate/benchmark results, but the per-phase measurement does not. This could be due to the influence created by the reluctance of the leakage flux path outside the leakage channel, thereby masking the changes in the leakage channel. In these cases, the results of the three-phase equivalent measurement should be used as a benchmark for future comparison.

For single-phase transformers, the per-phase test results should be within 3% of the nameplate value.

7.2.12.1.5 Test methods and procedures

Commercial automated leakage reactance test sets are available that can simplify the test setup, reporting, and data recording. When performing three-phase equivalent or single-phase leakage reactance tests using these instruments, refer to the appropriate test set manufacturer's literature for test procedure and setup. It is also possible to perform the test by using a voltmeter-ammeter.

In general, the short-circuit impedance, the reactance, and the impedance components can be calculated by Equation (2), Equation (3), Equation (4), and Equation (5):

$$Z = V/I \quad (2)$$

$$R = R_{P-dc} + R_{S-dc} + R_L \quad (3)$$

$$R = P/I^2 \quad (4)$$

$$X = \sqrt{Z^2 - R^2} \quad (5)$$

In Equation (2) to Equation (5), Z is the short-circuit impedance; X is the reactance; and R is the total resistance attributed to the primary winding dc losses (R_{P-dc}), secondary winding losses (R_{S-dc}), and eddy-current losses caused by leakage flux in the conductors, windings, and structural components (R_L).

When using a manual stand-alone voltmeter-ammeter, a power source is used to drive a current through the phase's winding under test. The current and the voltage across the impedance are measured simultaneously. The impedance is then given by the ratio of the measured voltage and current.

7.2.12.2 Leakage reactance: three-phase equivalent test

The three-phase equivalent test works for both delta- and wye-configured windings. In both cases, the neutral terminals are not used. The three LV-side or line-side bushing terminals are short-circuited together. Short circuits should be adequately sized, greater than No. 1 AWG, so that test results are not adversely influenced by resistance created by the small conductors. The short circuits should be as short as possible. Contacts should be clean and tight. Three sets of readings are taken on the three pairs of bushing terminals that correspond to the three legs of the transformer windings. For either a delta or a wye configuration, this would normally be from H_1 to H_2 , H_2 to H_3 , and H_3 to H_1 .

$$\%X = [(1/60) \sum X_m] [S_{3\Phi} / V_{LL}^2] \quad (6)$$

In Equation (6), the three individual reactance measurements are summed together to form $\sum X_m$. $S_{3\Phi}$ and V_{LL} are the base three-phase power and line-to-line voltage of the winding being measured. Three-phase power has the units of kVA, and line-to-line voltage has the units of kV.

7.2.12.3 Leakage reactance: per-phase equivalent test

The per-phase equivalent test can be very useful in further examining the individual windings. The three-phase equivalent test can be used to see the average of all three phases and compare them with the nameplate values. Given that the three phases are averaged, results can be masked that could be more closely examined in the per-phase test. In general, the three phases should compare very well. The difference in average is the difference between a single phase's per-phase result and the average of the three per-phase results. Normally, the average difference should be no greater than 3%.

The test is performed by measuring each individual core leg on the primary side with the corresponding core leg terminals on the secondary side short-circuited together. The test method differs from the three-phase equivalent in two ways: (1) only one core leg at a time is shorted on the secondary side and (2) a wye-configured winding is tested on a per-leg basis, not from line to line.

For a delta-connected winding:

$$\%X = [(1/30) X_m] [S_{3\Phi} / V_{LL}^2] \quad (7)$$

For a wye-connected winding:

$$\%X = [(1/10) X_m] [S_{3\Phi} / V_{LL}^2] \quad (8)$$

In Equation (7) and Equation (8), the three individual reactance measurements are summed together to form $\sum X_m$. $S_{3\Phi}$ and V_{LL} are the base three-phase power and line-to-line voltage of the winding being measured. Three-phase power has the units of kVA, and line-to-line voltage has the units of kV.

7.2.12.4 Manual test method

The manual test method is applicable to testing either single-phase or three-phase transformers.

True rms responding meters (voltmeter and ammeter) with accuracies of at least 0.5% and a sinusoidal 60 Hz (rated frequency) adjustable power source should be used for the measurements. The adjustable power source could be derived from the station service transformer through a variable autotransformer rated 0 to 280 V and at least 10 A. Alternatively, a completely isolated power amplifier with an internal 60 Hz oscillator rated at least 250 VA may be used. The adjustable power source should not be obtained directly from a portable gasoline engine generator since the output waveform is usually distorted and its frequency is not sufficiently stable.

7.2.12.4.1 Impedance test of a single-phase transformer

One of the two windings of the transformer (usually the LV winding) is short-circuited with a low-impedance conductor, and voltage at rated frequency is applied to the other winding. The energizing voltage is adjusted to circulate current on the order of 0.5% to 1.0% of rated current in the windings or 2 A to 10 A, depending on the rating of the transformer under test. Care should be taken to limit the test current so that it does not cause the energizing voltage waveform to become distorted due to overloading the power

source. An oscilloscope could be used to observe the voltage waveform during testing. The energizing voltage can be extremely small in comparison with the rated voltage of the winding without introducing significant errors. A typical arrangement is shown in Figure 8.

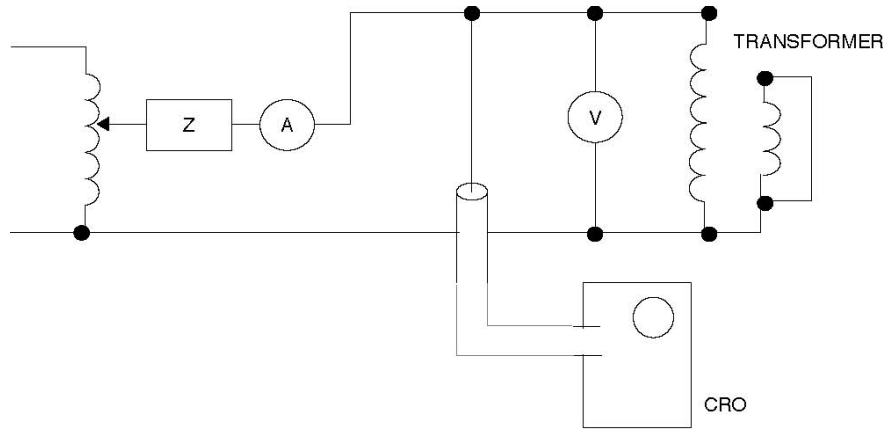


Figure 8—Short-circuit impedance measurement on single-phase transformer

For accurate measurements, the voltmeter should be connected directly to the transformer terminals to avoid voltage drop in the current-carrying leads. Meter ranges should be chosen so that their readings are in the upper half of full scale. The current and voltage readings should be read simultaneously.

The %Z of the single-phase transformer can be calculated using Equation (9):

$$\%Z_{\text{single-phase}} = (1/10) \cdot [(E_m/I_m) \cdot kVA_r / (kV_r)^2] \quad (9)$$

where

- E_m = measured test voltage
- I_m = measured test current
- kVA_r = the rating of the transformer in kilovolt-amperes
- kV_r = the rating of the winding being energized in kilovolts

7.2.12.4.2 Impedance test of an autotransformer

An autotransformer may be tested for impedance with its internal connections unchanged. The test is made by short-circuiting its LV terminals and applying voltage at rated frequency to the HV terminals. The same procedure is followed as that used for a single-phase transformer.

7.2.12.4.3 Impedance test of a three-phase, two-winding transformer

A three-phase transformer may be tested for impedance using a single-phase power source regardless of winding connection. The neutral terminals, if any, are not used. The test is made by short-circuiting the three line-leads of the LV windings and applying a single-phase voltage at rated frequency to two terminals of the other winding. Three successive readings are taken on the three pairs of leads, (e.g., H₁ and H₂, H₂ and H₃, H₃ and H₁), with the test current adjusted to the same level for each reading. Then the %Z of the three-phase transformer is given by Equation (10):

$$\%Z_{\text{three-phase}} = (1/60) \cdot [(E_{12} + E_{23} + E_{31})/I_m] \cdot [kVA_{3r} / (kV_{1r})^2] \quad (10)$$

where

E_{12}, E_{23}, E_{31} = measured test voltages
 I_m = measured test current
 kVA_{3r} = the three-phase rating in kilovolt-amperes
 kV_{lr} = the rated line-to-line voltage of the energized windings

7.2.12.4.4 Impedance test of a three-winding transformer

A three-winding transformer, which may be either single-phase or three-phase, may be tested for impedance by making two-winding impedance measurements with each pair of windings (which means three different impedance measurements) following the same procedure as that used for a two-winding transformer. The individual equivalent impedance of the separate windings may then be determined using the Equation 11, Equation 12, and Equation 13:

$$Z_1 = (Z_{12} - Z_{23} + Z_{31})/2 \quad (11)$$

$$Z_2 = (Z_{23} - Z_{31} + Z_{12})/2 \quad (12)$$

$$Z_3 = (Z_{31} - Z_{12} + Z_{23})/2 \quad (13)$$

where

Z_{12}, Z_{23}, Z_{31} = the measured impedance values between pairs of windings, as indicated, all expressed on the same kVA base

7.2.12.4.5 Impedance test of an autotransformer with tertiary winding

An autotransformer with tertiary winding, which may be either single-phase or three-phase, may be tested for impedance using the same procedure as that used for a three-winding transformer.

7.2.12.5 Interpretation of the impedance test

A change in the short-circuit impedance of the transformer indicates a possible winding movement. Since the overall measurement accuracy is no better than 1%, using 0.5% accuracy meters, changes of $\pm 2\%$ of the short-circuit impedance are usually not considered significant. Changes of more than $\pm 3\%$ of the short-circuit impedance should be considered significant. For example, a short-circuit impedance change from 5.0% to 5.4% should be considered significant since it indicates a change of 8%. For further information on impedance testing, see IEEE Std C57.12.90.

7.2.13 Insulation resistance

Insulation resistance tests are made to determine insulation resistance from individual windings to ground or between individual windings. Insulation resistance tests are commonly measured directly in megohms or may be calculated from measurements of applied voltage and leakage current.

The recommended practice in measuring insulation resistance is to always ground the tank (and the core). Short-circuit each winding of the transformer at the bushing terminals. Resistance measurements are then made between each winding and all other windings grounded.

Windings are never left floating for insulation resistance measurements. Solidly grounded windings must have the ground removed to measure the insulation resistance of the grounded winding. If the ground

cannot be removed, as in the case with some windings with solidly grounded neutrals, the insulation resistance of the winding cannot be measured. Treat it as part of the grounded section of the circuit.

7.2.13.1 Typical test connections

The following test connections should be used, depending on the type of transformer and the number of windings.

Two-winding transformer

- $(HV + LV) - GND$
- $HV - (LV + GND)$
- $LV - (HV + GND)$

Three-winding transformer

- $HV - (LV + TV + GND)$
- $LV - (HV + TV + GND)$
- $(HV + LV + TV) - GND$
- $TV - (HV + LV + GND)$

Autotransformer (two winding)

- $(HV + LV) - GND$

Autotransformer (three winding)

- $(HV + LV) - (TV + GND)$
- $(HV + LV + TV) - GND$
- $TV - (HV + LV + GND)$

The temperature of the windings and insulating liquid should be near the reference temperature of 20 °C. For diagnosis, a connection with guard circuit may be used for insulation resistance measuring.

Under no circumstances shall tests be made while the transformer is under vacuum.

7.2.13.2 Test voltages

Test voltages are typically 500 V, 1000 V, 2500 V, or 5000 V dc. Voltage should be increased in increments, typically 1 kV to 5.0 kV, and held for 1 min while the current is read. The insulation resistance may vary with applied voltage, and any comparison should be made with measurement at the same voltage.

The dc voltage applied for measuring insulation resistance to ground should not exceed a value equal to the rms low-frequency applied voltage.

The test should be discontinued immediately if the current begins to increase without stabilizing.

After the test has been completed, terminals should be grounded long enough to allow any trapped charges to decay to a negligible value.

7.2.13.3 Interpretation of results

The results of insulation resistance tests generally require some interpretation, depending on the design, type of insulating liquid, and dryness and cleanliness of the insulation involved. Transformers of higher insulation class typically have higher insulation resistance. Transformers filled with natural ester-based liquid generally have lower insulation resistance compared with ones filled with mineral oil. Comparison with factory results or previous field results is more significant than the absolute value of megohms. The winding and oil temperature also affects the reading, typically testing at higher temperature results in lower insulation resistance.

It is recommended that insulation resistance values be measured periodically (during maintenance shutdown) and plotted. Substantial variations in the plotted insulation resistance values should be investigated for cause.

When the insulation resistance falls below base line values, it can, in most cases of good design and where no defect exists, be brought up to the required standard by cleaning and drying the apparatus.

An insulation resistance of zero or a very low value indicates a grounded winding, a winding-to-winding short, or heavy carbon tracking. This possibility should be confirmed by additional tests such as polarization index, insulation PF, or moisture content of insulating liquid.

7.2.13.4 Polarization index tests

The polarization index is the ratio of the insulation resistance at the end of a 10 min test to that at the end of a 1 min test at a constant voltage.

The total current developed when applying a steady-state dc voltage is composed of the following three components:

- *Charging current*, due to the capacitance of the insulation being measured. This current falls off from maximum to zero very rapidly.
- *Absorption current*, due to molecular charge shifting in the insulation. This transient current decays to zero more slowly.
- *Leakage current*, which is the true conduction current of the insulation. The leakage current varies with the test voltage. It may also have a component due to the surface leakage that is due especially to surface contamination.

Since leakage current increases at a faster rate with moisture present than does absorption current, the megohm readings do not increase with time as fast with insulation in poor condition as with insulation in good condition. This results in a lower polarization index.

An advantage of the index ratio is that all of the variables that can affect a single megohm reading, such as temperature and humidity, are essentially the same for both the 1 min and 10 min readings.

$$\text{Polarization Index} = \frac{\text{10 min insulation resistance (megohm) reading}}{\text{1 min insulation resistance (megohm) reading}}$$

The following are guidelines for evaluating transformer insulation using polarization index values:

Less than 1.0 = Dangerous

1.0 to 1.1	= Poor
1.1 to 1.25	= Questionable
1.25 to 2.0	= Fair
Above 2.0	= Good

The polarization index method should not be used to assess insulation condition in new power transformers.

The polarization index for insulation liquid is always close to 1. Therefore, the polarization index for transformers with low conductivity liquids (e.g., new mineral oil) may be low in spite of good insulation condition.

7.2.14 Capacitance and dissipation factor (power factor)

7.2.14.1 Introduction: dissipation factor (power factor) and capacitance testing

Electrical DF testing goes hand in hand with capacitance testing. For the purposes of discussion, both PF and DF are considered functionally equivalent; however, there are differences in the calculations themselves. Typically oil analysis uses the DF term while electrical testing at least in North America calls it PF. Be aware that *tan-delta* is also another common way to refer to these functionally equivalent tests. For nearly all transformers, PF, DF, and *tan-delta* have the same calculated value up to two significant digits for most insulation systems below 10% PF and, therefore, can be used interchangeably for transformer insulation assessment.

The PF has long been known as one of the most effective methods of assessing the overall condition of a transformer and is central to a transformer condition-based maintenance program. The ac capacitance test is a subset of the PF test because the capacitance value and associated charging current are required to calculate the PF. Both values are routinely evaluated together due to their close association. In fact, these two measured values should always be analyzed together to ensure that the condition of the transformer insulation is being properly assessed.

PF testing of transformers can help determine whether the level of contamination is above acceptable risk standards or whether there is a possibility of mechanical damage due to bulk coil movement. The PF itself is one of the leading methods for detecting moisture and contamination within a transformer, but it can also be influenced by the condition of the bushings and testing environment. The capacitance measurement (as part of the PF test) can help in judging whether there has been a bulk movement of the coil or whether a layer of insulation has been shorted.

7.2.14.1.1 Principle of insulation isolation

This guide contains test procedures for the three most common transformer configurations (one winding, two windings, and three windings). These procedures help to verify that the insulation systems contained within the transformers are isolated for individual analysis. As a principle, insulation systems should not be evaluated in groups or sets; in other words, the test results that contain more than one set of test results should not be evaluated except to verify testing validity. It is possible that a good insulation system can mask poor test results from another insulation system, thus diluting the results.

7.2.14.1.2 Capacitance

Electrical insulation systems within a transformer can be generalized to simple parallel plate capacitor anode and cathodes. Three general variables contribute to changes in capacitance: dielectric constant, cathode/anode area, and distance between plates. When using capacitance as a diagnostic, the goal is to note movement of coils and leads within the transformer; therefore, changes in dielectric constant are not relevant. Neither is a change in dielectric constant significant in terms of capacitance testing unless there has been a fundamental change to the insulation composition. Both coils and leads contain a dielectric material (insulation) between two electrodes (conductors). The capacitance depends on the characteristics of the dielectric material and on the physical configuration of the electrodes. In electrical apparatus, if the insulating material characteristics or the conductor configurations change, a difference in the measured capacitance occurs. These changes are caused by deterioration of the insulation, contamination, or physical damage. The primary cause for changes in capacitance is changes to the distance between plate surfaces within the capacitive element.

Capacitance can be used to help assess the transformer for mechanical deformation. To properly analyze capacitance, a benchmark result is required. The first capacitance test performed serves as the benchmark. Subsequent tests are always compared to the benchmark results. To help ensure that this evaluation is valid, test conditions need to be consistent. Bushing or buswork changes can change the capacitance of the winding test because they are included in most field test cases.

In the field, transformer insulation systems should not change by more than 5% from the benchmark results. If the results are above 5% and below 10% change, an investigation needs to be conducted to determine the extent or severity of the issue. If the capacitance has changed by more the 10%, the transformer should not be returned to service.

7.2.14.2 Application

A PF testing program provides several important benefits. Initial tests on new equipment as it arrives from the manufacturer determine the presence of contaminants and overall material quality. Depending on type of material, voltage class, and insulating liquid, it is possible to have different acceptance criteria for PF limits. End users may want to adjust acceptance and field criteria for acceptable operation based on criticality of the unit and expected performance. Periodic tests performed during the service life of the equipment can indicate that the insulation is either aging normally or deteriorating rapidly. Diagnostic tests on suspect or failed equipment may disclose the location of a fault or the reason for failure.

Dielectric loss tests provide the greatest benefit when performed periodically as part of a complete maintenance program. To learn more about the theory behind PF and DF testing, refer to Annex A.

7.2.14.3 Test equipment

Dielectric loss is usually determined by a bridge measuring instrument, such as the Schering bridge, tan-delta bridge or transformer ratio-arm bridge. Instruments of this type normally have the means for determining the capacitance value as well as the loss factor of the insulation under test.

Along with the bridge, an ac power supply and standard capacitor (or equivalent) are required for measurement of loss factor. Portable test systems that include bridge, power supply, and capacitor in one enclosure are available for field testing. The operator of the test equipment should be completely familiar with the operation of the instruments and all safety procedures before attempting to perform these measurements. To obtain comparable benchmark test results, the instructions of the test device should be followed and the same instrument should be used.

7.2.14.4 Test voltage

Dielectric loss tests can be performed at any voltage within the normal operating range of the equipment under test. It may not be practical to perform a PF test at rated voltage on HV equipment in the field. To keep power supply requirements to a minimum, the test equipment is usually designed to perform the tests at reasonable voltage and current levels, allowing the test equipment to be portable using normal service power supplies.

Test voltages for a typical field test set range from below 100 V to as high as 12 kV. Field tests on most electrical equipment, however, are usually performed at rated voltage or a maximum of 10 kV. To provide comparability to factory tests, factory PF test results should also be available where the applied test voltage is no greater than 10 kV. Manufacturer's instruction manuals and appropriate test standards should be consulted for operating procedures. Test voltage should not go beyond the safe withstand voltage for a winding or other apparatus to be energized during test.

7.2.14.5 Environmental factors

It is important to record ambient conditions at the time of testing for reference when comparing test records. The loss factor of insulation can be sensitive to variations in temperature, in which case a correction factor needs to be applied to measured values. This is done to allow comparison of tests performed at different temperatures. The reference temperature used is 20 °C. Correction factors may be available from equipment manufacturers and test equipment manufacturers. (Additional correction factors were available in IEEE Std C57.12.90-2006, 10.10.5, which was removed in IEEE Std C57.12.90-2010.)

Testing at temperatures below freezing should be avoided, since this could significantly affect the measurement. Among the primary reasons for performing this test is the capability of detecting moisture in insulation. The electrical characteristics of ice and water are quite different, and it is much more difficult to detect the presence of ice than it is to detect water; sometimes it is impossible.

Other environmental factors, such as relative humidity and precipitation at the time of testing, should also be recorded for future reference. A very small amount of water vapor on the surface of external insulation could increase the amount of leakage current and appears as increased loss in the test results. This is especially a factor for lower voltage equipment where the bushing creep distance is short. For this reason, testing during periods of high humidity or precipitation should be done with care; otherwise, it makes proper evaluation of the test results very difficult.

Though not an environmental factor, connected apparatus such as bushings and insulators attached to leads can contribute to test results. This can become problematic when the PF of these support insulators varies a great deal from the PF of the transformer under test. These support apparatus influences should be considered during analysis.

7.2.14.6 Measurements

Each capacitor (insulation section) in a complex insulation system should be tested separately. Determining the characteristics of the individual components of a complex system is valuable in detecting and locating defective insulation in the system. Direct measurement should be made on each insulation system whenever possible.

7.2.14.7 Recommended test procedures

- Refer to Clause 6 for safety considerations in electrical testing.

- The electrical apparatus to be tested shall be isolated.
- The apparatus should be visually inspected to identify external damage or unusual conditions.
- The appropriate test procedure should be selected that accounts for every winding (a tertiary winding is considered a winding; thus, an autotransformer with a tertiary is considered a two-winding transformer for testing purposes). If a transformer greater than one winding (e.g., autotransformer or reactor) is being tested, test equipment must be used that is capable of performing a grounded specimen test (GST) and an ungrounded specimen test (UST) test circuit. Calculation of individual systems is not acceptable.
- The desired measurements should be performed following the operating instructions supplied with the test equipment. The lead connections may have to be changed several times, depending on the complexity of the apparatus and the test equipment.
- Apparatus nameplate data and all measurements should be recorded.
- Do not exceed withstand voltage of the winding or associated apparatus.
- If the unit is equipped with an LTC with a bypass resistor, the LTC should be placed off-neutral.
- Ground test leads should be connected securely (metal to metal) to the grounding cable coming immediately from the transformer under test. If a transformer is not grounded (spare), the transformer must have a temporary ground installed.
- The terminals on a winding should be shorted together if possible. If the windings are not shorted together, stray leakage can occur, thus artificially raising the PF of the winding.

The test procedures listed in Table 15, Table 16, and Table 17 indicate which winding should be energized at the prescribed test voltage. Connection for one or two LV leads is also included if testing a multiple winding transformer. Ground test leads are always connected to the ground cable coming immediately from the transformer. Insulation test systems are referred to as windings (CH, CL, CT) and inter-windings tests (CHL, CLT, CHT). Ground, guard, and UST refer to the various PF test circuit test lead connections. Test circuits are discussed in Annex A. In some cases, test procedures contain validation checks. Validation checks allow for checking the final test results and determining whether test connections were made correctly. These are mathematical checks to see whether measured and calculated insulation values match (if available). In general, the measured capacitance and calculated validation check values should be within 5% of previous values.

Table 15—One-winding transformer

Test	Energize	Measured	Notes
1	HV	CH	HV winding

Table 16—Two-winding transformer

Test	Energize	Ground	Guard	UST	Measured	Notes
1	HV	LV	—	—	CH + CHL	Validation check
2	HV	—	LV	—	CH	HV winding
3	HV	—	—	LV	CHL	Inter-winding
4	LV	HV	—	—	CL+CHL	Validation check
5	LV	—	HV	—	CL	LV winding
6	LV	—	—	HV	CHL	Inter-winding

Validation check		Calculation	Results
1		Test 1 – Test 2	CHL (calculated)
2		Test 4 – Test 5	CHL (calculated)

NOTE—Use Table 16 test values.

Table 17—Three-winding transformer

Test	Energize	Ground	Guard	UST	Measured	Notes
1	HV	LV	TV	—	CH + CHL	Validation check
2	HV	—	LV, TV	—	CH	HV winding
3	HV	—	—	LV	CHL	Inter-winding
4	LV	TV	HV	—	CL + CLT	Validation check
5	LV	—	HV, TV	—	CL	LV winding
6	LV	HV	—	TV	CLT	Inter-winding
7	TV	HV	LV	—	CT + CHT	Validation check
8	TV	—	HV, LV	—	CT	LV winding
9	TV	LV	—	HV	CHT	Inter-winding

Validation check	Calculation	Results
1	Test 1 – Test 2	CHL (calculated)
2	Test 4 – Test 5	CLT (calculated)
3	Test 7 – Test 8	CHT (calculated)

NOTE—Use Table 17 test values.

7.2.14.8 Mineral-oil-filled transformer power factor limits

The most common insulating liquid for power transformers is mineral oil. As such, the primary insulating dielectrics within a transformer are mineral oil and mineral-oil-impregnated paper. This type of insulation system has a well documented PF history. Given that higher voltage transformers are subjected to higher electrical stresses, it is reasonable that they should have even lower PFs. As shown in Table 18, the normal in-service and new PF limit for mineral-oil-filled power transformers < 230 kV is 0.5% PF at 20 °C, and the normal and new limit for transformers ≥ 230 kV is 0.4%. To help reduce the risk of catastrophic failure, the limit for serviceability of all mineral-oil-filled transformers is 1.0% PF at 20 °C. PFs between 0.5% and 1.0% at 20 °C require additional testing and investigation to confirm that a problem is not worsening. In some rare cases, selection of lower quality materials used in transformer manufacturing can lead to PF measurements greater than those previously mentioned. In these cases, end users need to discuss with manufacturers acceptable PFs. Arbitrarily raising PF limits because materials are unknown is not recommended because this may mask a degrading transformer without cause.

**Table 18—Nominal and serviceability service-aged limit:
power transformer insulation power factor**

Insulating liquid	kV rating	Nominal/new power factor limit	Serviceability aged limit
Mineral oil	< 230 kV	0.5%	1.0%
Mineral oil	≥ 230 kV	0.4%	1.0%
Natural ester	All	1.0%	1.0%

NOTE—All PFs are corrected to 20 °C except for natural esters, which at this time of writing the guide had no published temperature correction curves. Future work is needed to address this issue.

7.2.14.9 Natural-ester-filled transformer power factor limits

Natural ester liquids are known to be hydrophilic and therefore can hold more moisture in solution. As a result, the increased moisture being held by the ester chain is not considered detrimental to the power transformer's performance or life, but does lead to increased PFs. As shown in Table 18, natural ester transformers should generally have PFs below 1.0% whether they are new or in service. These numbers are provisional since they are based on preliminary industry findings.

7.2.14.10 Transformer power factor analysis

While standard or accepted values for dielectric loss have not been established for all types of electrical apparatus, there are established values for some apparatus such as mineral-oil-impregnated paper-insulated systems. Even with this, one of the most useful methods of evaluating test results is by trend analysis.

A meaningful evaluation includes comparison with previous test results on the same equipment, whenever available. This may include manufacturer's results taken at the factory and/or nameplate data. Comparison of test results with those for similar pieces of equipment, especially those tested under the same conditions, is also beneficial.

It is not advisable to energize a transformer received with a PF in excess of the values stated in Table 18 for new limit without complete internal inspection, consultation with the manufacturer, and drying or other correction, as indicated (IEEE Std C57.106-2006 [B39]).

The PFs recorded for routine overall tests on older apparatus provide information regarding the general condition of the ground and inter-winding insulation of transformers, regulators, and reactors. They also provide a valuable index of dryness and are helpful in detecting undesirable operating conditions and failure hazards resulting from moisture, carbonization of insulation, defective bushings, contamination of insulating liquid by dissolved materials or conducting particles, improperly grounded or ungrounded cores, etc. While the PFs of most older, mineral-oil-filled transformers are also $< 0.5\%$ (20 °C), PFs between 0.5% and 1.0% (20 °C) may be acceptable; however, PFs $> 1.0\%$ (20 °C) should be investigated.

7.2.15 Induced voltage test

7.2.15.1 General

The induced voltage test with PD measurement tests the transformer in the normal service condition at a voltage above that which would be seen in service. PD measurements made during this test are used to verify that the insulation in the transformer is in an acceptable condition and can withstand normal service conditions. In the factory testing, the induced voltage test is the last dielectric test and is used to confirm that the transformer passed all other tests and to confirm that there are no hidden dielectric problems or failures that were not detected in the previous tests. As such, successfully passing the field induced voltage test gives the user confidence that the transformer can operate in service without a significant risk of failure.

This test is recommended for the following cases when no other routine field diagnostic tests provide the confidence that the transformer is fit for service:

- Onsite test after repair or remanufacture
- For failure identification
- When DGA results indicate possible PD (H_2 gas as indicator for hermetically sealed transformers not free breathing transformers)
- If transformer may have been contaminated
- After failure of components (bushings, pumps, tap changers, etc.)
- After possible transportation damage as evidenced by impact recorder readings or diagnostic test (FRA) anomalies
- When unusual crackling or arcing sounds are heard from inside the unit

- When a transformer trips off line, especially when protective equipment (sudden pressure relay and differential relays, etc.) have operated
- When other dielectric test information, such as PF, DFR, insulation resistance, or insulating liquid quality indicates a problem

The induced voltage test is performed at a frequency above the rated frequency to prevent over-excitation of the core when above-normal voltage is used. The test may be done with a motor-generator set operating at a fixed frequency or with a variable frequency electronic power supply. When a generator is used, it is important to protect the transformer against self-excitation of the generator under the capacitive load from the transformer by the use of reactors and protective gaps during the setup process. When a variable frequency electronic power supply is used, no such protection is needed.

7.2.15.2 Preliminary test procedures

LV tests (insulation resistance, PF, ratio, insulating liquid dielectric, etc.) should be performed to determine whether the insulation of the transformer under test is suitable for energization. The insulating liquid should be sampled following prescribed procedures (refer to ASTM D923 [B3]), and its total dissolved gas-in-liquid level should be analyzed to ensure that it is acceptable. The moisture content of the insulating liquid should be tested to ensure that excessive amounts do not exist. A turns ratio test should be performed to confirm that the transformer's tap changer for de-energized operation is properly positioned and that shorted winding turns do not exist. For the interpretation, see Table 19.

Table 19—Recommended diagnostic characteristics

Procedure	New transformer	Service-aged transformer
Power factor(< 230 kV)	< 0.5%	< 1.0%
Power factor(>230 kV)	< 0.4%	< 1.0%
Total dissolved gas ^a	< 0.5%	< 0.8%
Moisture content	< 10 (ppm)	< 15 (ppm)
Turns ratio	Within 0.5% of nameplate	Within 0.5% of nameplate

^a For units equipped with nitrogen blankets, total dissolved gas should not exceed 2.0%. Values for natural ester transformers may vary and are not available at this time.

7.2.15.3 Special precautions before test

Every power cable lead, except the cables used to supply the energy from the main power supply, should be disconnected from the bushings before energizing the transformer. Sufficient clearances should be considered according to the estimated induced voltages at the enhanced level.

Any transformer-mounted surge arresters should be disconnected before energizing the transformer to avoid arrester damage and limitation of the test voltages due to arrester operation.

Internal current transformer secondary terminals must be short-circuited, and internal potential transformers must have their secondary winding terminals open-circuited for the test.

Corona at the tank of the transformer under test or at nearby grounded or energized objects may not only increase the PD background noise level, but also influence the results of withstand tests. Therefore, in preparation for the test, HV bushings, and in some cases, LV bushings of 69 kV and above should be fitted with corona rings of sufficient size to eliminate all possibilities of air corona discharges. To help prevent corona on the ground side, sharp edges and points on top of and close to the transformer tank should be masked by covering them with corona rings galvanically connected to the tank. The HV bushings should be carefully cleaned and dried. Immediately prior to the test, they should again be wiped dry. No conductive or semiconductive objects should be left ungrounded on the transformer or close to it because this may

produce discharges from floating objects. Therefore, they should be either taken away, when this is possible, or carefully grounded. Current-carrying connections should be made very carefully to ensure good electrical contact because contact arcing may produce unacceptably high PD levels.

Other energized equipment located in proximity to the transformer being tested may also contribute to high PD background noise levels. Arrangements may have to be made to de-energize this equipment for the duration of the test.

Notice: Prior to subjecting the transformer to the induced voltage test, it is recommended to consult with the original manufacturer to determine whether any specific limits on total dissolved gas in insulating liquid, particle count, or other dielectric values should be met.

7.2.15.4 Power factor pretest for excitation by motor-generator sets

After connecting the test set to the transformer under test, a pretest to determine the PF at the source should first be performed so that the amount of inductive compensation is sufficient that the load on the generator is not capacitive, because this could lead to dangerous overvoltages due to generator self-excitation. To perform this pretest, an HV bushing of the test transformer should first be temporarily fitted with an external spark gap adjusted to operate at about 50% of the transformer nominal voltage. During this pretest, the voltage should not be raised above 30% or preferably only to a level high enough to allow accurate PF measurement. The generator is capable of driving a slightly capacitive load provided that the power margin is sufficient; it is when this margin is exceeded that generator runaway occurs. Higher-than-required inductive compensation should first be used and it should be adjusted to a value that allows the test to be done at full test voltage without exceeding the generator limits, because this may cause generator protection tripping. The user should therefore be absolutely sure that the generator has a sufficient power margin to reach the maximum test level without danger of tripping. It is also important that the ratio of the step-up transformer used to match the voltage output of the generator to that of the transformer under test be as close as possible to the required value (optimum adaptation). This provides maximum power transfer from the generator to the transformer under test. After the pretest has been performed and the reactive compensation has been adjusted properly, the temporary spark gap should be removed and the voltage can then be taken to the test level.

7.2.15.5 Pretests for excitation by static frequency converters sets

Because there is no significant danger of self-excitation when using static frequency converters (SFCs), a special pretest for checking the PF is typically not necessary. In case of small test objects (light load on the SFC), the test sequence can simply be started after assembly of the temporary test field on site. But in case of large test objects (heavy load on the SFC), the ratio between the available test power and the needed test power should be checked in advance. It has to be taken into account that the transformer under test is a capacitive test object at high frequencies and an inductive one at lower frequencies. At the self-compensation frequency in between, the power demand is at its minimum. The SFC set should be adjusted to that frequency if it is high enough to reach the required test voltage.

NOTE—The ratio of the test frequency to the rated frequency of the transformer under test should be greater than or equal to the ratio of the test voltage to its rated voltage.

7.2.15.6 Test procedure and acceptance criteria

With some exceptions, the test procedures used in the factory induced voltage test may be used as a guide for testing in the field. The exceptions include the tap position, test voltage level, and acceptance criterion. Most often, the tap position used in the field test is the normal operating tap. Transformers with all new insulation may be tested at the standard test levels specified for new transformers. For transformers where all or some part of the insulation is not new, the test levels used are lower and should be established by agreement between the user, manufacturer, and testing company where applicable. IEEE Std C57.12.90

recommends testing transformers with insulation that is not new at no more than 85% of the IEEE test levels. When high PD is present, it may be necessary to limit the voltage to even lower levels than this to prevent damage to the transformer insulation (for example, at 110% of nominal voltage). In most instances for transformers with not-new insulation, the enhancement test is omitted and the test is performed at the reduced 1 h IEEE voltage level.

It is recommended to slowly raise the voltage at the beginning of the test and measure the PD at various voltage levels while the voltage is raised. If high PD occurs during this process, steps should be taken to diagnose and correct the cause if possible before raising the voltage, to prevent flashover either internally or externally due to unforeseen conditions.

PD measurements can also be made on the same levels while the voltage is decreased.

A comparison before and after the withstand level can supply useful information on the insulation condition.

What constitutes an acceptable test is a matter of agreement among the user, manufacturer, and test personnel and relies on the experience and judgment of those conducting and observing the test. In general and without other specifications, if the PD levels measured during the test meet the requirements of IEEE Std C57.12.90, the transformer is considered to have passed the test.

It may be difficult or impossible to achieve a sufficiently low PD background noise level on site. Therefore, various steps can be taken to reduce the background noise, such as measuring at frequency bands where there is lower background noise, making the PD measuring circuit as compact as possible, reducing grounding loops, and taking other measures for PD noise reduction.

If high PD levels are found during the tests, various diagnostic methods are available to help determine the cause and location of the PD. Some of these methods are as follows:

- Diagnostic step-voltage test
- PD phase-resolved pattern diagnosis and comparison to known causes
- Observation of the PD pulse shape with an oscilloscope
- Signal attenuation at various terminals of the transformer. In this case, measurements may be done at terminals not normally required for testing.
- Acoustic PD location
- UHF PD location

7.2.15.7 Performing the induced voltage test and interpreting the results

The induced voltage test should be performed on a clear day. Outside interferences such as operating cranes and motorized vehicles should be kept clear of the test site for successful interpretation of test results.

The test voltage level and duration for field tests are usually subject to negotiation and may vary from initial factory levels for new transformers to lower levels depending on the age and history of the transformer.

After connecting the test set to the transformer under test and calibrating the instrumentation, the voltage should be slowly raised to the test level. Instrumentation should be carefully observed throughout the test and PD or RIV levels recorded at 5 min intervals. Peaks observed between recording times should also be noted. Any wildly erratic readings may be cause to immediately terminate testing until the cause is determined.

If PD or RIV levels exhibit a tendency to continue to increase in the latter stages of testing, continue the test until the level stabilizes or begins a downward trend.

Interpretation: PD levels above 500 pC or RIV levels above 100 μv may be indicative of a substantial problem.

7.2.16 Partial discharge

7.2.16.1 General

PD in an insulation system occurs when an electric discharge that only partially bridges the insulation between the conductors creates local breakdown of the insulation medium and causes a transient redistribution of space charges within the insulation system. The breakdown PD is generally caused by localized field enhancements due to dielectric imperfections, such as voids in solid dielectrics, surface particle contamination, a design that does not withstand the dielectric stress where PD usually occurs at the insulation liquid interface, or water vapor in insulating liquid, and does not bridge the distance between the electrodes that set up the electric field. Generally, the PD events of interest occur within the insulation at the site of a void or foreign material such as water; PDs may also be ignited at a location where damage or misuse has occurred. This type of PD can cause further insulation degradation of the insulation in its vicinity and lead to eventual failure of the HV apparatus.

PDs generate low-amplitude current pulses, usually in the milliamperere range, that are of short duration, usually in the microsecond range or even below. Two different techniques are in common use to detect and measure these electrical signals. One technique consists of measurements with a radio noise meter. Levels are measured in microvolts and are referred to as *RIV signals*. The other method consists of measurements with a PD detector; these signals are measured in picocoulombs. The preferred method evaluates the PD level in terms of pC using the wideband method for measuring the apparent charge as specified in IEEE Std C57.113-2010 [B41]. The use of the RIV method that measures PD in terms of microvolts using the narrowband method is included as an annex in IEEE Std C57.113-2010 and is specified in NEMA 107 [B57].

The RIV and PD signals can be thought of as very small magnitude (low-amplitude), high-frequency spectrum pulses superimposed on the high voltage. Successful testing requires that suitable precautions be taken to reject electromagnetic noises to ensure that these small signals can be detected. Conducting objects that may be in the HV field should be solidly grounded, and objects that have sharp points or corners (e.g., bolts, tank headers, etc.) should be shielded with conductive material that has a smooth geometry. Current-carrying connections should be clean and secure.

7.2.16.2 Partial discharge—measuring the PD level

As in the case for RIV measurements, the PD signals to be measured are usually obtained from the bushing capacitance tap. A typical circuit arrangement is shown in Figure 9. The measurement impedance Z_m , as an essential part of the coupling device (see IEEE Std C57.113-2010 [B41]), is usually complex and represents a high-pass filter. The lower limit frequency is typically around 100 kHz. This circuit effectively serves to filter out the low-frequency ac test voltage frequency, and to transfer the high-frequency PD pulse signals, to be measured by the PD detector, to the PD measuring instrument, characterized by an upper limit frequency, typically around 300 kHz (see IEEE Std C57.113-2010).

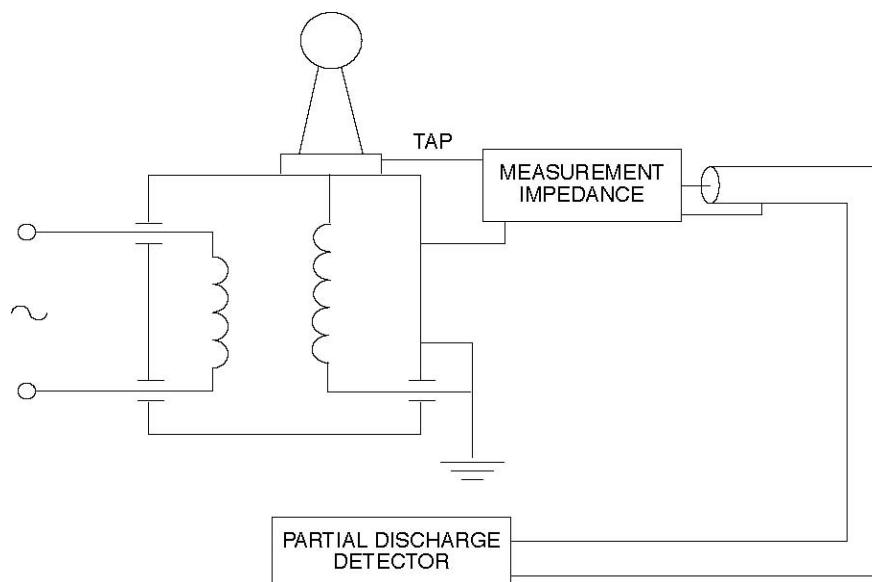


Figure 9—Partial discharge measurement circuit using bushing tap

The measuring sensitivity of the system depends on the capacitance of the coupling capacitor, the capacitance of the test object, the divider ratio given by the capacitive network of the bushing, and the stray capacitances of the test circuit. This requires the system to be calibrated for each test setup for accurate reproducible results. For this reason, commercial PD detector measuring instruments are supplied with a pulse generator to perform the PD calibration. The calibrator is equipped with a fast-step-pulse generator connected in series with a calibrating capacitor having a value typically around 100 pF. A common means of calibration is to inject a known amount of charge into the HV terminal and adjust the PD detector sensitivity of the PD measuring instrument to provide the proper reading. In practice, this is done by coupling a voltage pulse to the HV bus via a small (100 pF) capacitor.

The setup should be calibrated for the reasons previously discussed. The circuit is calibrated before it is energized. If an LV calibration capacitor is used, the circuit should be calibrated before it is energized and the LV capacitor should be removed during the test. The operator's manual for the PD detector measuring instrument being used should be consulted for detailed calibration procedures. For more information see IEEE Std C57.113-2010 [B41].

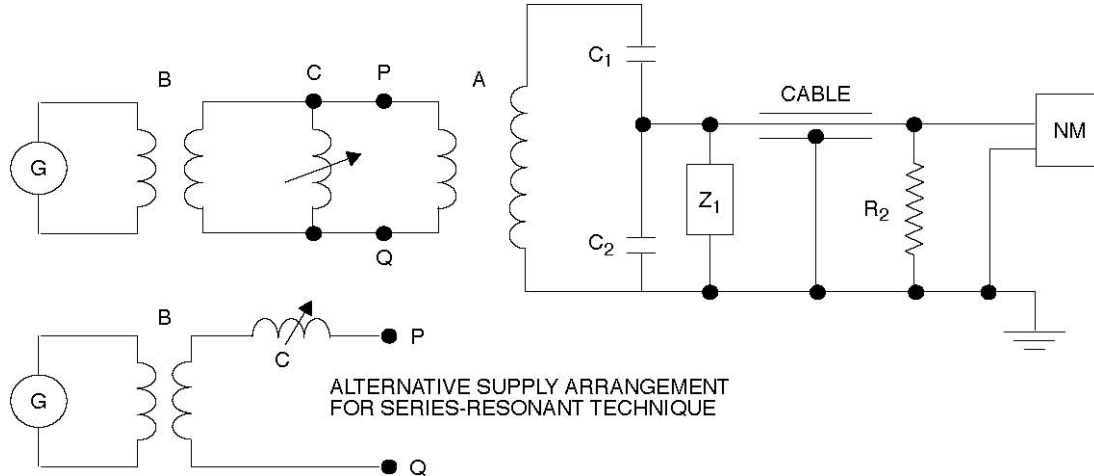
The HV source is normally turned on at zero to low voltage and slowly brought up to the desired test voltage. The reading of the PD detector measuring instrument should be monitored and recorded as the ac test voltage is increased. A reading substantially above the zero point of the indicating instrument may indicate noise or background interference. Noise on the PD detector display at low test voltage indicates interference, either radiated or coupled into the circuit from external sources. If this interference cannot be eliminated, it sets the limit of sensitivity for meaningful test results. In practice, an experienced operator can identify and ignore some noise if it originates from a known source other than the test object.

It is strongly recommended to visualize the output pulses of the PD measuring instrument in a phase-resolved manner using a scope or a computerized PD measuring system. This ensures not only a better assessment of critical PD events, but also the recognition of disturbing noises.

Typically, PD signals are not present at low voltage but appear suddenly at a test voltage level known as the *PD inception voltage*. As the voltage is raised beyond this inception voltage, more pulses appear and may grow in amplitude. When the voltage is reduced, a hysteresis effect may be noticed in which the PD pulses do not extinguish until the voltage has been reduced significantly below the inception voltage. The voltage at which PD has disappeared is known as the *PD extinction voltage*.

7.2.16.3 Measuring the RIV level

As in the case of the PD level measurement, the RIV signals are PD pulses that are usually obtained from the bushing capacitance tap as shown in Figure 10. The variable inductance Z_1 is tuned with the bushing tap-to-ground capacitance at the measuring frequency of the radio noise meter RIV measuring instrument. The coaxial cable, which may be of any suitable impedance, should be in the circuit when the tuning is performed. The purpose of the tuning is to minimize the dividing effect of the capacitive bushing network capacitance. The coaxial cable need not be terminated in its characteristic impedance.



Legend:

A = Transformer under test	CABLE = Shield cable
B = Auxiliary step-up transformer	G = Supply generator
C = Variable reactor	NM = Radio noise meter
C_1 = HV bushing capacitance	R_2 = Resistor
C_2 = HV bushing capacitance tap	Z_1 = Variable inductance

Figure 10—RIV measurement using bushing tap coupling mode

The system has to be calibrated for each test setup for accurate results. A common means of calibration is to apply to the HV terminal a radio-frequency signal of known amplitude and equal to the mid-band frequency of the detector RIV measuring instrument to be used. The detector instrument itself should be calibrated according to the manufacturer's recommendations. The bandwidth normally used is 9 kHz. The mid-band frequency in the range of 0.85 MHz to 1.15 MHz is normally used. However, other frequencies may be used if interference from radio broadcasting stations is present.

The circuit is calibrated before it is energized. The HV source is normally turned on at zero to low voltage and brought up to the desired test voltage slowly. The reading of the RIV measuring instrument detector should be monitored as the voltage is increased. Obtained readings should be recorded. Noise readings on the detector meter at low voltage indicate interference, either radiated or coupled into the circuit from external sources. If this interference cannot be eliminated by tuning the mid-band frequency accordingly, it sets the limit of sensitivity for meaningful test results. In practice, an experienced operator can identify and ignore some noise if it originates from a known source other than the test object.

Interpretation of RIV test results requires some experience with RIV tests in general and with the type of test object device being tested in particular. Since the RIV measuring instrument is based on a narrowband test technique, it is subject to possible resonances in the test object. The quasi-peak response of the detector also makes the response dependent on the repetition rate of the RIV impulse PD pulses, in particular for repetition rates below about 1000 pps. Interpretation of the results of this type of testing is best carried out in the context of previous measurements on the same piece of apparatus, including factory tests. Standards on particular classes of apparatus may offer some guidance on this subject.

7.2.17 Frequency response

Frequency response is covered in Annex F.

7.2.18 Dielectric frequency response

DFR is covered in Annex G.

7.2.19 Infrared

Infrared is covered in Annex C.

7.2.20 Furan test

Furan test is covered in Annex E.

7.3 Bushings

7.3.1 General

Bushings vary widely in construction and are essential elements of a transformer. They are relatively inexpensive compared with the cost of a transformer. However, their failure may result in the total destruction of the transformer. Failures of liquid-filled porcelain bushings often result in explosions, with broken pieces of porcelain and possibly an ensuing fire. Bushings should, therefore, be checked regularly and, if evidence of deterioration is found, they should be either repaired or replaced, depending on the type and degree of deterioration IEEE Std C57.19.100™-2012 [B36]. Although there are many different types of construction, many of the diagnostic tests are common and are described as follows.

7.3.2 Continuity

Continuity is covered in Annex B.

7.3.3 Capacitance, power factor, and dissipation factor

The capacitance and PF (or DF) of the C_1 main insulation and C_2 tap electrode insulation should be measured (see Annex B). Short-circuited capacitor sections can be detected by an increase in capacitance. The presence of moisture or other contaminants can usually be detected by an increase in PF. External moisture or surface contamination should be removed before performing PF measurements. Temperature corrections should be made during the measurements. When performing tests on the C_2 capacitance, care should be taken not to exceed the test voltage of the tap. It should be noted that the PFs of the C_1 and C_2 capacitances (typically measured in picofarads) may be considerably different from each other, and it is not uncommon for the C_2 capacitance to be 10 times greater than the C_1 capacitance. C_2 capacitance PF testing is difficult to perform correctly, and it is hard to reproduce the values on bushings without lower ground shields.

The *hot collar test* is a particularly useful test when the bushing is not equipped with a tap electrode and can be used to assess the condition of a specific small section of the bushing insulation between an area of

the upper porcelain and the central current-carrying conductor. It is performed by energizing one or more temporary electrodes (collars, usually of semiconducting rubber) placed around the outside of the porcelain, with the central conductor grounded. This type of test may be used to locate cracks in porcelain, degradation of insulation inside the upper section of the bushing, low compound or liquid level, and voids in the compound. Watt losses should be analyzed not power factor. The acceptable limit is 0.1 W (Gill [B27]). Hot collar tests can be performed in either grounded specimen test (GST) or ungrounded specimen test (UST) modes depending on conditions and preferences. The GST is described in Annex B. Performing the hot collar test in the UST mode can help to minimize any external influence that could be attributed to surface contamination. Any bushing that exhibits a history of continued power factor increase should be scheduled for removal from service and further investigation. The bushing manufacturer should be consulted for guidance. If any bushing exhibits an increase in power factor over a period of time, the rate of change of this increase should be monitored by more frequent tests. If the power factor measurement of a bushing increases from 1.5 to 2 times from its initial reading, then the test frequency should be increased or the bushing should be removed from service. If the power factor measurement triples the initial test reading, then the bushing should be removed from service.

Bushing capacitance should be measured with each power factor test and compared carefully with both nameplate and previous tests in assessing bushing condition. This is especially important for capacitance-graded bushings where an increase in capacitance of 5% or more over the initial/nameplate value is cause to investigate the suitability of the bushing for continued service. The manufacturer should be consulted for guidance on specific bushing IEEE Std C57.19.100-2012 [B36].

NOTE—On some bushings, the C_1 capacitance between the center tube and the tap, and the capacitance C_2 , between the test tap and ground are typically marked on the nameplate. The nominal capacitances, measured in the field may differ as it is highly dependent on the surrounding parts inside the transformer. As a resultant it may not be possible to obtain a nominal value valid for all service conditions.

7.3.4 Partial discharge in a bushing

7.3.4.1 General

Prolonged PD activity in the internal insulation of a bushing gradually reduces its dielectric strength and eventually results in failure. The presence of PD is detected by measuring either the PD or RIV level, which, of necessity, is done at a high voltage, usually line-to-ground voltage or higher (e.g., 130% to 150%).

These measurements may be made during an induced voltage test on the transformer. However, if PD is detected during this test, it is impossible to distinguish whether it originates from the bushing or from the transformer. For this reason, if PD or RIV measurements are required, it is preferable to remove the bushing from the transformer and test it alone. This PD test is performed with the bushing in a special tank using an HV test source.

7.3.4.2 PD test circuit

The circuit required for a PD test of the bushing by itself consists of an HV source, a PD measuring instrument or an RIV measuring instrument, a coupling capacitor connected to a coupling device equipped with the measuring impedance, and a PD calibrator, as well as the associated measuring connection leads (see IEEE Std C57.113-2010 [B41]).

The HV source may be a transformer or a series-resonant test set that is free of PD.

The purpose of the coupling capacitor in connection with the measuring impedance as part of the coupling device is to separate the high-frequency spectrum of the PD pulses of very low magnitudes from the HV

test voltage on the bus and to couple the PD pulses to the measuring impedance, from which they are routed to the PD or RIV measuring instrument. The coupling capacitor should be rated for the maximum ac test voltage and should be discharge free. Values of 1000 pF have been found to be satisfactory.

The PD activity can be measured by means of a PD measuring instrument or an RIV measuring instrument. Both types of instruments are commercially available. However, PD measuring instruments are usually supplied with an oscilloscope display on which the PD pulse activity can be observed in a phase-resolved manner, which may be useful in interpreting the results and for recognition of electromagnetic noises.

7.3.5 Infrared

Infrared is covered in Annex C.

7.3.6 Visual inspection

Some visual examination can be performed while the transformer is energized. The use of binoculars can reveal such defects as cracked or broken porcelains, leaking gaskets, and abnormal insulating liquid level. It is helpful to note the ambient temperature and, if possible, the load current at the time of observation.

More detailed visual inspection can be carried out when the transformer is de-energized. In addition to the items previously described, closer examination can reveal hairline cracks, deterioration of cemented joints, and surface contamination. If the porcelain housing is broken, how it is broken should be considered. A simple shed break is of minor concern. Unglazed porcelain does not constitute any immediate danger, since electrical grade porcelain is not porous and does not absorb moisture. If a crack or discontinuous, broken surface appears to enter or point to the main body of the bushing housing, closer examination is recommended. Cracks appearing to extend into the main body can grow and eventually cause failure. In such a case, the bushing should be replaced because no effective field repair is possible.

7.3.7 Liquid level

The liquid level should be checked in the sight glass or liquid gauge. Ambient temperature should be considered for proper assessment of liquid level. A common mistake is to add insulating liquid in colder temperatures to bring the insulating liquid level to normal levels. The normal level is usually made for an ambient temperature of 20 °C. Adding insulating liquid at lower temperatures results in overfill conditions when temperatures increase to summertime conditions. Insulating liquid filling plugs, etc., should be replaced and properly sealed after the inspection to prevent the entrance of contaminants. If the sight glass allows for the observation of the insulating liquid, it should be examined for waxing (X-wax formation) or contamination. The copper conductor should also be checked for any abnormal discoloration, which could be caused by overheating or by reaction with insulation liquid that contains corrosive sulfur.

7.4 Tap changers

7.4.1 General

The two types of tap changers in a power transformer are DETCs and LTCs.

WARNING

The construction of DETCs is such that they shall be operated only with the transformer de-energized. Failure to do so will result in severe equipment damage, personal injury, and possible loss of life.

DETCs are normally located in the higher voltage winding of a power transformer. LTCs are designed to be operated while the transformer is energized. LTCs may be located in either the HV winding or the LV winding, depending on the requirements of the user, the cost effectiveness of the application, and tap changer availability.

7.4.2 Load tap changers

7.4.2.1 General inspection procedures

In the LTC operating cycle, adjacent taps should be connected together at the point of transferring current from one tap to another. In an LTC, impedance is introduced between these taps to control the circulating current at the point where the taps are connected together. Either reactors or resistors are used as the transition impedances. In the load transfer operation, current is interrupted by a diverter switch. This switch may be an arcing-in-liquid switch or a vacuum switch.

Equipment used as a current-interrupting device requires periodic inspection and maintenance. The frequency of inspection should be based on time in service, range of use, and number of operations. The inspection intervals described below are indicative of frequently used values. However, the actual intervals to be used are those specified by the manufacturer unless previous operational experience or diagnostics indicate that more or less frequent inspections are necessary. An initial inspection should be made on a tap changer based on manufacturer's recommendations or at the end of the first year of operation. Subsequent inspections should be based on the results obtained from the initial inspection. Regardless of the measured contact wear, the inspection interval should not exceed 7 years.

LTCs may be supplied in a separate compartment, which is welded or bolted to the transformer tank, or they may be located within the transformer tank. Generally, reactor transition tap changers, whether with arcing diverter switches or vacuum diverter switches, are built into a separate compartment. Resistor transition tap changers are sometimes located in a separate tank and sometimes within the main transformer tank. Some tap changers located within the transformer tank have two main components. The first is a separate cylindrical insulating tank suspended from the transformer cover that contains the diverter switches and transition resistors. This tank is sealed so that the insulating liquid within it cannot mix with the insulating liquid in the main transformer tank. The diverter switch may have its own insulating liquid conservator; some designs share a common conservator with the transformer. A liquid filter is mounted in the piping to the diverter switch housing when there is a common conservator for the diverter switch and transformer. Directly under the sealed diverter switch tank are the tap selector and changeover selector switch. Since no arcing occurs on these switches except during operation of the changeover selector, they may be located in the insulating liquid of the main transformer tank. However, since they are located within the main transformer tank, these contacts cannot be inspected without removing the insulating liquid in the transformer tank. However, the diverter switches can be removed from this cylindrical tank for inspection without removing insulating liquid from the transformer tank.

Without de-energizing the transformer to inspect the LTC, a dissolved gas and oil quality sampling and analysis may be completed. ASTM D923 [B3] gives guidance for proper sampling techniques. See IEEE Std C57.139™-2010 [B43] for interpretation. LTC repairs are based on the interpretation of gas analysis.

While still in service, a separate LTC compartment may be inspected with an infrared scanner (see Annex C). Normally the temperature of the compartment may be a few degrees Celsius less than the main tank. Temperatures approaching or above that of the main tank may indicate an internal problem, although some LTC types can operate at temperatures above the main tank as part of normal operation. Prior to opening the LTC compartment, it should be inspected for external symptoms of potential problems. Such items as integrity of paint, bulges in tank, weld leaks, liquid seal integrity, dehydrating breather desiccant, pressure-relief device, and liquid level gauge should be inspected prior to entering the LTC.

Following de-energization, the separate LTC compartment should be drained of insulating liquid for internal inspection. Upon opening the LTC compartment, the door gasket should be inspected for signs of deterioration. The compartment floor should be inspected for debris that might indicate abnormal wear, and sliding surfaces should be inspected for signs of excessive wear.

Transformer testing performed after maintenance and insulating liquid samples taken from the LTC compartment prior to service restoration may include the following:

- Transformer turns ratio
- Excitation current
- Winding insulation PF
- Winding resistance
- Contact resistance
- Continuity test (for LTC in wye-connected winding)

7.4.2.2 Specific inspection procedures

A periodic visual in-service inspection of the LTC components should include the following:

- a) Any looseness
- b) Motor drive mechanism
- c) Heater function
- d) Counter reading
- e) Dehydrating breather desiccant
- f) Ground fault protection, if provided
- g) Liquid level, as required
- h) Insulating liquid filter, if provided
- i) Pressure relief indicator

The following checkpoints should be addressed, and the manufacturer's manual should be consulted for details to identify problems and improve operation in the future.

- 1) Inspection and maintenance of resistance-type and reactance-type load tap changing equipment (arc switching type) mounted in a separate compartment
 - Signs of leaks
 - Integrity of seals
 - Color of dehydrating breather desiccant
 - Liquid level indicator(s)
 - Operation of insulating liquid filtration, if provided
 - Function of pressure relay/ pressure relief device/oil-flow relay (whatever is provided)
 - Function of control switches
 - LTC stopping on position

- Fastener tightness
- Signs of moisture such as rusting, oxidation, or free-standing water
- Clean tap changer compartment and insulating parts
- Mechanical clearances as specified by manufacturer's instruction booklet
- Operation and condition of tap selector, changeover selector, and arcing-transfer switches
- Drive mechanism operation condition of devices and lubrication
- Counter operation
- Position indicator operation and its coordination with mechanism and tap selector positions
- Limit switch operation
- Mechanical block integrity
- Proper operation of hand-crank and its interlock switch
- Physical condition of tap selector
- Freedom of movement and lubrication of external shaft assembly
- Extent of arc erosion on stationary and movable arcing contacts
- Inspection of barrier board for tracking and cracking
- Insulating liquid quality (to include insulating liquid dielectric breakdown voltage and water content)
- Contact mid position of movable contacts (alignment)
- Resistance of transition resistors (if applicable)

Finally, carbonization that may have been deposited should be removed and the tap changer compartment should be flushed with clean transformer liquid. After filling with insulating liquid, a manual cranking should be performed throughout the entire range.

- 2) Inspection and maintenance of reactance-type load-tap changing equipment (vacuum switching type) mounted in a separate compartment or resistive type with vacuum diverters mounted in the transformer compartment.

The preceding checklist should be followed as applicable. In addition, the following should also be checked:

- Vacuum interrupter wear (contact erosion) and presence of vacuum
NOTE—An ac high-potentiometer test should be performed if vacuum integrity is suspect.
- Vacuum monitoring system operation (if applicable)
- Coordination of vacuum interrupters with selector mechanism (switching sequence)
- Coupling of tap selector; check "NEUTRAL" position (if applicable)

There should be only very minor amounts of carbon. Dielectric strength and water content of the insulating liquid should be tested (see 7.2.5.1 and 7.2.5.4 for details). The color should be generally clear if the LTC has been operating properly. Refer to the manufacturer's instruction book for details on insulating liquid filling of the compartment. Some vacuum-switching-type LTCs require insulating liquid filling under vacuum after maintenance using degassed insulating liquid.

- 3) Inspection and maintenance of resistance-type load-tap changing equipment (arc switching type) mounted inside the transformer compartment

The preceding checklist should be followed as applicable. In addition, the following should also be performed:

- Inspect and clean filter in insulating liquid conservator pipe (if applicable)
- Check coupling of tap selector; check “NEUTRAL” position (if applicable)

7.4.3 De-energized tap changers

7.4.3.1 General

WARNING

Failure to de-energize the tap changer before operating will result in violent equipment failure and may cause severe personal injury.

The DETC is normally located in the higher voltage winding of a power transformer. Its purpose is to adjust the turns ratio between the primary and secondary windings. Since this device is basically a switch or a link board, few tests are available with regard to its proper operation. Malfunction is generally indicated by the generation of excessive combustible gases in the insulating liquid. These gases would be indicative of hot metal in the insulating liquid without cellulosic involvement.

DETCs are located within the transformer tank. Therefore, to inspect these devices, it is necessary to drain the insulating liquid to such a level that the tap changer is available for inspection. Refer to IEEE Std C57.93-2007 for internal inspection in confined space.

Diagnostic checks normally involve operation of drive mechanism, freedom of movement of drive shaft, verification of contact alignment, contact pressure, checking of DETC monitor switches and contacts, and visual inspection. Tests involving operation of the tap changer for de-energized operation should be performed with the equipment de-energized.

7.4.3.2 Diagnostic check procedures

The diagnostic checks are performed as follows:

- a) *Alignment:* After operation, correct positioning should be verified by performing a turns ratio test. This check is to determine the proper alignment of the DETC contacts without entering the transformer tank. Improper alignment of the contacts may cause high contact temperature and ultimately result in failure of the power transformer. This is normally the first test to be performed on the tap changer.

A transformer turns ratio tester is connected to the HV and LV windings of the phase to be tested. After nulling the meter, the tap changer operating handle is slowly moved in one direction until the null is lost. The position of the handle is marked on the face of the selector plate. The operating handle is then moved in the opposite direction until the null reappears and is subsequently lost again. This new position is also marked on the selector plate. The operating handle is then restored to the ON position. The final location of the handle should be halfway between the marks. Any significant deviation is indicative of misalignment and requires repairs before the transformer is re-energized. The preceding procedure should be repeated for all tap settings.

- b) *Contact pressure:* Any of the techniques described in 7.2.7 may be used to measure resistance. The measured resistance values should be corrected to factory values. Any substantial deviation (increase over factory values) could be indicative of improper contact pressure. In addition, if the transformer also has an LTC, the LTC switch should be in the NEUTRAL position to compare the measured resistance readings with factory values. In single-phase or wye-connected transformers,

any phase that has a significantly higher resistance has a suspect contact. In a delta configuration, the single winding between the bushings where the significantly higher reading was obtained has a suspect contact. The other readings in the delta are affected, but to a lesser degree. The measurements should be performed on each tap position of the DETC.

If any of the resistance measurements are abnormal, the insulating liquid should be removed and the tap changer switch should be isolated. Resistance measurements should be repeated across the isolated switch to confirm the suspected defect before undertaking repair.

- c) *Visual*: Failure of alignment and contact pressure tests to reveal a problem with a DETC requires the performance of a visual inspection. This test should be performed as a last resort since insulating liquid is removed from the transformer. Extraordinary efforts may be required to observe the tap changer if access is difficult. For example, a flexible fiber-optic viewing device may be required to view the tap changer. The tap changer should be examined for signs of burning or tracking. Any such damage should be corrected before restoration.

When a DETC is operated, inspection and testing should be performed.

Transformer testing performed after maintenance and insulating liquid samples taken prior to service restoration may include the following:

- Transformer turns ratio
- LV excitation
- Winding insulation PF

7.4.4 Voltage regulators

A regulator is a device that maintains a preselected voltage level on a regulated system regardless of load fluctuations within its rated capabilities. The main components in a step-voltage regulator are a tapped autotransformer, an LTC, and a control system.

Maintenance checks are normally performed in two stages: while the equipment is energized and while the equipment is de-energized. The details of these checks are described as follows:

- a) *Energized*. While still in service, the following checks may be made:
 - 1) The liquid level should be read from the liquid level gauge on the side of the unit.
 - 2) The operation of the regulator's control system may be checked by using the manual mode of operation and running the regulator to a position outside the voltage bandwidth in the raise direction. The controls should then be switched to the AUTOMATIC setting. After the time delay programmed into the control expires, the regulator should return within bandwidth (which is normally the same as the starting position unless the incoming voltage is constantly varying).
 - 3) The temperature of the regulator should be checked by means of the top liquid thermometer and winding temperature indicator (if supplied) or by infrared scanning techniques (see Annex C). A comparison can be made between identical units on different phases.
 - 4) If water leakage into the tank is suspected, the moisture content of the insulating liquid and its dielectric strength should be checked.
- b) *De-energized*. After disconnecting from service, the following measurements should be made:
 - 1) Insulation resistance of winding
 - 2) Insulation PF
 - 3) Winding ratio

- 4) Winding resistance
- 5) Dielectric breakdown strength of insulating liquid and moisture content

Bushings should be visually inspected for signs of cracks and insulating liquid leaks. This is difficult when energized. An inspection may be performed with an infrared scanner; any temperature rise above ambient in excess of 25 °C should be investigated.

If the regulator has to be untanked following the preceding procedures, the following items should be checked:

- All connections should be examined for tightness.
- All contacts should be examined for wear following guidelines from the manufacturer's manual.
- The manufacturer's guidelines should be followed for the retanking procedure.

The following tests should be performed on the regulator after maintenance has been performed and insulating liquid samples have been taken, but prior to service restoration:

- Winding insulation PF
- Winding resistance

7.4.5 Infrared for LTC

Infrared for LTC is covered in Annex C.

7.5 Ancillary equipment

7.5.1 Cooling system

Large power transformers are fitted with some type of cooling system. Cooling systems generally consist of combinations of heat exchangers/radiators, fans, and pumps.

7.5.1.1 Cooling system heat exchangers

Basically, three types of heat exchangers are used to dissipate heat generated by power transformers:

- a) *Water cooler*: This heat exchanger consists of a set of tubes installed inside the equipment's tank and immersed in the equipment's insulating liquid. Fresh water is pumped through these tubes to carry off excess heat from the insulating liquid.
- b) *Oil-water cooler*: This heat exchanger is a type of cooler found on older equipment at generating plants.
- c) *Forced-air, forced-oil cooler*: This heat exchanger is found primarily at generating plants and on large, extra-high-voltage transformers. It is characterized by small, usually vertical, tubes wrapped with thin fins. The tubes are encased in a shell that is open on one side and has a fan shroud on the other side. Due to the closeness of the fins, this type of device is very efficient but is also susceptible to clogging by debris. Reduced air flow has a dramatic effect on the device's efficiency.

All other transformers are equipped with radiators. This type of cooling is characterized by long, broad, flat, hollow tubes mounted between two large pipes called *headers*. The insulating liquid flows into the upper header, is cooled, and flows naturally back through the lower header into the equipment's tank.

The type of cooling device the equipment is fitted with should be identified, and the appropriate instructions below should be followed.

Water coolers: Water flow rate should be observed so that this type of cooler can operate at its maximum efficiency. Inadequate water flow or higher water temperatures reduce the efficiencies of this type of system. Insulating liquid samples should be taken from the equipment's tank quite often (weekly) with this type of cooler to determine whether it is leaking water into the equipment's tank.

Interpretation: The pressure regulator or water pump output (or both) should be adjusted for proper water flow. Any amount of visible water found in the insulating liquid sample calls for immediate removal of the equipment from service until the water leak has been repaired and the spilled water removed from the equipment's tank.

Air coolers: Visual observation should be made through the cooler from one side to the other. It may be necessary to hold a strong light source on the opposite side to allow for inspection of trapped debris. The surfaces of the cooler fins should be examined for signs of contamination. On coolers where the fan action pulls air through the cooler, a single sheet of copier paper (standard weight) should be placed on the air inlet side. The paper should be held in place by the force of the air flow. For more precise measurements, an anemometer may be used to measure air flow at several points on the cooler for comparison with a serviceable cooler of the same specifications and size.

Radiators: Air flow is generally not a problem due to the relatively wide spacing between the tubes.

Interpretation: Care should be taken to remove any debris that becomes lodged between the fins or tubes of air coolers and radiators. In addition, any contamination buildup should be removed from the fins or tubes, when practical, to prevent a reduction in the device's efficiency.

7.5.1.2 Fans

7.5.1.2.1 Rotation of cooling fans

Cooling fans are designed to move air at ambient temperature across the radiator or cooler and provide heat transfer from the equipment's insulating liquid to the surrounding atmosphere.

The rotation of the fan blades should be observed to confirm that the air flow is in the proper direction for the type of device involved. Observation may be facilitated if it is performed at a lower-than-normal speed, either during startup or immediately after switching off.

Interpretation: Corrections to rotation should be made as indicated by inspection.

CAUTION

When examining fans, care should be taken not to contact the blades while they are rotating.

7.5.1.2.2 Visual inspection of cooling fans

Cooling fans are designed to enhance the transfer of heat generated by electrical equipment to the atmosphere. Assuming that the fans are properly dimensioned with respect to the design of the cooling

system, the inspection should include confirming that they are operating at their design speed, that airways are not blocked, and that guards and blades are not damaged.

At least two fans should be observed while they are running. Thus, any fan running at less than its design speed will be obvious to the naked eye. For more precise measurements, a tachometer or other type of timing device may be used, but this is rarely necessary. Visual inspection should be made for any trash or debris that could reduce the air flow onto the heat exchanger surface. The fan guard and blade should be examined for signs of distortion or other damage.

Interpretation: Improper air flow can reduce cooling system efficiency, cause overheating, and result in damage to electrical equipment. Fans not running at design speed should be replaced. After stopping fans, any obstructions to air flow should be removed and any damaged fan guards or blades should be replaced or repaired.

7.5.1.2.3 Cooling fan controls

Cooling fan controls are designed to operate both manually and automatically. The automatic function is generally related to load or energization (or both). If the transformer is single rated, the cooling equipment should operate when the transformer is energized, since this type of transformer has no self-cooled rating and will otherwise severely overheat. Triple-rated transformers have a self-cooled rating, as well as two other stages of cooling. These stages of cooling can be initiated by either insulating-liquid-temperature-controlled switches or a device sensitive to the transformer loading, such as a winding temperature indicator, which has become the preferred method.

Checks should be carried out according to the procedure described as follows.

Procedure: The type of cooling control system installed on the transformer should be determined to ascertain what inspections or tests are required. Running high speed pumps by manual control may, under certain conditions in some cooling systems, result in static electrification failure of a power transformer. The manufacturer's up-to-date recommendations should be referred to.

Manual control: This should be turned on briefly to confirm that each stage has sufficient voltage to operate. Fan operation should be observed. Insulating liquid pumps should be checked by observing their flow gauges. The manufacturer's recommendations should be referred to.

Temperature control: The temperature bulb should be removed from its well on the side of the transformer. The master control should be set to the AUTOMATIC position. Using a temperature-controlled calibration instrument, the temperature of the bulb should be slowly raised and observed for proper calibration (operation).

Load control: The secondary current of the controlling current transformer (CT) should be checked to confirm that it is operating properly. After shorting out the secondary of the CT (if transformer is energized) the secondary lead should be removed from the control circuit. Current should then be injected into the control circuit and the level of this current varied to observe proper operation.

Interpretation: Any improper operation should be corrected for satisfactory performance of the transformer.

WARNING

Extreme caution should be observed when performing operations or with the secondary of an energized CT. If the secondary of the CT becomes open-circuited (no burden) while the CT is energized, catastrophic results can occur without warning.

7.5.1.3 Cooling system pumps

Large power equipment is commonly equipped with liquid pumps to increase the cooling system's efficiency. These pumps consist of three-phase or single-phase motors, usually rated in the 230 V to 480 V range. Motor size and pump capacity vary. Most pumps are equipped with sleeve-type thrust bearings as opposed to ball or roller bearings. The power equipment's insulating liquid flows through the motor's winding and carries off heat generated by the losses of the motor.

7.5.1.3.1 Bearings of cooling pumps

Bearing wear is a cause of pump failure. For the most part, the only method of determining whether excessive bearing wear exists is by removal of the pump for a visual inspection. Abnormal vibration or noise when the pump is running may indicate a need for further investigation, but is far from conclusive. State-of-the-art fiber optic wear indicators have been installed on some newer cooling pumps and eliminate the need for pump removal to determine whether wear is excessive.

After removing the pump from the system, the end play of the shaft should be measured. The impeller and impeller housing should be examined for any wear.

Interpretation: The manufacturer's guide should be consulted to determine whether excessive bearing wear exists as indicated by amount of shaft end-play observed. Any indication of wear on the impeller and impeller housing is indicative of excessive thrust bearing wear.

Precautions: Removing a cooling pump requires a very precise knowledge of the arrangement of the cooling system. Equipment and cooling pump shall be de-energized. The cooling system surrounding the pump shall be effectively isolated from the remainder of the equipment's cooling system. Isolating valves should be closed and the system should be drained before the pump is removed. It is recommended that blanking plates be installed after the pump is removed. Pumps should never be run without complete immersion in insulating liquid.

7.5.1.3.2 Electrical problems of cooling pumps

Due to the integral relationship between a cooling pump and the power equipment's insulating liquid, electrical problems in the pump motor can give false indications of the power equipment's condition when using gas chromatography. Partially shorted motor windings and other electrical problems with the pump motor cause the generation of combustible gases in the equipment's insulating liquid since it should flow directly through the pump motor during normal operation.

Routinely, or after detecting abnormal levels of combustible gas in the power equipment's insulating liquid, the current flowing to each electrical terminal of each pump should be accurately measured while the pump is operating.

Interpretation: Any significant imbalance of current between terminals > 15% to 20% is indicative of a problem with the pump motor. Differences between current ranges for like pumps on the same piece of electrical equipment should be compared. Any significant difference may be indicative of a restriction in the area of the cooling system where the pump with the higher current drain is located or a problem within the pump itself.

7.5.1.3.3 Rotation of cooling pumps

Cooling pumps are, for the most part, centrifugal pumps that pump some liquid regardless of their direction of rotation.

The pumps should be manually turned on and off and the action of each pump's flow gauge should be observed while the pump is coming on. The insulating liquid flow should have ceased or be at a bare minimum before energizing the circuit. Higher viscosity alternative insulating liquids may require positive displacement pumps, particularly for installations in colder ambient temperatures.

Interpretation: Pumps that have their direction of rotation reversed build up flow at a visibly slower rate than normally operating pumps. Sluggish movement of the flow gauge flag where three-phase motors are used is an indication of reverse rotation. Reverse any two electrical leads supplying the suspect pump and re-energize. The movement of the flow gauge flag should now be much prompter.

7.5.2 Fault gas detector relay

In general, only conservator-equipped power transformers are equipped with fault gas detector relays. The gas detector relay detects the presence of free gas liberated from the insulating liquid, indicating a level of gas generation beyond the dissolved gas saturation limits of the insulating liquid. Air leaking into the transformer, usually during extremely cold ambient conditions, occasionally can also register on the gas detector relay.

The accumulated gas should be analyzed per the manufacturer's instructions whenever the gauge indicates any value above zero. Dissolved-gas-in-insulating-liquid analysis would also be appropriate at this time.

Some devices in use on transformers also perform some limited online, dissolved-gas-in-insulating-liquid analysis. The purpose of these devices is to alert the user when gas generation rates exceed predetermined limits. When this alert is received, more detailed laboratory gas-in-insulating-liquid analysis can be performed.

7.5.3 Fault pressure relay

There are two types of sudden-pressure relays. The most common type is mounted under the insulating liquid. The other type is mounted in the gas space. Internal arcing in liquid-filled electrical power equipment generates excessive gas pressure that can severely damage equipment and present extreme hazards to personnel. The sudden-pressure relay is intended to minimize the extent of damage by quickly activating protection systems.

The manufacturer's recommendations should be referred to for adjustment, repair, or replacement of improperly operating devices.

7.5.4 Gauges

7.5.4.1 Flow gauge

All power equipment cooling pumps should be equipped with a cooling pump flow gauge. This device is used to determine whether insulating liquid is flowing through the pump. It is not indicative of the velocity of the insulating liquid or the condition of the pump.

After making sure that insulating liquid cooling pumps are on, the flow gauge should be observed for indication of flow. The pump should then be turned off momentarily to check that the gauge position changes to the OFF (no flow) position.

Interpretation: If the pump is on and no flow is indicated, the sending unit may be defective. If the pump is turned off and the flow gauge continues to indicate flow, the gauge is probably stuck in the flow position.

and the sending unit or entire gauge may need to be replaced. It is normal for the gauge to continue to indicate flow for a brief period before indicating off (no flow), after turning off a pump. This is due to conservation of momentum with respect to the insulating liquid.

Precautions: Lack of flow from insulating liquid cooling pumps during operation may be indicative of imminent failure. Necessary corrective action should be taken immediately.

7.5.4.2 Liquid level gauge

Knowledge of the insulating liquid level in a transformer tank is of paramount importance. Most tanks are equipped with a liquid level gauge normalized for 25 °C operation. As the temperature of the liquid changes, the level rises or falls correspondingly. The equipment's nameplate may state the increment of increase or decrease in liquid level for each 10-degree variation in liquid temperature. This specification may also reference the distance from the tank cover to the liquid at a specific reference temperature (usually 25 °C). The common gauge is a float type with a round face and is generally equipped with one or two alarm contacts. One contact indicates low liquid level, while the second, if supplied, indicates high liquid level. The face is usually marked at the 25 °C (or normal) point, high, and low. The last two indications are relative and have no specific relationship to any real value.

The indication of the needle on the face of the liquid level gauge should be observed. This reading should be reasonably normalized with respect to the top insulating liquid temperature reading.

Calibration of this gauge should never be required. If the gauge is out of calibration, replacement is recommended.

Precautions: It is important to maintain proper insulating liquid level throughout the entire temperature operating range of the equipment. Failure to do so may result in loss of cooling and, in severe cases, damage to equipment.

7.5.4.3 Pressure gauge

The internal pressure of a power equipment tank is a function of liquid temperature and gas generation. This pressure is measured by a pressure gauge that should be calibrated periodically per appropriate standards.

On an LTC compartment or regulator there should be a small positive pressure relative to that in the power transformer tank. If the LTC is the vacuum-bottle type, there should never be any pressure buildup. In an LTC with a sealed compartment, pressure builds up with every tap change operation. These compartments are supplied with a pressure-relief valve that opens at about 3 psi and reseals at about 1 psi. This prevents any ingress of moisture into the tap changer compartment.

Precautions: High pressure can be indicative of extremely serious operating conditions and should be investigated immediately.

7.5.4.4 Temperature gauge

Liquid temperature and hot spot temperature gauges are important for proper operation of the transformer. These gauges not only indicate temperature, but also operate the fans and coolers by means of micro-switches that can be adjusted for various temperature settings. These gauges should be calibrated regularly on site with portable devices or in the laboratory.

- a) *Calibration of top insulating liquid temperature gauge:* The gauge should be removed from the transformer, and the sensing bulb should be placed in a controlled hot insulating liquid bath. Calibration should be checked at several points on the gauge.
- b) *Calibration of winding hot spot temperature gauge:* Based on measured temperature rises, or data from tests of a thermally duplicate transformer, bias current to the heating coil of the winding temperature indicator is factory adjusted to simulate the same gradient in degrees Celsius over top liquid rise as is experienced by the hottest spot in the transformer windings.

Current in the heater circuit is adjusted by the transformer manufacturer; the magnitude of this current should be known to allow calibration of the unit to be verified. The calibration curve of the heater current vs. hottest spot gradient is available from the transformer manufacturer.

7.5.5 Bushing CTs

Bushing current transformers are instrument transformers and are discussed in IEEE Std C57.13-2008 [B34]. Polarity, ratio, and phase relation testing is also described in IEEE Std C57.12.90.

Perform a test to determine that each bushing current transformer (BCT) is installed in the proper orientation. The test may be performed by applying direct current to the bushing while observing the deflection of a voltmeter connected to the BCT secondary.

Perform a ratio test of each BCT. For a multi-ratio BCT, it may be useful to provide test values for all five available taps. Secondary measuring points should be the terminal blocks in the control cabinet. It is suggested that the short-circuiting jumper be replaced on each BCT terminal block upon completion of test.

NOTE—The DC supply equipment must be at zero current before the equipment is turned off and disconnected to prevent saturating the BCT.

An insulation resistance test may be performed on each BCT to ground. Set the test set for an output of 500 V.

8. Diagnostic chart

For the purpose of this guide, diagnostic tests are described with reference to principle categories of systems that constitute the transformer, reactor, or regulator (e.g., windings, bushings, insulating liquids, tap changers, core, tanks, and associated devices). For each category, the quantities measured are shown in the diagnostic test chart (Table 20) for ease of reference. In some cases further subdivision is necessary.

Not all tests are necessarily performed by any single user. In addition, the specific tests carried out vary according to the regular practice of the user and may depend on the history of the apparatus.

The establishment of benchmark values on a new piece of electrical equipment is very important when considering evaluation of future test results. Benchmark values are the first measurements taken on a piece of new or used equipment. Subsequent test results from tests on the same unit or from similar tests on similar equipment, when compared to these initial values and similar tests on similar equipment, may indicate a trend.

Table 20—Diagnostic test chart

Component	First sub-component	Second sub-component	Test	Transformer	Reactor	Regulator
Windings			Insulation resistance	X	X	X
			Ratio/polarity/phase	X		X
			Excitation current	X	X	X
			Leakage reactance	X		
			Winding resistance	X	X	X
			Capacitance	X	X	X
			Power factor (dissipation factor)	X	X	X
			Induced voltage/partial discharge/RIV	X	X	
			Frequency response analysis	X	X	X
Bushings			Capacitance	X	X	X
			Dielectric loss	X	X	X
			Power factor (dissipation factor), C_1 and C_2 PF (DF)	X	X	X
			Partial discharge	X	X	X
			Temperature (infrared)	X	X	X
			Insulating liquid level	X	X	X
			Visual inspection	X	X	X

Table 20—Diagnostic test chart (continued)

Component	First sub-component	Second sub-component	Test	Transformer	Reactor	Regulator
Insulating liquid			Water content	X	X	X
			Dissolved gas	X	X	X
			Dielectric strength	X	X	X
			Particle count	X	X	X
			Dielectric loss	X	X	X
			Dissipation factor	X	X	X
			Interfacial tension	X	X	X
			Acidity	X	X	X
			Visual inspection	X	X	X
			Color	X	X	X
			Oxidation stability	X	X	X
			Furan	X	X	X
			Corrosive sulfur	X	X	X
Tap changers	Load		Contact continuity	X		X
			Temperature (infrared)	X		X
			Ratio	X		X
			Motor currents	X		X
			Limit switch	X		X
	De-energized		Contact pressure	X		
			Centering	X		
			Ratio	X		
Core			Visual inspection	X		
			Insulation resistance	X	X	X
Tanks and associated devices	Tank		Ground test	X	X	X
			Pressure	X	X	X
			vacuum	X	X	X
			Dew point	X	X	X
			Temperature (infrared)	X	X	X
	Conservator		Visual inspection	X	X	
			Visual inspection	X	X	
			Total combustible gases	X	X	
	Inert air system		Visual inspection	X	X	
			Calibration	X	X	
	Gauges		Visual inspection	X	X	X
			Calibration	X	X	
	Fault pressure relay		Continuity	X	X	
			Continuity	X	X	
	Cooling system	Heat exchanger	Air flow	X	X	
			Visual inspection	X	X	X
			Cleaning	X	X	X
		Fans	Rotation	X	X	X
			Controls	X	X	X
			Visual inspection	X	X	X
		Pumps	Rotation	X	X	
			Currents	X	X	
			Bearings	X	X	
			Bearings	X	X	
Current transformers			Ratio	X	X	X
			Polarity	X	X	X
			Resistance	X	X	X

Annex A

(informative)

Power factor measurements

A.1 General

A.1.1 Background

Insulation power factor (PF) is one of the most common tests performed on transformers, regulators, reactors, bushings, and other support equipment and should be conducted as part of factory, acceptance, and routine assessment. Though dielectrics have inherent losses due to construction materials, PF measurement is most effective at detecting the relative levels of moisture and contamination. Evaluation of the capacitance measurement is effective in detecting physical defects that lead to changes in the dielectric's geometry.

A.1.2 Imperfect dielectric

The properties of true dielectric insulations are often simplified and expressed as imperfect dielectrics. An imperfect dielectric is a dielectric where the energy necessary to create an electric field is not returned to the electric field when the energy is removed. The energy is converted into heat in the dielectric. Conversely, a perfect dielectric has zero conductivity, and there are no absorption effects. A high vacuum is an example of a perfect dielectric. Most dielectrics tested are considered imperfect dielectrics due to the presence of moisture, contaminants, and other inherent polar molecules.

The two common methods of representing the imperfect dielectric are the series and parallel circuit. These two circuits include two elements, a capacitor and a resistor.

The resistor represents the loss component of the insulation, while the capacitor represents the geometric and physical properties, such as the dielectric constant.

Figure A.1 displays the two common methods of representing the imperfect dielectric.

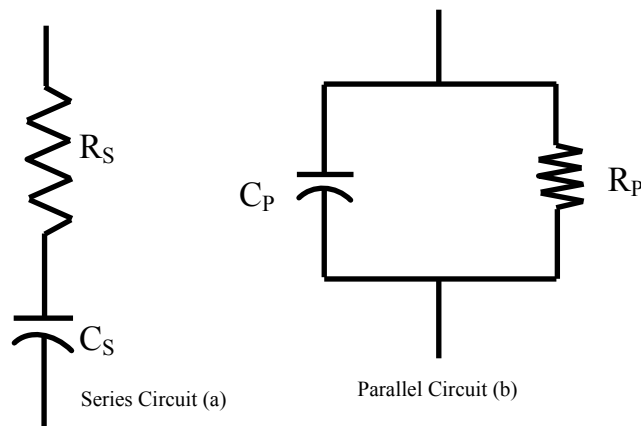


Figure A.1—Imperfect dielectric circuit: (a) series circuit; (b) parallel circuit

Either circuit is adequate for explaining the effects of PF within a dielectric specimen. In the series circuit, R_s represent the series ac resistance. The parallel circuit shows R_p as the equivalent parallel resistance. Though both can be used in discussing and defining PF, the parallel circuit is the most common. In both cases, the resistance represents the presence of moisture, contaminants, lossy partial discharge (PD) activity, and inherent polar materials.

A.1.3 Power factor

PF is an indication of loss per unit volume. As such, it is a dimensionless value that indicates the ratio of loss associated with the dielectric under test. It is an inherent value property and is independent of volume. The PF for an ac circuit is defined as shown in Equation (A.1):

$$PF = \text{Watts}/(E \times I) = \cos(\theta) \quad (\text{A.1})$$

Where *Watts* is the real power generated, E is volts, and I is the total current in amperes supplied to the dielectric circuit. Phase angle θ (*theta*) is the angle formed between the voltage applied (E) and charging current I . Understanding the PF is more evident when shown in a phasor diagram. Figure A.2 shows the phasor diagrams of both the series and parallel circuits.

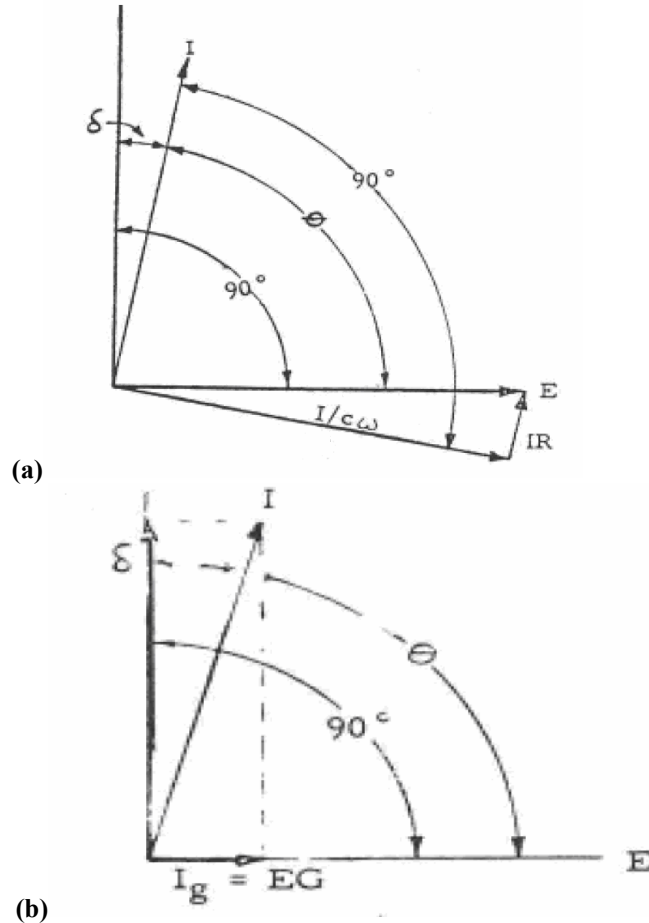


Figure A.2—Phasor diagrams: (a) Series dielectric; (b) parallel dielectric

In both diagrams of Figure A.2, the PF angle is the angle θ . The voltage E is applied across the specimen's impedance with total charging current of I . Depending on the model chosen, current flows across the capacitor as well as the resistor. In the parallel circuit, current I_g represents the current created by admittance G from resistance R_p . Most dielectrics have a θ PF angle nearly equal to 90 degrees. Another method used to express dielectric loss is phase defect angle δ (*delta*), which is the complement of θ . DF is calculated by taking the tangent of δ and has values very nearly equal to PF for most dielectrics with typical θ values close to 90 degrees. Mathematical comparison of DF and PF shows that the two values are nearly identical up to PF and DF of 0.10.

Since θ is derived by the combination of the capacitive charging current and resistive current, it represents a ratio of real to reactive currents. Thus, it represents a dielectric efficiency and can be used to gauge the quality of a dielectric. For simplicity, the following discussions use the parallel imperfect dielectric model and relate the ratio relationships of power factor [see Equation (A.2), Equation (A.3), Equation (A.4), and Equation (A.5)].

$$\text{Watts} = E \times I_{Rp} \quad (\text{A.2})$$

$$\text{Watts} = E \times IT \times \cos(\theta) \quad (\text{A.3})$$

$$PF = \cos(\theta) = \text{Watts}/(E \times IT) \quad (\text{A.4})$$

$$= (E \times IR)/(E \times IT) = IR/IT \quad (\text{A.5})$$

Finally, it is common to discuss PF in terms of percentage. To calculate percent power factor use Equation (A.6).

$$\%PF = PF \times 100 \quad (\text{A.6})$$

PF can be used to compare dielectric materials in terms of loss. By removing the units of volume, it is possible to compare similar construction dielectric materials in a large statistical group. Thus, comparison of apparatus is easier because better statistical analyses are possible.

Finally, it should be stated that the effects of temperature on PF are well documented in other sources. Generally, PF is corrected to 20 °C to ensure that true comparisons are being conducted. Please refer to appropriate manufacturer literature on appropriate temperature corrections.

A.2 Test equipment

Most modern dielectric loss/PF test sets are equipped with selectable test modes that simplify the testing of complex insulating systems. The two basic test modes are grounded specimen test (GST) and ungrounded specimen test (UST). Test equipment should also include additional guard circuitry that allows for variations on these two modes, thus allowing each section of complex insulating systems to be tested separately. It is important for individual sections of insulation to be tested separately if possible, to prevent large sections from concealing the deterioration in small sections.

A.3 Test modes

PF test equipment allows for the relative positioning of the power source, guard, and ground. Rearranging these various positions allows for the creation of the GST and UST circuit (see Table A.1). Additional test leads can be used to modify the measurement circuit and ground or guard additional legs of the dielectric circuit under test. Modification of the test circuit with additional leads allow for the creation of three basic test modes: UST, GST-Guard, and GST-Ground (see Table A.2).

Table A.1—Test circuits

Test mode	Description
GST	Grounded Specimen Test—The GST measures current flowing to ground via the meter circuit.
UST	Ungrounded Specimen Test—The UST measures current flowing to an ungrounded (floating) meter circuit.

Table A.2—Test modes

Configuration	Reference	HV test lead	LV lead	Tank ground	Measuring
GST—Ground	Figure A.1	H1, H2, H3	Meter in	Meter in	$C_{HL} + C_H$
UST	Figure A.4	H1, H2, H3	Meter in	Meter out	C_{HL}
GST with Guard	Figure A.5	H1, H2, H3	Meter out	Meter in	C_H

A.3.1 Grounded specimen test

The GST configuration permits testing of a grounded insulation specimen through the specimen's ground. All current flowing to ground is measured via the meter circuit. The configuration is illustrated in Figure A.3.

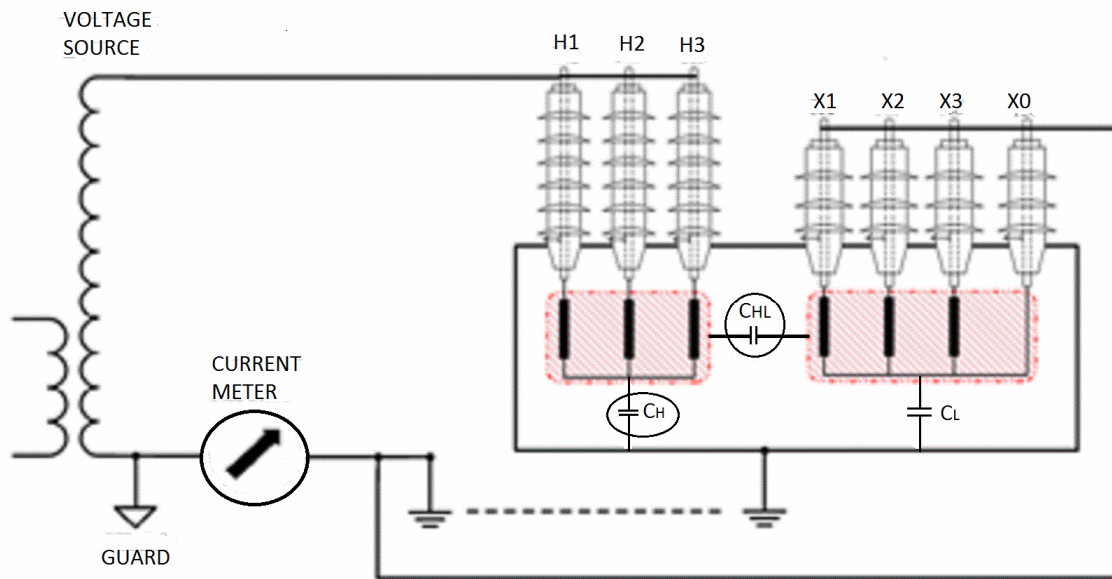


Figure A.3—Grounded specimen test circuit

A.3.2 Ungrounded specimen test

The UST configuration is used for measurements between two terminals of a test specimen that are not grounded or that can be removed from ground. In the UST configuration, current flowing in the insulation between the voltage lead and the measuring lead of the instrument is measured and current flowing to ground is not measured. The test configuration also shifts the ground of the test circuit to the guard point to the left of the meter, allowing the ground current to bypass the metering circuit. This configuration is illustrated in Figure A.4.

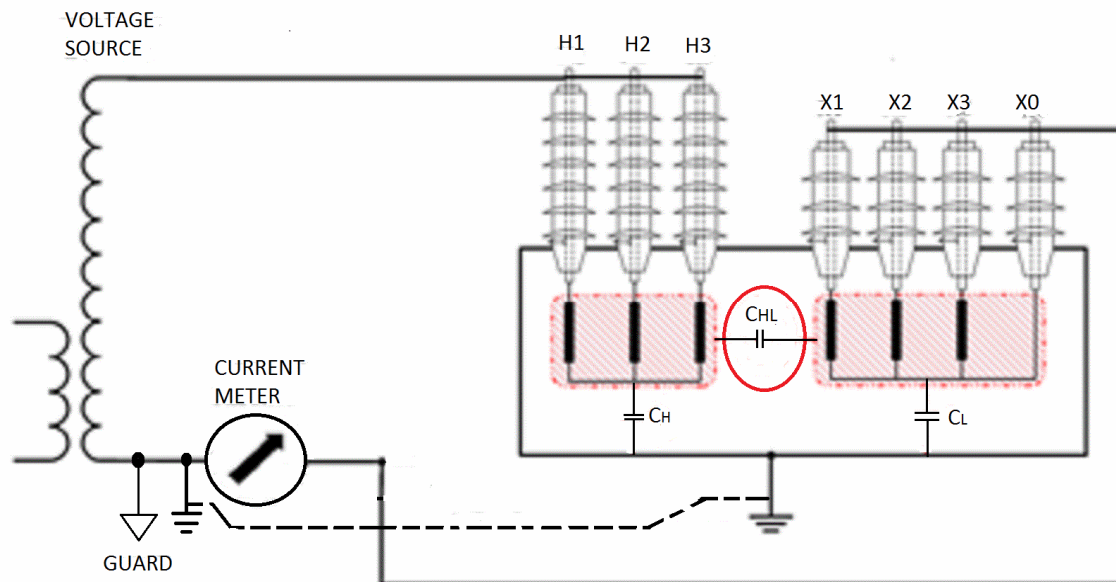


Figure A.4—Ungrounded specimen test circuit

A.3.3 Grounded specimen test with guard

The GST-Guard configuration allows unwanted currents to bypass the measuring circuit and enables smaller sections of insulation to be tested individually. Only the ground currents are measured using a GST-Guard configuration. Current flowing to terminals with the guard connection is not measured. This configuration is illustrated in Figure A.5.

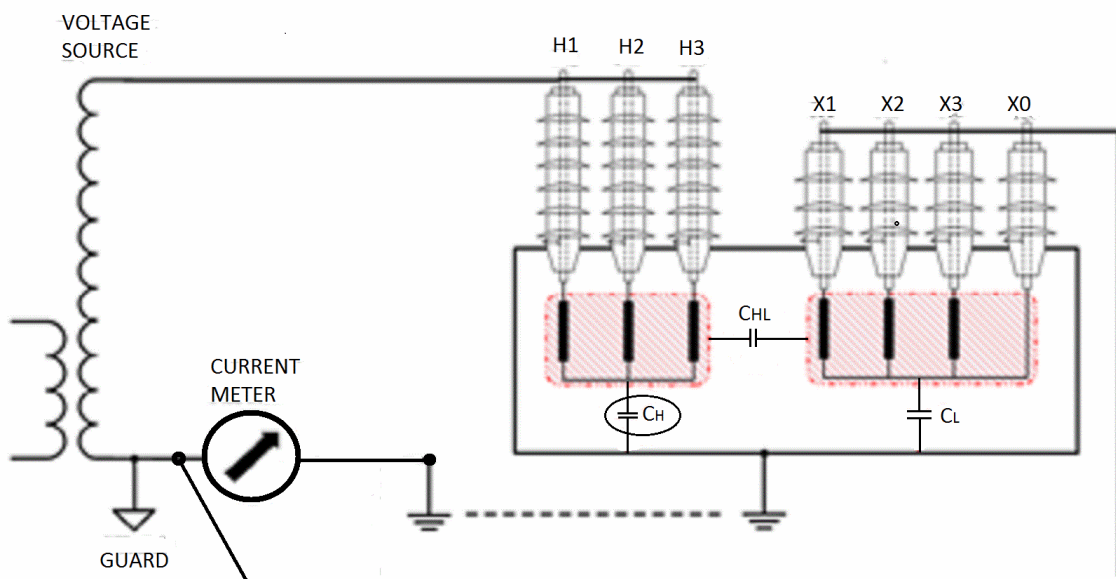


Figure A.5—Grounded specimen with guard test circuit

A.3.4 Complex UST and GST test circuits

Test circuit configurations can also be made more complex by adding a second LV test lead. Using these configurations, it is possible to create the following GST circuits: GST-Ground Guard, GST-Ground Ground, and GST-Guard Guard circuits. UST circuits can be modified to include UST-Ground and double UST circuits (where two leads are measured in parallel). These configurations allow complex systems to be tested.

A.4 Simple and complex insulating systems

A.4.1 Simple system

A simple insulating system consists of two terminals separated by insulation and is represented as a single capacitor. An example of a simple system is the bushing overall test, with its center conductor and mounting flange as the two electrodes.

A.4.2 Complex system

A complex insulating system consists of three or more terminals insulated from each other. A three-terminal system can be represented by a network of three capacitors, and a four-terminal system by six capacitors. Two-winding transformers and three-winding transformers are complex systems. Figure A.6 illustrates a complex system. PF calculations should not be used to determine the integrity of insulation if the *measured* current is less than 0.3 mA. At low *measured* currents, PF calculations are susceptible to large swings, which could be misleading. *Therefore, in those cases, the test results should be evaluated based on current and loss readings.*

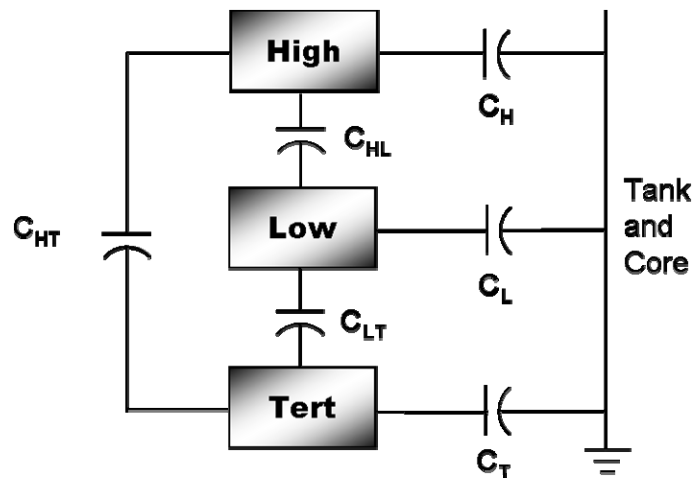


Figure A.6—Three-winding transformer dielectric circuit

For references, see Doble Test Procedures [B22] and Dielectric Theory and Practice [B23].

Annex B

(informative)

Bushings

Bushings may be classified generally by design as follows:

- a) Condenser type
 - 1) Insulating liquid-impregnated paper insulation, with interspersed conducting (condenser) layers or insulating liquid-impregnated paper insulation, continuously wound with interleaved lined paper layers
 - 2) Resin-bonded
 - 3) Resin-impregnated paper insulation, with interspersed conducting (condenser) layers
- b) Noncondenser type
 - 1) Solid core or alternate layers of solid and liquid insulation
 - 2) Solid mass of homogeneous insulating material (e.g., solid porcelain)
 - 3) Gas filled

For outdoor bushings, the primary insulation is contained in a weatherproof housing, usually porcelain or compound type with silicone rubber sheds. The space between the primary insulation and the housing is generally filled with an insulating liquid or compound (also, plastic and foam). Some of the solid homogenous types may use insulating liquid or tar to fill the space between the conductor and the inner wall of the housing. Bushings may also use gas such as SF₆ as an insulating medium between the center conductor and outer housing.

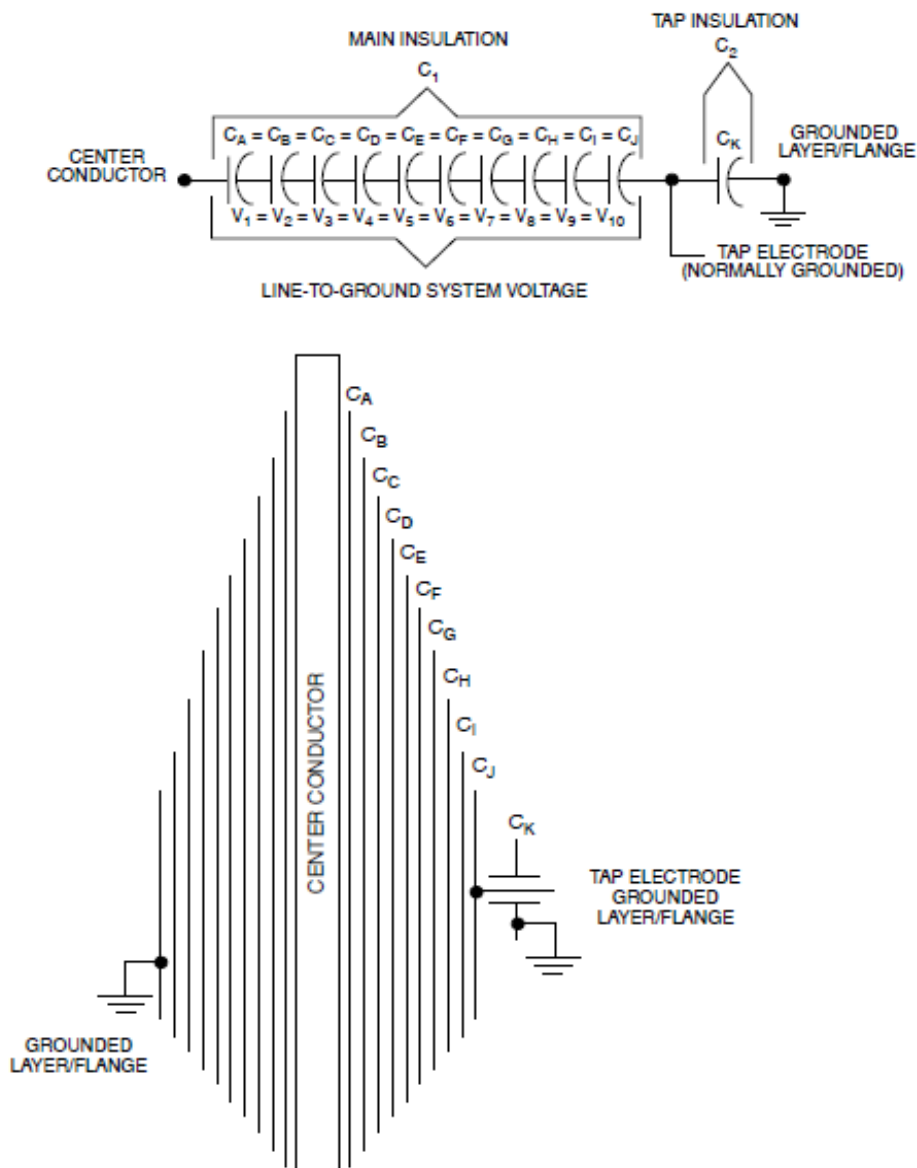
Bushings may be further classified generally as being equipped with a tap electrode or not equipped with a tap electrode. Based on IEEE Std C57.19.00-2004 [B35], the following two types of taps have been defined:

- *Test tap* for 350 kV BIL and below bushings; the withstand level is 2 kV/1 min.
- *Voltage tap* for 350 kV BIL above bushings; the withstand level is 20 kV/1 min.

The bushing, without tap electrodes, is a two-terminal device that is generally tested overall (center conductor to flange) by the grounded specimen test (GST) method. If the bushing is installed in an apparatus, the overall GST measurement includes all connected and energized insulating components between the conductor and ground.

A condenser bushing is essentially a series of concentric capacitors between the center conductor and the ground sleeve or mounting flange. A conducting layer near the ground sleeve may be tapped and brought out to a tap electrode to provide a three-terminal specimen. The tapped bushing is essentially a voltage divider and, in higher voltage designs, the tap potential may be used to supply a bushing potential device for protection relays and other purposes. In this design, the voltage tap also acts as an LV power factor (PF) test terminal for the main bushing insulation, C₁. Refer to Figure B.1.

Modern bushings rated above 350 kV BIL are usually equipped with voltage tap. Typical design of a voltage tap involves connecting the outmost (ground) layer of the bushing core. With the grounding cap removed, the tap electrode is available as an LV terminal for an ungrounded specimen test (UST) measurement on the main bushing insulation, C₁, conductor to tapped layer.



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NOTE 1—Equal capacitances, C_A through C_J , produce equal distribution of voltage from the energized center conductor to the grounded condenser layer and flange.

NOTE 2—The tap electrode is normally grounded in service except for certain designs and bushings used with potential device.

NOTE 3—For bushings with potential taps, the C_2 capacitance is much greater than C_1 . For bushings with PF tap, C_1 and C_2 capacitances may be same order of magnitude.

Figure B.1—Condenser bushing

Insulating Liquid Sampling—Insulating liquid samples may be taken from liquid-filled bushings for dissolved gas analysis (DGA) of the insulating liquid. While this is not yet a common practice among the utilities in North America, DGA on bushing mineral oil has been proven to be a good diagnostic technique in detecting internal gassing problem of bushings. There is currently no IEEE standard with established limits on the gas levels. IEC 61464:2003 [B31] where mineral oil is the impregnating medium of the main insulation (generally paper), provides the normal limits on gas levels shown in Table B.1. Lower limits for gases may be used at the user's discretion. Some engineers suggest that no detectable levels on acetylene should be accepted since the presence of acetylene is a sign of arcing. Since a bushing is a hermetically sealed body using in most cases a porcelain insulator, a bushing with arcing signatures should be replaced.

Table B.1—Normal limits on gas levels

Type of gas	Concentration (ppm)
Hydrogen (H ₂)	140
Methane (CH ₄)	40
Ethylene (C ₂ H ₄)	30
Ethane (C ₂ H ₆)	70
Acetylene (C ₂ H ₂)	2
Carbon monoxide (CO)	1000
Carbon dioxide (CO ₂)	3400

Annex C

(informative)

Infrared temperature measurements

C.1 General

Infrared (IR) temperature measurement systems can provide an effective noncontact means of detecting the localized temperature anomalies associated with power apparatus. The use of IR emissions to measure object temperatures is based on the fact that IR emissions increase predictably with temperature. Therefore, IR detectors “see” heat in the IR spectrum in the way that light can be seen in the visible spectrum. The systems offered by manufacturers include spot radiometers, line scanners, pyroelectric vidicon tube imagers, solid-state detector imagers, and radiometers. These systems are available with different levels of sophistication in controls and data presentation.

C.2 IR temperature measurement

IR temperature measurement instruments allow the user to detect the thermal anomalies associated with many faults in power apparatus. Thermal variations in power apparatus result from increased electrical resistance due to component failure, fatigue, and mechanical misalignment. The emission of IR energy from an object increases as a function of the object temperature. The IR instruments collect the energy emitted by the object of interest and present to the user a qualitative and/or quantitative representation of the object temperature. This annex is intended to highlight some of the parameters that should be understood when performing an IR measurement as part of a maintenance program.

Every object radiates energy. The amount of radiated energy is a function of the object temperature and the emissivity of the surface. The emissivity is a parameter that specifies how well the surface emits radiation. The value varies from 1.0 to 0.0, where 1.0 is a perfect emitter and 0.0 is a perfect reflector. The value of the emissivity is equal to one minus the reflectivity if the object does not transmit. For example, if an object has an emissivity of 0.9, it emits 90% of the IR energy emitted by a perfect emitter, while it reflects 10% of the energy incident upon its surface.

An IR system cannot distinguish between emitted and reflected energy. The user is only interested in measuring the target’s emitted energy, which is a function of the object’s temperature. Many IR temperature measurement systems allow the user to mathematically compensate for the reflected IR energy by entering an estimated emissivity value. The user should always keep in mind that the source of the reflected IR energy can have a significant impact on the absolute accuracy of the temperature measurement. Some systems allow the operator to specify the temperature of the reflected source, while others use a nominal ambient temperature value. The emissivity value is best determined experimentally by collecting representative values of the emissivity for various objects of interest. Emissivity data provided by the manufacturer can also be satisfactorily used. As a general rule, most painted, dirty, or corroded objects have a high emissivity value (0.7 to 0.9). Severe corrosion, while highly emissive, can form an insulating layer that can conceal the true target temperature. For painted objects, the gloss or shine of the coating is more indicative of the IR emissivity than is the color. As a general rule, color does not affect IR emissivity. Shiny metals generally have a low emissivity value.

The geometry of the measurement setup (angle of incidence) is important because it defines the source of the reflected IR energy. To a lesser extent, it also influences how the surface reflects the IR energy. Regardless of the angle of incidence, the user should note what source is reflected by the object of interest. When measuring temperatures outdoors, care should be taken to eliminate reflections from the sun.

Reflected IR energy is not to be confused with actual solar gain, where the sun's radiance actually increases the object's temperature. Discrimination of reflections can be accomplished by moving the point of observation 90 degrees.

Round or cylindrical objects can be especially difficult to measure. Depending on the surface, an accurate temperature may be available only over a small portion of the object. This effect is clear when using an imaging system, but spot and line scan systems make it very difficult for the user to visualize the geometrical effect. An extremely valuable practice is to measure the temperature from several different positions to minimize the chance of error.

The maximum distance between the IR instrument and the target of interest is determined by the instrument configuration, the stand-off distance, and the size of the target. The IR systems discussed in this annex have a minimum target size for which the temperature can be accurately measured.

For spot radiometers, the measurement region is relatively large and is delineated in the optical viewer of the sensor or described in the specifications. The manufacturer's recommendations should be followed so that only the object of interest is measured and the surrounding background is not averaged in with the desired object's temperature.

Imagers and line scan systems have a relatively small portion of the field of view defined as the pixel, resolution element size, or instantaneous field of view. This small element is similar to the "spot" discussed above for spot radiometers except that it is very small in comparison. Even though it is small, it may view more than one object at one time. Many systems are provided with a manual that discusses the importance of measuring temperatures with several pixels aligned on the target. Any such guidelines should be followed to maximize measurement accuracy. Even though a thin or small object such as a bushing connector can be seen in the image of the instrument, it does not follow that the measured temperature is accurate unless enough pixels view only the target and not the target and background together. This becomes more important as the distance between the instrument and the object of interest increases. Therefore, high spatial resolution is very desirable.

The temperature comparisons shown in Table C.1 between similar components under similar loading and temperature rises above ambient have been found to be practical during IR inspection according to Table 100.18, Thermographic Survey Suggested Actions Based on Temperature Rise (see NETA ATS [B59]) *[with clarifying inserts for this guide]*.

Table C.1—Temperature comparisons between similar components

Temperature difference (ΔT) based on comparisons between similar components under similar loading in $^{\circ}\text{C}$	Temperature difference (ΔT) based on comparisons between components and ambient air temperatures in $^{\circ}\text{C}$	Recommended action
1 to 3	1 to 10	<i>[Normal to]</i> possible deficiency; warrants investigation
4 to 15	11 to 20	Indicates probable deficiency; repair as time permits
<i>Greater than 15</i>	21 to 40	Monitor until corrective measures can be accomplished
Greater than 15	Greater than 40	Major discrepancy; repair immediately <i>[For top liquid temperature rises of 65 $^{\circ}\text{C}$ there may be cases where up to 65 $^{\circ}\text{C}$ is normal]</i>

Temperature specifications vary depending on the exact type of equipment. Even in the same class of equipment (e.g., cables) there are various temperature ratings. Heating is generally related to the square of the current; therefore, the load current has a major impact on temperature difference (ΔT). In the absence of consensus standards for ΔT , the values in Table C.1 provide reasonable guidelines.

An alternative method of evaluation is the standards-based temperature rating system as discussed in 8.9.2, Conducting an IR Thermographic Inspection, in Gill 107.

It is a necessary and valid requirement that the person performing the electrical inspection be thoroughly trained and experienced concerning the apparatus and systems being evaluated, as well as knowledgeable of thermographic methodology.

Annex D

(informative)

Dew point test

D.1 General

A number of measuring techniques are used to determine the amount of moisture remaining in the transformer insulation. Dew point testing is a measurement of surface moisture and not the average moisture in the insulation. Dew point testing is a reliable technique to estimate moisture. The dew point temperature is the temperature at which water or dew and water vapor present in gas in the transformer tank are in equilibrium at a given gas pressure. In other words, at this temperature liquid water evaporates at the same rate at which it condenses.

Moisture content determined by measuring the dew point is the “average” surface moisture. It is very important to know that the average temperature of the insulation needs to be reasonably accurate to minimize the magnitude of the error produced in the conversion process.

D.2 Dew point test

When to perform dew point measurement:

- a) Upon initial rail or truck inspection if the transformer pressure gauge indicates a loss of pressure. A dew point measurement should be taken to determine whether moisture has entered the unit.
- b) After assembly of the transformer and prior to vacuum filling the unit, dew point should be measured to confirm that the unit is sufficiently dry.

D.2.1 Procedure

Seal the unit and evacuate to 2 mm Hg for 4 h. Relieve the vacuum in the transformer tank with dry air to 3 psi and then allow it stand for 12 h so that the vapor pressure in the insulation and the gas approaches equilibrium.

The dew point of the gas in the tank is then measured along with the temperature of the transformer insulation and the pressure inside the tank. From these measurements the moisture remaining in the insulation is estimated using available curves. If the measurement is within the acceptable range on the dew point limit curve then proceed with vacuum filling. If the measurement is not within the acceptable range then additional processing is required to remove moisture.

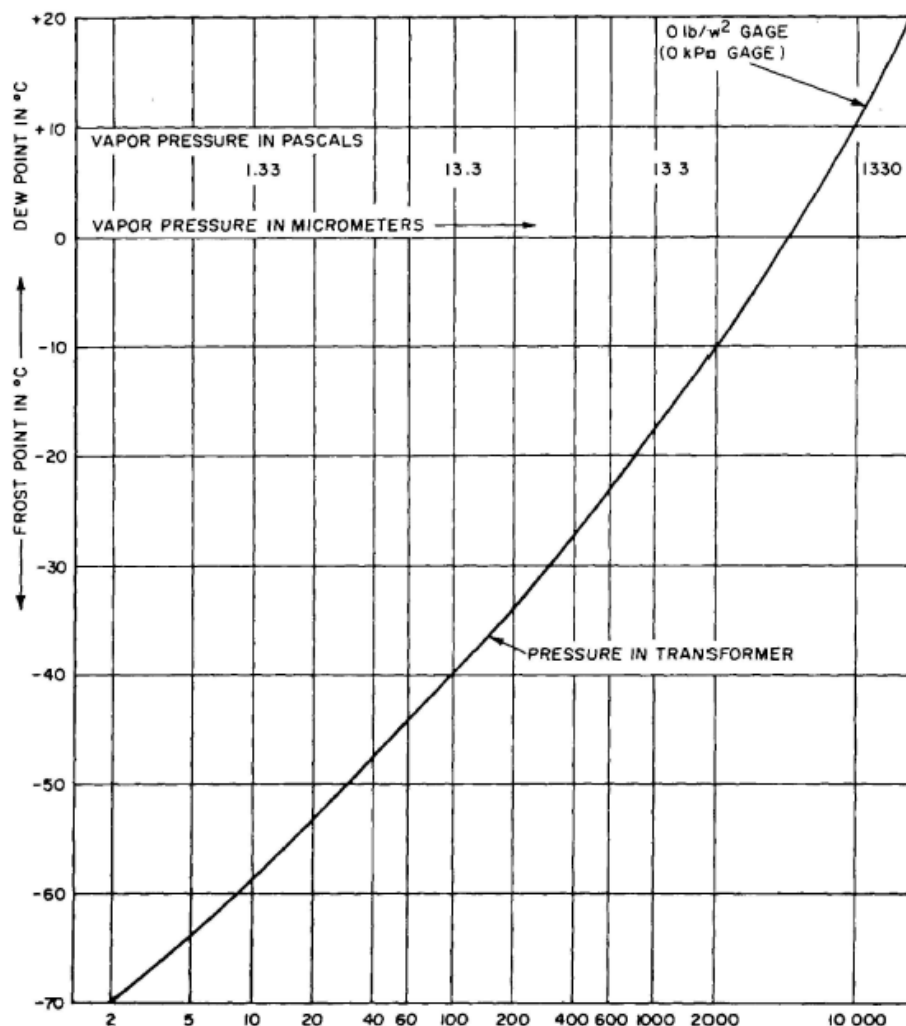
D.2.2 Example

The following example clarifies the procedure:

Measured dew point = $-45\text{ }^{\circ}\text{C}$
Insulation temperature = $20\text{ }^{\circ}\text{C}$
Pressure in tank = 2.0 psig (13.8 kPa)
Atmospheric pressure = 14.7 psi (101.325 kPa)

On Figure D.1, read the vapor pressure corresponding to a dew point of -45°C as $60\text{ }\mu\text{m}$ (8 Pa). Correct this vapor pressure for the overpressure in the tank of 2.0 psig (13.8 kPa) as follows:

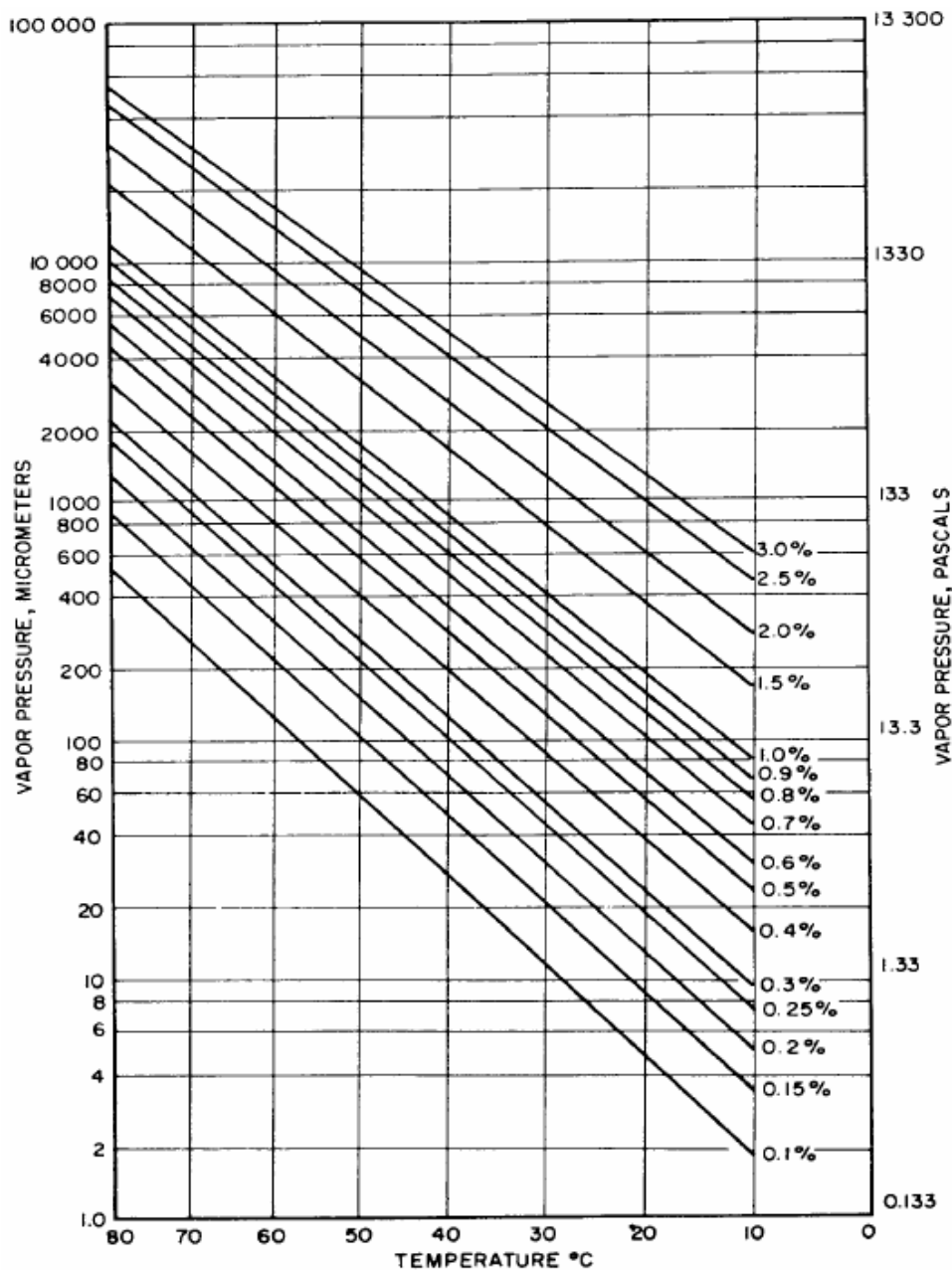
$$[(14.7 + 2.0) \times 60] / 14.7 = 68\text{ }\mu\text{m} \text{ or } [(101.325 + 13.8) \times 8] / 101.325 = 9.1\text{ Pa}$$



Source: IEEE Std C57.93.

Figure D.1—Conversion from dew point or frost point to vapor pressure

Now using the moisture equilibrium chart, Figure D.2, find the intersection of 20°C insulation temperature and $68\text{ }\mu\text{m}$ (9.1 Pa) of vapor pressure. Read the moisture content of approximately 0.6% from the diagonal lines labeled in percent moisture content.



Source: IEEE Std C57.93.

NOTE—This chart was prepared using information from a Piper Chart (see Piper [B65]), which was extrapolated and interpolated from published data on cotton paper. Piper gave a multiplication factor of 1.7 to use for Kraft paper (non-thermally upgraded). This chart incorporates the 1.7 factor, and the values obtained need to be corrected. It should be noted that equilibrium moisture content of cellulose is dependent on whether equilibrium is approached by absorption or desorption, since there is a hysteresis effect.

Figure D.2—Moisture equilibrium chart (with moisture content in percent of dry weight of insulation)

Annex E

(informative)

Furan testing

Paper is the major solid dielectric material within a transformer, used either as conductor wraps and impregnated with insulating liquid, or as barrier boards, wraps, spacers, and clamps in compressed or resin-bonded forms. The major constituents of the paper are cellulose (about 90%), lignin (about 6%), and the remainder (about 4%). Each cellulose molecule has a different length, and these different groups of cellulose molecules are held adjacent to each other via the hydroxyl group (-OH). Each cellulose molecule consists of a linear glucose molecule polymer. These glucose molecules, or the D-anhydro-glucopyranose units, are held together through a β -1, 4-glycosidic bond. A single cellulose fiber is formed from a number of these chains held together by hydrogen bonds as indicated in Figure E.1.

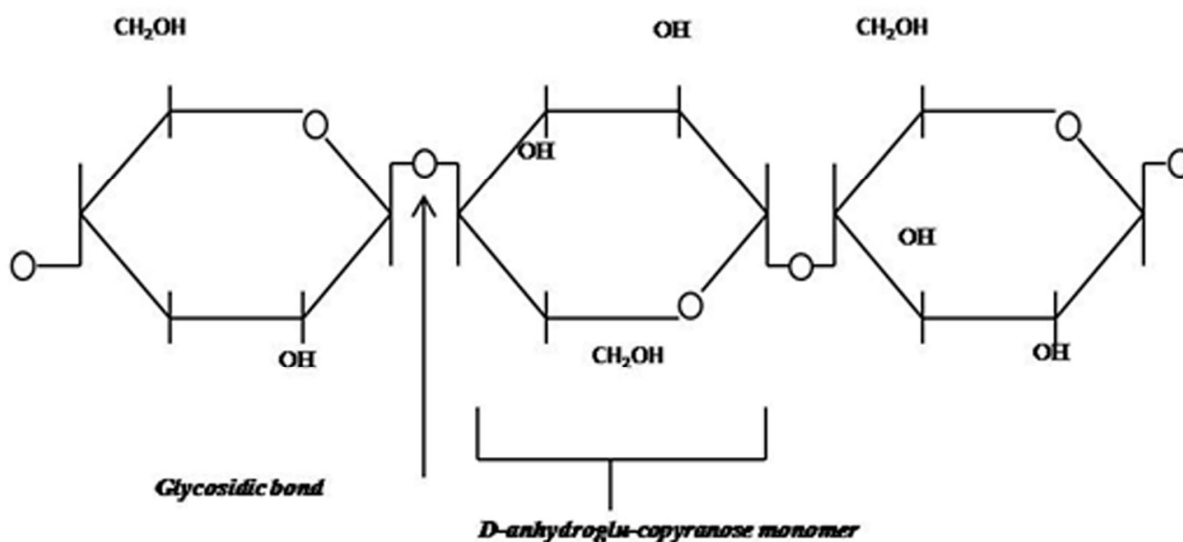


Figure E.1—Structural formula of cellulose

The three most common degradation factors of cellulose have been identified as thermal, oxidative, and hydrolytic. When cellulose is subjected to a temperature of 200 °C, the beta linkages (glycosidic bonds) tend to break and open the glucose molecule rings and thereby lose mechanical strength. Byproducts of this reaction include free glucose molecules, moisture, CO, CO₂, and organic acids. The presence of oxygen promotes oxidation, and cellulose molecules are prone to oxidize. The reaction of oxidation on these cellulose molecules causes the glycosidic bond to weaken and can cause scission of the cellulose molecule chain. The oxidation of hydroxyl produces carbonyl (aldehydic) and carboxyl (acidic) compounds. Moisture is also a byproduct from this oxidative reaction. With water and acids present, the glycosidic bond is exposed to slicing that, in turn, produces free glucose. Hydrolytic degradation can be initiated out of thermal and oxidative degradation since both of these degradations produce moisture and acids. From previous discussion, it can be observed that the immediate byproducts related to paper degradation are CO, CO₂, moisture, organic acids, and free glucose molecules. The free glucose degrades further into aromatic components known as furans. The presence of moisture and organic acids in the insulating liquid can further degrade the free glucose molecule into 5-hydroxymethyl-2-furfuryl or 5H2F.

5H2F is an unstable compound and can decompose further into other furans as follows:

- 2-furaldehyde (2FAL)
- 5 methyl-2-furaldehyde (5M2F)
- 2-acetylfuran (2ACF)
- furfuryl alcohol (2-FOL)

All five compounds previously mentioned are known as furan derivatives. It has been shown that all of these compounds except 2FAL are not very stable under operating conditions found in transformers. These compounds apparently form and then further degrade to 2FAL over a time span of a few months. 2FAL is apparently stable for several years under the same conditions. The molecular structure of this compound is shown in Figure E.2.

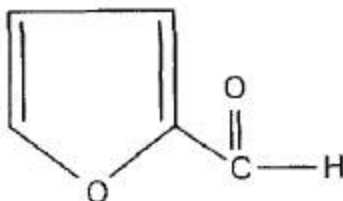


Figure E.2—Molecular structure of 2FAL

Furans are currently detected by manual extraction of the compounds from insulating liquid samples taken from the transformers prior to insulating liquid treatment with Fullers earth. The method of sampling should comply with ASTM D923 [B3] to ensure sampling integrity. The furanic compounds can be extracted from the insulating liquid samples by way of liquid/liquid extraction or solid-phase extraction. A sample of insulating liquid mixed with methanol or acetonitrile is shaken and is allowed to settle until the solvent and insulating liquid layers are completely separated. An alternative method that can be used is to dissolve the insulating liquid in n-pentane solution and filter the mixture through a silica-packed disposable cartridge. Once the filtering process is completed, the silica retains the polar constituents of the insulating liquid, including the furanic compounds. Next, the furanic compounds are recovered by applying methanol or acetonitrile. Once the furanic compounds have been extracted via either one of the above methods, the furan concentration is then measured in a laboratory using high performance liquid chromatography (HPLC) with an ultraviolet detector. The technique for testing is described in ASTM D5837-2012 [B17], but there remains no guidance for interpreting the results.

NOTE 1—The IEEE Transformers Committee is forming a working group to prepare interpretation.

The concentration of the furanic compounds gives an indication of the condition of the paper in terms of the degree of polymerization, while the rate of change of furan concentration can indicate the rate of aging of paper. *Generally, the total concentrations are less than 0.5 ppm and, in some cases, these levels may be maintained throughout transformer life.* The types and concentrations of furans in the insulating liquid sample can also indicate the occurrence of abnormal stresses in a transformer, whether short duration overheating of the insulation or prolonged general overheating.

The main advantage of using this technique as a diagnostic tool is that these furan compounds are degradation byproducts specific to paper which are soluble in oil but cannot be produced by the oil itself.

NOTE 2—Another source of furans in new insulating liquid is the residual tetrahydrofuran (THF) used in European refining as a solvent. The THF process is not used in US refined insulating liquid.

Annex F

(informative)

Frequency response testing

F.1 General

Frequency response analysis (FRA) is a diagnostic technique for detecting geometric change(s) related to the internal characteristics of a power transformer. The FRA measurement produces a transfer function from the resistive, capacitive, and inductive elements that represents the mechanical geometry of a power transformer. Detecting mechanical change or damage to transformer windings is one of the main interests of FRA test measurement. Such changes can result from various types of electrical or mechanical stresses (shipping damage, seismic forces, loss of clamping pressure, short-circuit forces, etc.). The measurement is performed over a wide range of frequencies, and the results are compared with a reference “signature” or “fingerprint” to make a diagnosis.

F.2 Purpose

Detecting mechanical change or damage to transformer windings is one of the main interests of FRA test measurement. Such changes can result from various types of electrical or mechanical stresses (shipping damage, seismic forces, loss of clamping pressure, short-circuit forces, etc.).

There are several distinct reasons to generate diagnostic FRA measurements, as follows:

- After factory short-circuit testing
- Relocation and commissioning validation
- Post-incident: lightning, external through-fault, internal short circuit, seismic event, etc.
- Routine diagnostic purposes
- Condition assessment of older transformers
- Evaluation of used or spare transformers

The FRA test has been driven by the desire to detect mechanical failures within a transformer. Failure modes are not exclusive to geometric variations within a transformer and can include variation in the core’s magnetic circuit and contact resistance. FRA test variations can be caused by a single type of failure or a combination of two or more.

FRA is known to be useful in detecting the following failure modes:

- Radial “hoop buckling” deformation of winding
- Axial winding elongation “telescoping”
- Overall bulk and localized movement
- Core defects
- Contact resistance
- Winding turn-to-turn short circuit

- Open-circuited winding
- Winding loosen due to transportation
- Floating shield

F.3 Test equipment

Test equipment should produce a frequency response measurement with the following characteristics:

- The test should be made over a wide range of frequencies so as to be able to diagnose problems in the core, clamping structure, windings, and interconnections.
- Successive measurements should have adequate resolution to give unambiguous diagnosis.

The test equipment should have the following attributes:

- The equipment should be calibrated to a traceable standard.
- The output power of the excitation source should provide adequate power over the entire frequency range to allow for consistent measurement of the transfer function across the frequency range.
- The test set should be capable of measuring sufficient dynamic range over the frequency range to accommodate most transformer test objects.
- The test set should be capable of collecting a minimum of 200 measurements per decade, either spaced linearly or logarithmically.
- The test system (set and leads) should provide a known and constant characteristic impedance. The test set and lead characteristic impedances should be matched.
- A three-lead system, signal, reference and test, should be used to reduce the effect of leads in the measurement.
- Test leads should be coaxial cables as close to the same length as possible and less than 30 m (100 ft) long. Shielded test leads should have the ability to be grounded at either end.
- Both the magnitude and phase angle of the calculated transfer function should be presented.

F.4 Test procedure

F.4.1 Safety

A frequency response measurement should be done in a safe and controlled manner irrespective of test location.

- Any transformer under test should be completely isolated from any HV source or power system source.
- The transformer tank should be grounded.
- All instrumentation shall be grounded appropriately for the specific test setup and isolated from any high voltage source or power system source. Avoid subjecting the test instrument, test leads, or power supply to station wiring surges and external interference, including transferred potentials.
- During the test, there should be strict adherence to local safety regulations and guidelines.

F.4.2 Preparation

It is recommended that the transformer be as close to “in-service” condition as possible.

External bushing connections should be disconnected, and, whenever possible, test lead connections should be made directly to the bushing terminals. This includes phase connections, neutral connections, and tertiary grounds. Any extra conductor length included in the test circuit path influences the FRA test result.

F.4.3 Tap positions

It is recommended that the LTC be in the extreme raise position.

It is recommended that the DETC be in the position dictated by in service conditions.

Transformers in service occasionally have problems due to DETC movement; it is not recommended that the DETC position be altered for an FRA test.

F.4.4 Test connections

The following three leads should be used:

- Excitation “source”
- Specimen input “reference”
- Specimen output “measure”

These leads should ideally be as close to the same length as possible and have characteristic impedance matched to the test set. As a minimum, the “reference” and “measure” leads should be identical.

Recommended and alternative test connections instructions are provided in IEEE Std C57.149™-2012 [B47].

F.5 Analysis of results

Since transformer designs and applications vary, the FRA results inherit diverse properties and characteristics. However, a FRA trace over specified frequency ranges has a degree of predictability for low-frequency core effects, main winding effects, and short-circuit responses. These expectations can be used to identify basic problems that may exist within a transformer.

Trace comparison is the primary method for the analysis of FRA results. Comparisons can be made against the following:

- Baselines
- Similar units
- Across phases

Traces can also be examined for expected patterns.

Depending on the test connections, the FRA results can be used to confirm the results of other diagnostic tests. These tests include the following:

- Single-phase excitation current
- Turns ratio
- Short-circuit impedance (leakage reactance)
- DC winding resistance

Annex G

(informative)

Dielectric frequency response

G.1 General

Dielectric frequency response (DFR) refers to a measurement of the dielectric properties, expressed as, e.g., capacitance (C) and power factor (PF), of an insulation system as a function of frequency. This is also known as *frequency domain spectroscopy* (FDS), which is an advanced diagnostic test for the field. Any factory testing is only for a signature and not an acceptance test for the power transformer. The effect of moisture and ionic contamination on the dielectric properties of the insulation system is more pronounced at low frequencies. For a mineral oil/cellulose insulation system used in transformers, the elements involved in this analysis include the moisture in the cellulose material, the conductivity of the oil, and the presence of contaminants or other materials that affect the capacitance or dielectric loss of the system.

G.2 DFR test procedure

The DFR test is performed by applying a varying frequency low-voltage signal to the insulation system under test and measuring the applied voltage, current, and phase angle to determine the specimen capacitance and PF over the frequency range of interest. The connections for the test are the same as those used for the standard capacitance and PF measurements; see Figure G.1. The elements in the transformer that are tested generally include the insulation between isolated winding sections and between the windings and ground. The measurements between windings and ground generally include the insulation between the windings and the core or other grounded parts of the transformer; the bushings; and the insulation of tap changers, reactors, and other accessories connected to line potential.

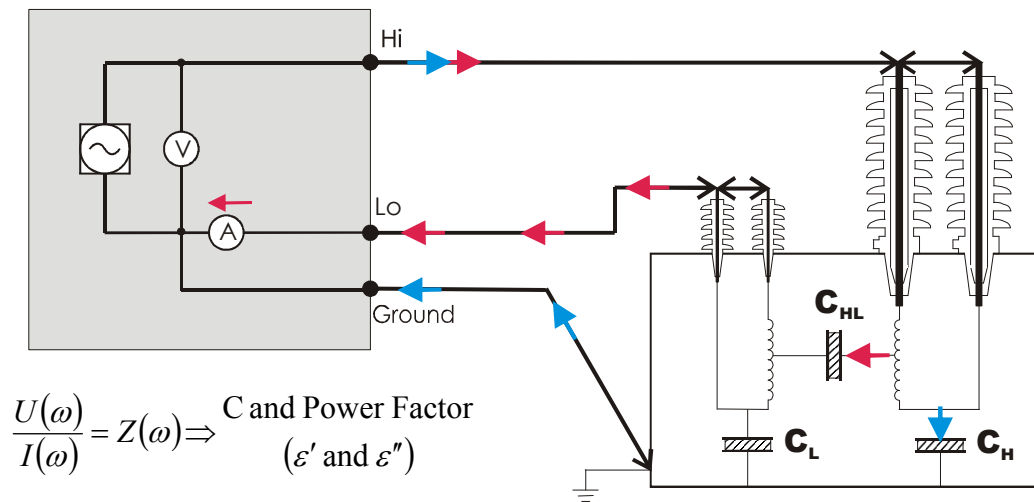


Figure G.1—DFR measurement of capacitance and power factor of insulation between high voltage winding and low voltage winding, labeled C_{HL} .

In Figure G.2, the discrete measurement data points from a typical measurement on a new power transformer are plotted. In Figure G.2(a), the capacitance and the dielectric losses are plotted as function of frequency. The plot in Figure G.2(b) shows the PF as function of frequency.

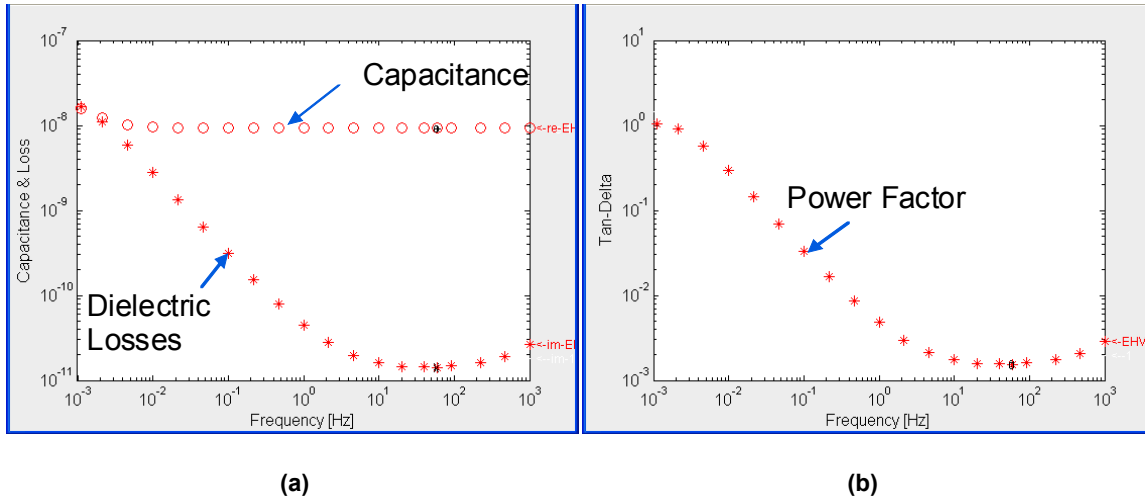


Figure G.2—DFR measurement between high voltage windings and low voltage windings of a new power transformer

G.3 DFR test analysis

The DFR interpretation is model based as recommended in Gubanski et al. [B29] and based on the X-Y model. See Figure G.3. The geometrical properties used in the X-Y model include the geometric capacitance, which is calculated based on the area of the windings in the inner winding insulation and the spacing between windings, and the relative amounts of insulating liquid and cellulose material, both in the direction between windings (barriers, washers, etc.) and tangential to this direction (spacer sticks or blocks). This method has also proven to be useful for other winding configurations and shell-type designs. the geometrical properties used to calculate the X-Y model can be taken from the design data or they can be estimated using the modeling. It is noted that approximating the X-Y parameters may lead to inaccuracies in the “measured” results.

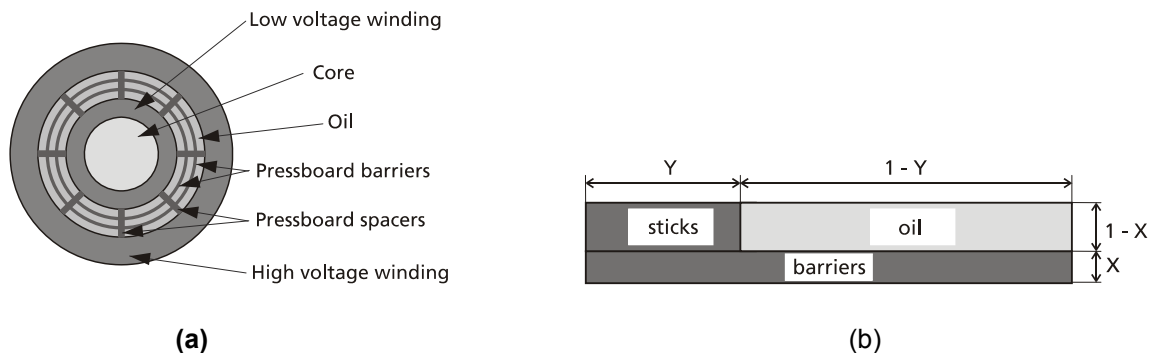


Figure G.3(a)—Drawing of high voltage and low voltage winding and its separation by use of barriers and spacers and oil in oil ducts; (b) the relationships of pressboard barriers, spaces, and free insulating liquid represented by the geometrical parameters X and Y

The interpretation software basis is to compare the measurement results with a combination of the two materials at an approximate ratio (X-Y) and at given temperature. From an optimization, the estimated moisture content, insulating liquid conductivity (or insulating liquid PF), and any abnormalities at given temperature are given as output. See Figure G.4.

The insulating liquid is modeled with insulating liquid permittivity and insulating liquid conductivity, which is proven to be an adequate assumption.

The cellulose paper is modeled by a database of measurements obtained from laboratory samples at different moisture contents. Additional sets of data can be loaded to the software.

NOTE—Database of measurements are not openly available to the industry and are not based upon service aged equipment. The temperature effect is taken into account using the Arrhenius formula, with activation energy in the range of 0.9 eV to 1.1 eV.

See Figure G.5.

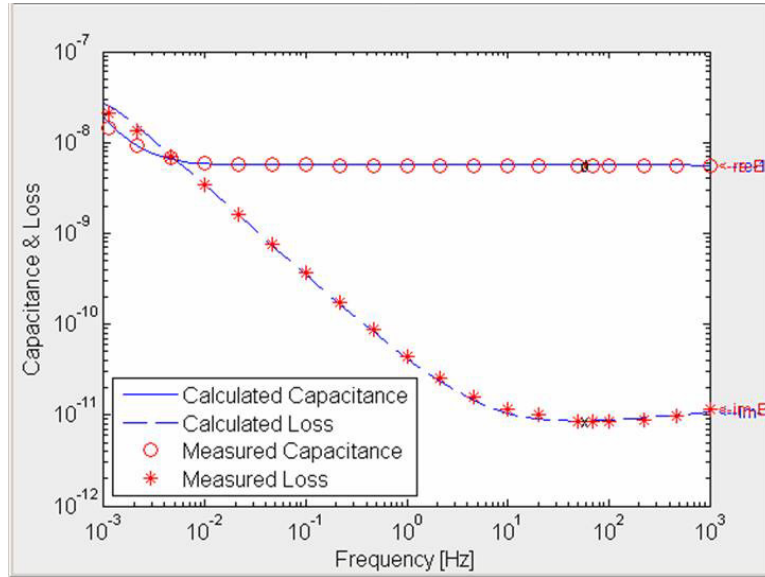


Figure G.4—Mode curve (blue line) matched to measurement points (red stars) by adjusting moisture content and insulating liquid conductivity

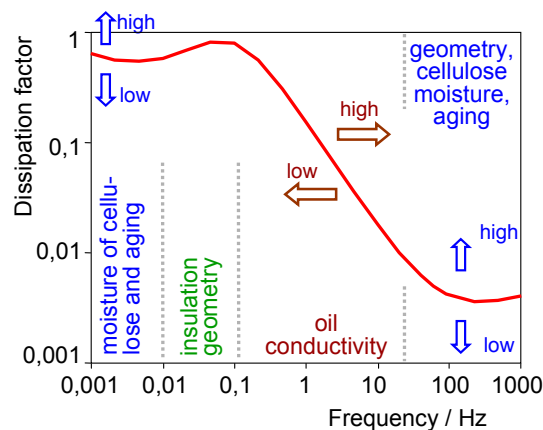


Figure G.5—Interpretation scheme for DFR providing discrimination between influences of moisture, aging, insulating liquid conductivity, and insulation geometry

From Figure G.5, it becomes clear that the:

- Losses at medium frequencies (below 10 Hz in this example) are determined by the insulating liquid conductivity.
- Lowest frequencies (below 0.01 Hz in this example) reflect the moisture concentration in the solid insulation.

The frequency ranges of Figure G.5 may vary depending on the insulation condition. In particular, the characteristic S-shape curve shifts towards lower frequencies for transformers at colder temperatures and towards higher frequencies for hotter temperatures. At the same time, the moisture concentration and aging by-products can change the shape of the characteristic curve.

Annex H

(informative)

Other methods to verify polarity from previous field test guide revisions

H.1 Verification of transformer polarity

A number of commercial transformer turns ratio test sets are available from manufacturers serving the power industry. These instruments, when operated in accordance with the manufacturer's instructions, provide convenient and accurate readings of power transformer ratios and polarities.

If a commercial test set is not available, then transformer polarity may be measured and interpreted using the procedures in H.1.1 and H.1.2.

H.1.1 Verification of transformer polarity by inductive kick

Polarity by inductive kick may be measured using two dc voltmeters and a source of dc current. For safety reasons, it is preferable to apply the dc source across the HV winding. Figure H.1 illustrates the technique.

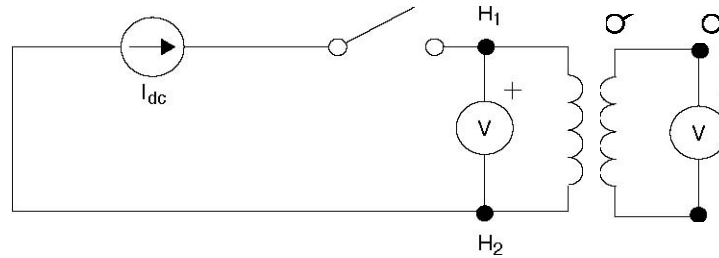


Figure H.1—Polarity by inductive kick

A dc voltmeter should be placed across the H_1 – H_2 leads, with the positive lead connected to the H_1 terminal, and a dc voltmeter should be placed across the X_1 – X_2 leads, with the positive lead connected to the X_1 terminal.

An LV source, such as a battery, should be connected to the H_1 – H_2 terminals, thus causing a small but noticeable deflection of the dc voltmeter connected across the H_1 – H_2 terminals. The connection of the dc source should be such that the dc voltmeter indication is positive. The magnitude of the deflection is not of concern.

The direction of the deflection of the dc voltmeter connected across terminals X_1 – X_2 should be observed as the excitation is broken. If the deflection is positive, then the transformer is additive. If the deflection is negative, then the transformer is subtractive. The polarity, not the magnitude of deflection, is of concern.

This test should be repeated for each phase of a polyphase transformer.

H.1.2 Verification of transformer polarity by alternating voltage

If the transformer ratio is < 30 , then polarity may be measured by using a convenient ac source with an ac voltmeter, as shown in Figure H.2.

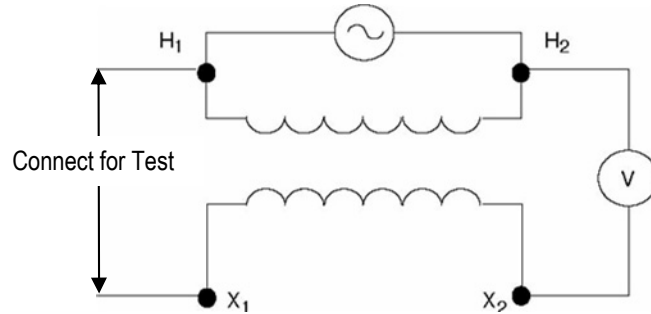


Figure H.2—Polarity by ac method

The transformer should be connected as shown in Figure H.2. Connect H_1 to X_1 for testing.

A small alternating voltage (as measured in the tens of volts) supplied by a fused variable transformer should be applied to the H_1 – H_2 leads.

If the ac voltmeter indicates a value less than the source voltage, then the polarity is subtractive. If the voltmeter indicates a value greater than the source voltage, then the polarity of the transformer is additive.

H.1.3 Polarity of polyphase transformers

Each phase of a polyphase transformer should have the same relative polarity when tested in accordance with either of the methods described in H.1.1 and H.1.2 or with a commercial instrument.

H.2 Voltmeter method for ratio

Two ac voltmeters are used, one connected to the HV winding and the other connected to the LV winding. The HV winding is excited to a voltage not exceeding the rating of the voltmeter. Both voltmeters are read simultaneously. A second set of readings should be taken with the instruments interchanged. The values indicated should be averaged to calculate the ratio.

A meaningful ratio measurement may be made using only a few volts of excitation. The transformer should be excited from the highest voltage winding to avoid possibly unsafe high voltages. Care should be taken during the application of voltage and during the measurement. It is important that simultaneous readings of both voltmeters be taken.

The voltmeters used should have accuracies commensurate with the requirements of a 0.5% ratio calculation.

H.3 Ratio measurement using a capacitance and power factor bridge

Transformation ratio may be measured with a capacitance and power factor bridge (sometimes called a *dissipation factor* or *DF bridge*). This method provides good results with power transformers as well as with potential transformers, where the phase angle error can also be measured. In addition, higher voltage tests may be performed, up to the rating of the instrument, which is frequently 10 kV or 12 kV. Several excellent instruments are available for this purpose. The manufacturer's instructions should be consulted for the exact procedure for the bridge used.

Annex I

(informative)

Particle count

I.1 Sampling

The sampling procedure is critical to avoid contamination and proper procedures, ASTM D6786 [B18] and ASTM D923 [B3] must be followed as a minimum requirement. One consideration is to obtain commercially prepared bottles according to ISO 3722 specifications for particle sample collection. ASTM D6786 [B18] defines bottle cleanliness or clean as a bottle which contributes less than 1% particles, of the lowest expected results, to the sample. Special care should be taken to keep airborne particles from entering the bottle. The sample should be kept out of direct sunlight and stored in a cool, dark location until the analysis is completed.

I.2 Particle count interpretation

The particle count test (ASTM D6786 [B18]) determines the number and size of particles present in a mineral insulating oil sample. The sample should be at least 100 ml. The data that is reported is the number of particles greater than a specific particle reference size (μm) in a specified volume of liquid (ml). The most modern testing procedures use automatic particle counters (APCs). It is important to understand the calibration method used by the APC for testing.

Prior to 1999, the calibration method used was ISO 4402:1991 [B50]. This method of calibration used General Motors, Air Cleaner Fine Test Dust (ACFTD), often called *Arizona Test Dust*. This calibration method will produce a count for particle size in μm and a count of all particles exceeding the size, as follows: $>2\ \mu\text{m}$, $>5\ \mu\text{m}$, $>10\ \mu\text{m}$, $>15\ \mu\text{m}$, $>25\ \mu\text{m}$, $>50\ \mu\text{m}$, and $>100\ \mu\text{m}$. ACFTD is no longer manufactured, which led to the development of a new calibration standard.

In 2000, a new standard, ISO 11171:2010 [B52], used a method of calibration using Medium Test Dust (MTD). This calibration will produce a count for particle size in μm and a count of all particles exceeding the size, as follows: $>4\ \mu\text{m}$, $>6\ \mu\text{m}$, $>10\ \mu\text{m}$, $>14\ \mu\text{m}$, $>21\ \mu\text{m}$, $>38\ \mu\text{m}$, $>70\ \mu\text{m}$ followed by a (c). The “c” indicates that the particle counter is calibrated per ISO 11171. The National Institute of Standards and Technology (NIST) have set a traceable calibration standard for MTD or standard reference material (SRM) 2806 or ISO 12103-A3.

Depending on the lab being used for testing, it is important to understand to which standard the test instrument has been calibrated. The laboratory performing the test should follow ASTM D6786 [B18] or ISO 4406:1999 [B51]. Both of these testing methods have been updated to reflect ISO 11171 calibration method using MTD.

Some laboratories may still have testing instruments that are calibrated to ACFTD. If so, the ACFTD sizes of $>2\ \mu\text{m}$, $>5\ \mu\text{m}$, $>10\ \mu\text{m}$, $>15\ \mu\text{m}$, $>25\ \mu\text{m}$, $>50\ \mu\text{m}$, and $>100\ \mu\text{m}$ correspond nearly to MTD size of $>4\ \mu\text{m}$, $>5\ \mu\text{m}$, $>6\ \mu\text{m}$, $>10\ \mu\text{m}$, $>14\ \mu\text{m}$, $>21\ \mu\text{m}$, $>38\ \mu\text{m}$, and $>70\ \mu\text{m}$ (c), respectively, and may be used to convert from one method to the other.

If testing has been done according to ASTM D6786 [B18], the results are reported as the average particle count runs as the cumulative number of particles **per ml** $>4\ \mu\text{m}$, $>6\ \mu\text{m}$, $>10\ \mu\text{m}$, $>14\ \mu\text{m}$, $>21\ \mu\text{m}$, $>38\ \mu\text{m}$, $>70\ \mu\text{m}$ (c). If the device is calibrated to ASTM D6786 [B18] using ACFTD, the results are reported in the average particle count runs as the cumulative number of particles per ml $>2\ \mu\text{m}$, $>5\ \mu\text{m}$,

>10 μm , >15 μm , >25 μm , >50 μm , and >100 μm . Some laboratories may report particle count per 10 ml or 100 ml and division by the amount of ml reported will be necessary to obtain a per ml base.

As an alternate, the testing laboratory may report the results in the ISO 4406:1999 [B51] format. If the laboratory uses this system, the user should obtain a copy of ASTM D1275-2006 [B7]. Table I.1 provides guidelines obtained from manufacturers', laboratory tests of transformers, and CIGRE [B21]. One other published document (Service Handbook for Power Transformer [B69]) provides some guidelines for both the ACFTD and MTD calibration methods. Action points for microscopic counting (CIGRE [B21]) are provided in Table I.1.

It should be noted that some manufacturers of transformers have particle count limits for prior to energization and these limits may differ significantly from in service transformers. Further, the particle limits for EHV transformers >242 kV may be as much as half the action points indicated in Table I.1. It is recommended that for units >242 kV the manufacturer be contacted if the limits are unknown or have not been provided.

No empirical limits exist with respect to condemning insulating liquids for operation of equipment or continued use solely based on particle count. Particle counting is a diagnostic tool, similar to other fluid diagnostics that are used in conjunction with other tests to evaluate insulating liquids. There are no grounds to reject a transformer based on these particle limits alone. If marginal particle action points are reached or exceeded and water content (ASTM D1533 [B11]), PF (ASTM D924 [B4]), and dielectric breakdown (ASTM D1816 [B13]) are not within specified limits for the transformer voltage class, further evaluation should be done to determine the particle make up to ensure corrective action is appropriate. One method is Analytical Ferrography that yields information on the following types of particles:

- Carbon particles
- Nonferrous metals
- Ferrous metals
- Sliding wear particles
- Cutting wear particles
- Arcing spheres
- Overheated metal particles
- Insulation particles and fibers
- Dirt and debris
- Film particles or “varnish”

Processing the liquid may improve the factors but trending should be initiated to insure the problem has been solved.

NOTE—IEEE has not established any limits. These suggested points are from CIGRE, laboratory research, and manufacturers as previously referenced.

**Table I.1—In-service transformer suggested particle count action points
in particles per mL by μm**

	MTD			Microscope		ACFTD		
	$>4\ \mu\text{m}$ (c)	$>6\ \mu\text{m}$ (c)	$>14\ \mu\text{m}$ (c)	$\geq 5\ \mu\text{m}$	$\geq 15\ \mu\text{m}$	$>2\ \mu\text{m}$	$>5\ \mu\text{m}$	$>15\ \mu\text{m}$
Normal	1500	150	3	320	40	1500	150	3
Marginal	2500	160	5	1300	160	2500	160	5
High	5000	320	10	2500	320	5000	320	10

NOTE—Some believe that the quantity of particles and their composition are significant only in their relation to previous levels and types. Trends observed may be significant to determine if excessive cooling pump bearing wear or insulation degradation is being experienced. Some particles, not suspended in the insulating liquid, are rarely seen since they have a tendency to fall to the bottom of the equipment's tank and are not available to be sampled and are removed during valve flushing.

Annex J

(informative)

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